Mass Exit in the Market for Governance

A Behavioral Foundation for Theories of State Development[*]

Shusuke Ioku[†]

October 15, 2026

Abstract. State development theories posit that political fragmentation lets subjects exit to other authorities, spurring jurisdictional conflict that drives institutional development. I provide a behavioral micro-foundation for this mechanism by developing a theory of exit through alternative authority and testing its implications in early-modern Japan. In Tokugawa Japan, villages practiced mass exit to gain bargaining leverage against their home rulers through assistance from outside authorities. I exploit domain conversions to central rule, which brought central authority closer to neighboring villages, as a natural experiment. Difference-in-differences analysis shows a 4.3 p.p. increase after conversions in the probability of exit among nearby villages under other lords. The analysis further shows that proximity to neighboring capitals predicted more exit and that domains with easier access to neighboring rulers tended to set lower tax rates, especially with economically attractive destinations. These findings strengthen the behavioral foundations of theories linking political fragmentation to state development.

Total Words (Main Text): 13,824

[*]I thank my advisors, Scott Abramson, Casey Petroff, and Hein Goemans, for their guidance. I also thank Matteo Bertoli, Amy Catalinac, Nathan Feldman, Mark Fey, Fransisco Garfias, Jenny Guardado, Yukari Iwanami, Daiki Kishishita, Alexander Lee, Bingru Li, Chiaki Moriguchi, Emily Sellars, Yu Xia, and all participants of JPOSS (Japanese Politics Online Seminar Series), the University of Rochester Graduate Research Seminar and Brownbag Seminar, and JSQPS 2024 Summer Meeting for their helpful comments and suggestions. Finally, I thank three anonymous reviewers and the two editors for their valuable comments.

[†]Ph.D. Candidate, Department of Political Science, University of Rochester

1. Introduction

Many influential theories of state development rest on a common premise: political fragmentation gives subjects access to alternative authorities, and that access disciplines extraction and shapes institutions. This premise does the central causal work in leading explanations for the Great Divergence (Cox 2017; Fernández-Villaverde et al. 2023; Jones 2003; Mokyr 2007); in accounts linking Europe's polycentric structure to constrained taxation (Abramson 2017; Dincecco and Wang 2018; Sng and Moriguchi 2014; Stasavage 2016); in fiscal-federalism arguments that competing jurisdictions tame Leviathan (Qian and Weingast 1997; Rodden 2003; Tiebout 1956; Weingast 1995); and in recent theories arguing that the modern territorial state system cartelized the *market for governance* by restricting subject mobility (Acharya and Lee 2018; 2023). These accounts differ in scope, but they share one mechanism: fragmentation gives subjects credible exit options, and rulers compete or cooperate when subjects threaten or use exit. There is still little direct evidence on the behavioral link: whether access to authority outside the home jurisdiction shaped how subjects resisted. I test this.

Subjects have used mass exit to resist rulers in many historical contexts. In medieval Europe, migration to towns offered serfs a way to resist their lords (Adas 1981, 217). In precolonial Java and Burma, cultivators escaped excessive demands by transferring their allegiance to rival patrons (Adas 1981, 229–231). Gandhi described an Indian village whose inhabitants began leaving in protest against their prince, prompting him to withdraw his command (Gandhi and Parel 1997, 95). In Tokugawa Japan, villages fled collectively into neighboring domains and petitioned neighboring rulers for permission to remain (Moore 1993, 255). These episodes illustrate exit as a bargaining tactic against the home ruler, often with return once concessions are granted, rather than passive or permanent relocation. I define such resistance tactics as *exit through alternative authority*: any action that moves subjects or their grievances beyond the home ruler's exclusive control, whether by flight to another authority, a petition routed through a neighboring ruler, or an appeal over the home lord's head to a superior.

A common context for such resistance combined horizontal rivalry among local rulers with vertical oversight by a common superior; Section 2.1 develops this architecture.

In this paper, I develop a theory of how exit through alternative authority shapes the mode of popular resistance and constrain extractive institutions in the horizontal–vertical structure of authority. When subjects exit, a neighboring ruler can shelter them and provide protection or mediation, allowing them to reside outside their home jurisdiction during bargaining. Appealing to a superior authority may also bring investigation or discipline from above, posing an additional threat to the local ruler's position. These horizontal and vertical routes can operate independently or jointly, making temporary departure a credible bargaining tool, even though it usually entails considerable cost for subjects. The credibility of the exit threat also depends on the incentives of both neighboring rulers and the higher authority to intervene.

Tokugawa Japan (1615–1868) provides an ideal test case for this mechanism of exit through alternative authority. Nearly 300 feudal domains exercised local autonomy, each with its own legal code, tax system, and border controls, under the Tokugawa shogunate as a common superior. In this order, horizontal rivalry among domain lords coexisted with vertical oversight from above. The shogunate retained supra-domain arbitration, attainder power, direct peasant-appeal authority, and the power to assume direct rule (Moore 1993; Ravina 1999; White 1995). In the empirical analysis, these actions appear as collective flight into a neighboring domain (*chōsan*), petitions routed through a neighboring lord, and appeals over the home lord's head to the shogunate (*osso*).[3]

This context helps address two central empirical challenges. First, episodes of mass exit are often more subtle and covert than outright rebellion, which means they rarely leave extensive historical records (Bailey 2014, 76). As a result, most studies must infer exit only indirectly. In Tokugawa Japan, however, newly digitized records of peasant resistance compiled by Aoki (1975) provide systematic documentation of both direct confrontation and exit via alternative authorities at the village level. Second, Tokugawa Japan offers considerable variation in access to outside authority, all within a single

[3]In Tokugawa Japan, mass exits were often acts of resistance rather than permanent relocation. Peasants presented demands to home-domain officials and at the same time sought permission from neighboring rulers to cross the border; documented cases include return once concessions were granted. Moore (1993, p. 255) relies on Borton's catalog to document 106 such mass desertions across the period, citing them as evidence of "the solidarity of the Japanese village." My catalog extends that count and covers the full archipelago.

legal, linguistic, and religious framework. By contrast, region-level comparisons—such as Europe versus China, a common qualitative approach—struggle to isolate the effect of access to outside authority, since regions also differ in law, kinship patterns, religion, bureaucratic recruitment, and geography. In Tokugawa Japan, nearly 300 domains varied in size, wealth, and their villages' proximity to rival rulers, yet all operated within a shared institutional order. Furthermore, the peace established after 1615 dampened the recurring war-finance pressures that Rosenthal and Wong (2011) argue overwhelmed the disciplining effects of exit in early modern Europe. This long period of stability thus allows the analysis to focus more cleanly on the vertical relationship between rulers and their subjects, rather than on conflict between rulers, in shaping institutions (Moore 1993).

The analysis leverages cases in which domains were converted to direct shogunal rule (*tenryo*), which brought shogunal authority geographically closer to neighboring villages. Applying a difference-in-differences approach, I find that following these conversions, villages situated near the new tenryo but governed by other lords became 4.3 percentage points more likely to opt for exit through external authorities. Critically, there was no corresponding increase in direct confrontation, and event study estimates suggest no pre-existing differences between groups before the conversion.

To probe this relationship further, I construct a grid-year panel that records the creation, abolition, and relocation of neighboring domain capitals for villages in each grid cell. This analysis strengthens the central finding: geographic exposure to overlapping jurisdictions, measured by the ratio of distances to the two nearest domain capitals, is robustly linked to the incidence of exit, but not to direct confrontation. Further examining data from cases in which rebelling villages chose between exit and confrontation, I find that villages situated closer to a foreign domain capital than to their home capital are more likely to exit, while those closer to their home capital are more likely to confront. Two additional results—first, that proximity to the nearest foreign domain capital is a stronger predictor of exit than proximity to the home capital; and second, that mere nearness to a jurisdictional border does not consistently predict exit rates—suggest that it is genuine access to an alternative authority capable of offering protection or mediation, that matters most in driving this relationship.

Finally, cross-sectional analyses from later periods indicate that villages with greater geographic exposure tend to report lower tax rates, especially where neighboring domains could plausibly absorb migrants, though estimates are subject to some uncertainty.

This paper advances several key contributions. First, it offers direct, micro-level evidence for a foundational behavioral mechanism shared across several strands of political economy: each tradition posits that access to outside authority enables exit, thereby shaping institutional outcomes. Specifically, these include the Great Divergence argument that competition among rulers for mobile subjects curbed extraction and fostered European development (Jones 2003; Mokyr 2007); state-formation theories in which the threat of exit compelled rulers to bargain and expand representation (Dincecco and Wang 2018; Stasavage 2016); fiscal federalism's claim that jurisdictional competition disciplines Leviathan through taxpayer and labor mobility (Rodden 2003; Tiebout 1956; Weingast 1995); and cartel theories of the territorial state, which hold that modern borders were designed to restrict subject exit (Acharya and Lee 2018; 2023). Much existing work infers exit only indirectly from migration and flight beyond state reach (Carneiro 1970; Herbst 2000; Peterson 1992; Pruett 2024; Scott 2009; Tiebout 1956), jurisdictional havens and fiscal responses (Guardado and Franco-Vivanco 2025; Kleven et al. 2020; Matranga and Natkhov 2025), or related voice-and-exit dynamics (Hirschman 1993; Lueders 2025; Michel, Miller, and Peters 2023; Pfaff and Kim 2003; Schoppa 2022). This paper instead provides direct evidence: village-level records that capture exit behavior itself. The findings demonstrate that greater access to external authority systematically encourages subjects to employ exit, thereby substantiating the crucial first link in this shared causal chain.

This paper brings exit, typically regarded as a background phenomenon in studies of state development, to the forefront, highlighting it as a bargaining and resistance tactic —not mere relocation or preference sorting—by which populations can check authority. Drawing on Hirschman (1970), most analytical traditions emphasize "voice": revolution and rebellion as engines of regime change (Acemoglu and Robinson 2001; Skocpol et al. 1982), collective mobilization as the driver of state formation (Hintze 1975; Tilly 1978), and the threat of revolt as a key constraint on extraction (Besley and Persson

2009; Fearon 2011). When exit is considered, it is usually relegated to a secondary role, acknowledged mainly for its influence on incentives (Besley and Persson 2009; Fearon 2011) or its role in long-term sorting (Jones 2003), and is rarely directly observed. Yet, historians of everyday resistance have long demonstrated that exit, not revolt, often predominated (Adas 1981; Bailey 2014; Camp 2004; Diouf 2014; Kolchin 1990; Scott 1976; 1985; Sreenivasan 2004). This paper adds empirical weight to that view by directly documenting exit as a distinct and observable form of resistance.

Finally, the paper contributes to the state development literature by showing how much an underrepresented non-European setting can add. Comparative work on fragmentation, exit, and institution building has drawn heavily on early modern Europe, while Tokugawa Japan has remained largely peripheral to that conversation. The Japanese case nonetheless offers distinct advantages for testing the shared behavioral premise. Village-level resistance records supply unusually granular data on exit and direct confrontation (Aoki 1975). Nearly 300 domains differed in size, wealth, and villagers' access to rival rulers, yet all lay within a single common political order and shared a basic institutional framework, such ascommon language, law, and broader religious order, so the analysis can compare divergent local units without the confounds of cross-regional comparison. The long peace after 1615 also dampened the recurrent war-finance pressures that Rosenthal and Wong (2011) argue overwhelmed exit discipline in early modern Europe, allowing sharper focus on how access to alternative authority shaped resistance and extraction. Overall, the study demonstrates the value of a Japanese context that this literature has largely overlooked.

2. Exit through Alternative Authority

War prompted rulers to build states in leading accounts of state development. Rulers, to finance armies, had to extract men, money, and supplies from those who controlled capital, land, and labor; the bargains they struck produced tax bureaus, representative assemblies, and ultimately citizenship (Tilly 1985; 1992). Much of the empirical literature since has studied how conflict shaped state building (Dincecco and Wang 2018; Gennaioli and Voth 2015; Spruyt 2017).

War incentivized rulers to extract greater revenue, but their ability to do so was constrained by their subjects' resistance. Scholars have argued that exit was often a more effective tactic of popular resistance than direct confrontation (Adas 1981; Bailey 2014; Moore 1993; Scott 1985). Tilly himself also states that mobility was a crucial limit on rulers' power, albeit primarily for elites. Merchants and financiers in the capital-rich cities of Europe could "escape with their capital to another business site if the ruler's demands became unbearable," prompting rulers in places like Genoa, Amsterdam, and the Hanseatic cities to compete for their loyalty by offering better terms (Tilly 1992, 53). By contrast, in coercion-intensive regions, rulers and landlords sought to bind peasants to the land: Russia's Assembly Code of 1649 and Prussian serfdom emerged precisely because peasant flight threatened rulers' revenues (Tilly 1992, 22–23, 141). In general, over-taxation could cost a ruler both taxpayers and, if subjects could exit, invite intervention from rival authorities. The competition between political authorities created by the possibility of exit helped curb arbitrary rule (Jones 2003) and shaped the emergence of the territorial state system with defined sovereign borders (Acharya and Lee 2018).

I refer to the form of exit studied here as *exit through alternative authority*. In this mode of exit, subjects submit themselves or their claims to the jurisdiction of another than their own ruler through fleeing to another authority, petitioning through a neighboring ruler, or appealing to a higher power. This is a different concept than exit as sorting. In the form of Tiebout (1956), people tend to move to the jurisdiction that best fits their preferences for taxes and public goods, and this move often ends their relationship with their previous jurisdiction. By contrast, exit through alternative authority serves as a tactic of resistance. Subjects may bring demands to their original ruler using the shelter, mediation or intervention of another authority and may choose to return after achieving concessions. I here develop a theory of how subjects use exit through alternative authority to constrain their rulers.

2.1. Horizontal and Vertical Authority

Exit through alternative authority exploits the leverage that political fragmentation affords subjects. Throughout history, one widely seen form of fragmentation combines

horizontal rivalry among local rulers with vertical oversight by a common superior. In these arrangements, subjects can turn to neighboring rulers or appeal upward to the superior, which increases their bargaining power. The pattern spans feudal monarchies and principalities—the lordships and towns of Languedoc under the early modern French crown (Gerring et al. 2011, 405) and the territories of the Holy Roman Empire under the emperor (Cantoni, Mohr, and Weigand 2024); empires and provinces, such as county jurisdictions within Qing China's provincial and imperial hierarchy (Sng and Moriguchi 2014, 446–447); colonial systems of indirect rule, such as princely states under the British Raj (Iyer 2010) and emirates of northern Nigeria under colonial supervision (Gerring et al. 2011, 399); and federal governments (Gerring et al. 2011, 377–382, 405–407).

Local autonomy and subjects' access to outside authorities varied across these settings. Local rulers taxed and governed their territories, but a superior authority could settle disputes, hear appeals, or intervene in cases of misrule. When dissatisfied, subjects normally have two main options: to seek redress from a neighboring local ruler (horizontal option) or to appeal to the superior (vertical option). Turning to a neighboring ruler threatens the home ruler's tax base, and appealing to the superior further threatens the ruler's position. These two routes may also operate together; for example, a neighboring ruler might grant shelter to subjects and then petition the superior on their behalf.

2.2. Actors and Choices

The key players in this dynamic are two actors: the ruler who relies on cultivators for tax revenue, and groups of subjects who pay taxes collectively. Higher taxes could mean more revenue from subjects who comply, but also could prompt others to push back. Subjects want to ensure their livelihoods and get concessions from the authority that taxes them. The cost of exit is particularly great for peasants with land. Their right to farm is typically tied to their household's claim or to membership in a landholding village, and the same land anchors their tax liability, unlike the rent tenants pay to village landlords. Distance, lost earnings, and the threat of losing land rights therefore make exit costly, even when a new authority is not far away. Land tenure arrangements shape

how large this cost is but rarely remove it (Section 3.2 documents the arrangements in Tokugawa Japan).

The rulers are aware that subjects compare the life they have at home with the life they can get somewhere else. When trouble comes, the first step is usually to appeal to the home ruler for relief. If that plea is denied, subjects can accept their lot (compliance), confront the ruler openly (direct confrontation), or leave (exit). Compliance preserves post-tax income without the costs of resistance. Confrontation might involve forceful demands, property damage, or even open rebellion; such actions can win concessions through collective strength but also expose subjects to severe reprisals. Exit instead allows them to escape extraction while causing fiscal and reputational damage to their rulers without incurring the costs of violent revolt (Adas 1981, 217–219; Scott 1976, 66). Research on peasant resistance highlights the importance of avoidance protest, including exit, as tactics that have traditionally been more prevalent and frequently more successful than aggressive protest (Adas 1981).

Exit can come in many forms, from individual sorting across jurisdictions to flight beyond the reach of the state (Scott 2009; Tiebout 1956). The form examined here is exit to alternative authority, where subjects or their grievances are brought under the jurisdiction of an outside authority other than their own ruler. Such an exit in the structure described in Section 2.1 exists in three forms: leaving the territory together, petitioning through a neighboring ruler, or appealing to the superior. An appeal to the superior counts as exit here, because it takes the dispute out of the hands of the home ruler, just as when subjects flee. Exit offers the possibility of outside help or intervention, but only if subjects can find an alternative authority to appeal to and can survive a difficult and uncertain process.

Exit and confrontation are different tactics but they often depend on the same people and the same collective organization. Departure can weaken confrontation in three ways: it reduces the number of participants, it increases the cost of mobilizing those who remain, and it makes the holdouts less confident in their ability to assemble enough support (Sellars 2019). Within one phase of a dispute, exit tends to replace confrontation if leaving pulls out the very people needed to act together at home, and if outside help cannot

make up for that loss. This logic is strongest in instances of mass exit and confrontation, the central cases this paper considers. Appeals by a few representatives, in contrast, can leave most subjects available for confrontation at home.

But sometimes exit and confrontation reinforce each other. Migrants may send money home, advice or organizational help (Pfaff and Kim 2003). The rest might have some bargaining power, knowing their peers have found alternatives (Karadja and Prawitz n.d.). The availability of an outside option alone can embolden additional collective demands (Clark, Golder, and Golder 2017; Hirschman 1993). Having recognition, protection, or political support from outside also makes it easier to protest at home. Exit and confrontation can therefore occur together or at different stages of the same dispute. The 1853 Morioka records, for example, contain end-run appeals and confrontation at different points in a broader cycle of resistance (Aoki 1975). The distinction here concerns individual actions rather than entire disputes; the empirical analysis accordingly treats these actions as separate event records.

2.3. How Exit Constrains Extraction

Exit limits how much a ruler can extract from subjects because it removes people and resources from the ruler's control, or at least establishes a credible threat that this will be the case. There are many ways this happens. When a mass exodus happens, the ruler immediately loses labor and tax revenue. If subjects seek help from a neighboring local ruler, the home ruler risks the neighbor taking them in and offering protection or mediation, or at least publicizing the home ruler's misrule. It could damage the reputation and standing of the home ruler. Or, if subjects appeal to a higher authority, their complaint may be investigated or disciplined from outside. Sometimes they combine these strategies, for example seeking a neighbor's protection and then filing a complaint to a higher authority. In any case, the key is subjects' access to an alternative authority, which strengthens their bargaining power.

It is crucial to distinguish between the ease of reaching another authority and the willingness of that authority to offer help. The former is shaped by geography and the relative position of jurisdictions: how many alternative authorities are actually accessible,

the distance subjects must travel to reach them, and the degree of cultural and legal difference between jurisdictions. When cultural and legal differences are minor, moving becomes substantially less costly. In early modern Europe, relocating could entail learning a new language, adopting a different religion, or living under unfamiliar laws. The latter, whether the receiving authority will provide help, depends on that authority's interests: the demand for new labor, and its willingness to risk antagonizing a neighboring ruler. Merely being close to a border does not guarantee support; proximity alone does not ensure access to shelter.

The incentives of the receiving ruler condition the willingness of an outside ruler to accept runners. Local rulers can determine taxes and welcome newcomers, which gives outsiders incentives. But not all local rulers are that independent. For instance, Qing China's counties easily slotted into a provincial and imperial system, and appointed magistrates typically served only a few years, giving counties less autonomy than Tokugawa Japan's domains, where lords ruled with a great deal of independence (Sng and Moriguchi 2014, 446–447). Where local rulers have it, keeping new cultivators means more revenue, sending refugees back only bolsters a rival's coffers. Reporting a neighbor's misrule can even get the target punished by the superior. These factors make it hard for rulers to form perfect cartels that shut down alternatives for their subjects (Acharya and Lee 2018). Cooperation among rulers nonetheless remains possible, and rulers could gain from sending migrants home; shelter costs money, rulers often owe each other favors, and pacts to repatriate fugitives can override competitive temptations (Section 3.2 describes both incentives in Tokugawa Japan).

If local rulers are subject to a greater authority, the superior adds another dimension. In the case of a common superior, an outside ruler may not only offer asylum but could also pressure the superior to punish the home ruler. Higher authorities can investigate, discipline or even punish the bad behavior of local rulers when they get involved. Superiors also benefit: petitions can alert them to crop failures or abuse that subordinates would otherwise hide. But hearing a complaint does not force the higher authority to help peasants win. Superiors could investigate abuses and still affirm unpopular taxes, or even punish those who appealed without permission (Vlastos 1986, 42–43, 53). Subjects

sometimes contact the superior directly; sometimes they ask a neighbor to back them up, which can put them in a stronger position. In any event, the home ruler's actions are observed from above and his freedom to solve matters his own way is curtailed. In late medieval England, subjects sought the Crown's protection against their lords' demands (Bailey 2014). In precolonial Burma and Java, peasants who found their taxes too high or their lord abusive could petition a higher official or the monarch himself (Adas 1981, 229), and villages in Tokugawa Japan took their grievances over their lords' heads to the shogunate (Section 3.2).

2.4. Observable Implications

Where external authorities are more accessible, we should see more exit and, if after-tax incomes in neighboring domains are attractive enough, lower tax rates as well. Supporting Information (SI) Section A sets out the logic for this effect of access to outside authority through a simple model of tax competition between two rulers governing mobile subjects.

The model offers predictions on both behavioral and fiscal outcomes. First, easier access to outside authorities should make exit more frequent even after rulers respond by reducing taxes. While lower taxes keep some subjects who might have left, they also reduce revenues from those who would have complied anyway. This tradeoff prevents rulers from simply cutting taxes enough to erase exit entirely. Under the model's conditions (SI Proposition 1), easier exit always leads to a greater likelihood of exit, no matter how rulers adjust. When improved access forces a tax cut, the share of exit among subjects who resist rises, because the tax cut pulls would-be confronters back into compliance while exit remains comparably attractive.

The overall effect of better access to outside authority on rulers' taxation depends on how appealing neighboring jurisdictions are, after taxes. When outside options offer high after-tax income, rulers have a strong incentive to lower taxes to compete for subjects. If the outside option is not so appealing, reduced confrontation at home may allow rulers to increase taxes instead (SI Proposition 2). In short, greater ability to exit leads to lower taxes only when neighboring jurisdictions make that option attractive enough.

3. Empirical Evidence

The empirical analysis is structured around three main components. First, the study investigates the phenomenon of exit through domain conversions to the central rule and spatio-temporal variation at the grid-year level. Second, it looks at the composition of resistance, what strategies villages used, exit or direct confrontation. Third, it evaluates the fiscal impact by comparing tax outcomes across domains.

3.1. Setting: Tokugawa Japan

During the Tokugawa period (1603–1868), nearly 300 domain lords (*daimyō*) governed Japan under the authority of the shogunate. Each domain maintained its own government and military forces and set its own tax rates (Ravina 1999, 1000). The shogunate also ruled its own territories, known as *tenryō*, and could transfer, reduce, or confiscate a lord's domain. It also arbitrated disputes between lords and heard petitions against them (Steele, Paik, and Tanaka 2017, 3–5). Villagers could thus seek assistance from the shogunate directly or indirectly through an neighboring lord (Section 3.2).

Villages paid land taxes in aggregate to their lord, and village leaders collected the necessary dues from individual households (Saxonhouse 1995; Walthall 1997, 89; White 1995). The function of the village as the unit of tax collection also allowed collective resistance, through legal petitions, confrontation, or departure. Section 3.2 describes the costs of such actions in terms of landholding and tax liability.

The capital of a neighboring domain provided the villagers with access to officials who would hear their grievances. Lords assembled their administration in castle towns, where officials levied taxes, dispensed justice, and enforced public order (Hall 1955). Villagers could cross a domain border and put themselves out of reach of their own lord's immediate control, while local officials might provide shelter or mediation. Villagers in the capital could bring their disputes before central domain officials who might offer official support or pass complaints on to the shogunate in Edo (Hall 1955; Ravina 1999). Being near a capital, therefore, offered opportunities for support other than immediate border refuge. Section 3.4 investigates this distinction by considering distances to capitals, borders and local offices.

Access to other authorities within this political order differed from village to village, and over time. Domains were of different sizes and wealth, and the location of villages and capitals made some villages much closer to a neighboring ruler than others. The creation and abolition of domains altered which rulers villagers could approach. The power available in the vicinity changed with conversion to direct shogunal rule even when the administrative center remained in place. Figure 1(a) displays the 264 capitals that were active in 1868 and Figure 1(b) displays the variation in their number over time. These differences allow us to compare the extent to which villages have access to outside authority in a common institutional setting. The resistance records collected by Aoki (1975) allow us to witness how those opportunities were utilized by the villagers (Section 3.2).

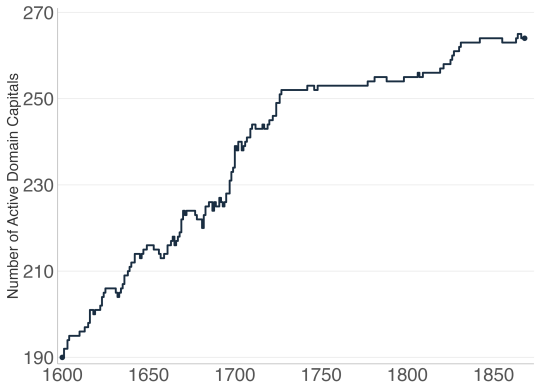
Tokugawa Japan provides a conservative testing ground for the exit mechanism. The shogunate and domain lords would not allow a peasant to leave without permission. Villagers who did run away faced punishment. However, mere flight to the territory of another lord did not guaranty safety, as neighboring lords might agree to return the fugitives (Section 3.2). Villagers therefore had to find a ruler who was willing to shelter them despite rules designed to keep them under control of their own lord. Evidence that access to neighboring authorities influenced resistance even in these constraints suggests that the mechanism can generally function in many other settings.

The long peace during the period further makes it easier to study the effects of access to other rulers by reducing the fiscal pressures of recurrent warfare. The settlement after 1615 largely prevented wars between domains until the regime's collapse in the 1860s.[4] While domains maintained military forces and defense obligations, they did not engage in continuous warfare against one another. This absence of persistent conflict reduces a source of fiscal pressure that could otherwise obscure the effects of exit (Rosenthal and Wong 2011, 169–184).

[4]The Shimabara Rebellion of 1637–38 was suppressed by a shogunal coalition, rather than fought between rival *daimyō* (Vlastos 1986, 8). The Chōshū campaigns of 1864–1866 and the Boshin War of 1868–1869 marked the return of large-scale domestic warfare near the period's end. Section 3.5 addresses the coverage of tax data for this late period.



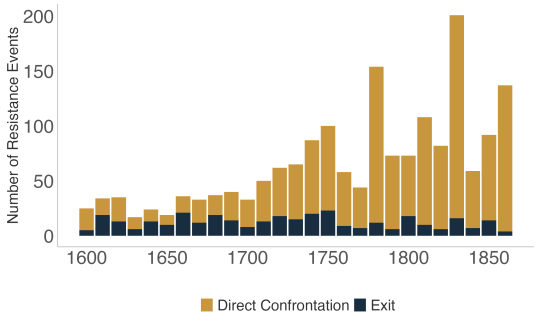
(a) Domain Capitals in 1868 (N = 264)



(b) Number of Active Domain Capitals, 1600–1868



(c) Spatial Distribution of Recorded Resistance



(d) Recorded Resistance over Time

Figure 1: The Tokugawa Domain System and the Record of Peasant Resistance

Note: Tokugawa Japan was divided into roughly 300 domains, each governed from a castle town that served as its capital and housed the officials who received petitions. (a) Locations of the 264 capitals active in 1868, plotted over terrain; the archipelago runs about 1,900 km southwest to northeast. (b) The number of active capitals each year, 1600–1868, which changed as the shogunate created, merged, transferred, and abolished domains. (c) Villages recorded as resisting, colored by whether they exited (left the domain or petitioned an outside authority) or confronted their lord directly; gray points are villages present through the end of the period with no recorded resistance. (d) Recorded episodes per decade.

3.2. Peasant Resistance In Tokugawa Japan

Aoki (1975) classified peasant resistance in Tokugawa Japan into five tactics: coercive appeals (*gōso*), destruction (*uchikowashi*), rebellion (*hōki*), desertion (*chōsan*), and end-run appeals (*osso*).[5] I map these tactics onto the distinction developed in Section 2.2, classifying the first three as *direct confrontation* and the last two as *exit through*

[5]Later studies follow this classification. White (1995) analyzes protest repertoires and their evolution; Vlastos (1986) treats end-run appeals as a distinct form that dominated the mid-Tokugawa period; Walthall (1997) translates primary documents from all five categories.

alternative authority.[6] Figure 1(c) maps the recorded incidents, and Figure 1(d) plots their frequency over time.

Direct confrontation took several different forms like coercive appeal, property destruction, and armed rebellion with varying intensity. Coercive appeals, the most frequent type of direct resistance, were mass marches on domain offices or castles, where villagers presented demands backed by an implicit threat of violence. Participants wore straw raincoats and carried bamboo spears as symbols of collective resolve (Walthall 1997, 131–132). Destruction targeted houses and storerooms belonging to officials, merchants, or wealthy villagers associated with oppressive policies (Walthall 1997, 89–130). Rebellion involved armed uprising against domain authority and accounted for roughly 1% of recorded incidents (White 1995, 142). These tactics sometimes secured concessions but also exposed participants to severe repression.

Villagers resisted through external authorities employing two kinds of exit strategies: desertion and end-run appeals. Section 2 shows that desertion was the horizontal path, as villagers left their home domain for refuge under a neighboring lord. The vertical path in Section 2, end-run appeals, sought to bypass the home ruler in seeking redress from another authority, often the shogunate. These appeals often blended the horizontal and the vertical routes, with a neighboring lord taking petitions under his wing or supporting their claims before the shogunate. I categorize desertion and end-run appeals as one type of exit for two reasons. First of all, they have one basic characteristic in common: both freed the villagers or their claims from the exclusive control of the home ruler. Secondly, they were often used together by the villagers. For example, in Aizu in 1642 they fled into neighboring domains and petitioned the shogunate (Vlastos 1986, 35–38).

3.2.1. Exit as a Bargaining Tactic

Peasants used exit to bargain over conditions at home, and several documented disputes ended in concessions and return (Brown 1993, 138; Steele, Paik, and Tanaka 2017, 356). Kaga domain officials, for example, attempted to bring farmers back by waiving arrears,

[6] "Direct confrontation" is more accurate than "fight" because coercive appeals were ritualized and often only implicitly violent.

offering rice provision, and promising to correct abuses (Brown 1993, 138). The negotiated return then allowed villagers to obtain relief and resume cultivation at home.

Landholding gave peasants a reason to seek concessions and return rather than permanently relocating. Households held cultivation rights conditional on paying land taxes (Nakabayashi et al. 2020, 158, 175), and passed their holdings down within the family (Kumon 2025; 2026, 158, 160; Nakabayashi 2025). Permanent departure could mean giving up such valuable rights on land. Where villages held land collectively, departure could likewise jeopardize access to fields (Brown 1993, 94–95, 107–108; Nakabayashi et al. 2020, 175–176). Negotiated return was thus an ideal outcome allowing villagers to press for relief sustaining their landholding rights.

Villagers could enhance their bargaining position through the two strategies outlined in Section 2.3. The horizontal strategy involved seeking protection from a neighbouring lord while advancing their demands. The vertical strategy entailed appealing to the shogunate, which could subject their home lord to sanctions. The subsequent subsections illustrate how villagers employed these sources of leverage, often in conjunction with one another.

3.2.2. Neighboring Protection

Neighboring protection provided villagers with bargaining leverage by imposing fiscal and reputational costs on their home lord. Legally, the shogunate and domain lords prohibited unauthorized peasant departure(Kuromasa 1927, 69–71). Also, neighboring domains often had agreements for the reciprocal return of fugitives (Miyazaki 1990, 38–39). Contrary to these restrictions on migration, domain lords also had a strong incentive to accept migrants, since their settlement brought increased revenue to his domain (Miyazaki 1995); they often sheltered migrating villagers, mediated their demands, or even accepted them as their subjects, as illustrated by the cases below.

Pressure from a neighboring ruler often assisted villagers in securing concessions. In 1841, 631 villagers from Yamashirodani left the Tokushima Domain and entered neighboring Imabari Domain, presenting sixteen demands related to monopolies and trade restrictions. They insisted on not returning without concessions and protection from punishment. The lord of Imabari supported their demands and warned that he

would consider them his own subjects if Tokushima did not comply. His officials provided food and accommodated the villagers in a temple and nearby homes while Tokushima dispatched officials to negotiate. After ten days, the villagers returned with a written agreement that accepted their demands and guaranteed no punishment (Aoki 1999; Okamura 2009). In 1741, rising rice prices, declining tea prices, and obligations related to paper production burdened villagers in the mountainous Kuma district of Matsuyama. An initial appeal yielded no effective relief. A second mobilization spread across twenty-six villages, leading villagers to cross into Ōzu, where they refused to return unless their complaints were addressed. Matsuyama sought the assistance of the abbot of Daihōji to mediate. The resulting settlement canceled proposed increases in tea and other levies, alleviated paper-related debts, and offered protection to protest leaders from investigation, although it did not fulfill every demand. Villagers returned in August, and senior domain officials faced exile (Ehime Prefectural History Compilation Committee 1986; Nakagawa 1980).

3.2.3. Shogunal Intervention

The shogunate carefully examined direct appeals from not only its own territories but also daimyō domains and sometimes punished domain lords for their over-exploitation or other types of bad governance (Kurushima 2002, 185–198; Vlastos 1986, 35–38, 55–59), providing villagers with leverage along the vertical path described in Section 2.3. The Buke Shohatto (Laws of Military Houses) required domains to be "well-governed," and failure gave the shogunate grounds to confiscate, reduce, or transfer a lord's territory (Steele, Paik, and Tanaka 2017, 3–5). While prohibiting direct appeals by peasants in general, the shogunate permitted ways to do so legally. The 1633 rules permitted appeals to the shogunate after domain-level petitions failed while threatening punishment for unauthorized fiscal petitions (Vlastos 1986, 42–43). The shogunate also introduced in 1720 a petition-box system that sent complaints about local officials directly to the shogunate for investigation (Sng and Moriguchi 2014, 447).

Vertical appeal to the shogunate was often practiced in conjunction with horizontal exit. During the Aizu uprising of 1642, several thousand villagers crossed into neighboring domains, publicly petitioning the shogunate and requesting permanent resettlement.

Their lord, being afraid of the shogunate's intervention, partly canceled arrears (Vlastos 1986, 35–38). In 1690, repeated floods and heavy taxation prompted villagers in Nobeoka Domain to seek relief from their lord. After he rejected their appeal, about 1,418 peasants crossed into Takanabe Domain. They stayed there for ten months while negotiations proceeded through the Takanabe lord (Steele, Paik, and Tanaka 2017, 356). Takanabe officials sheltered them, guarded their representatives, and helped take the dispute to the shogunate when negotiations stalled. Nobeoka accepted all demands in the end, including tax relief, and then the villagers returned without punishment (Ishikawa 1981). In cases known as daihyō osso (representative end-run appeals), village representatives filed an appeal to the shogunate on behalf of their villages while the rest of the villagers sought refuge in a neighboring domain (Miyazaki 1986, 305; White 1995, 158–159).

3.3. Exposure to Outside Authority and Exit

I examine whether access to outside authority affects the occurrence of mass exit using two designs. The first is a difference-in-differences (DiD) design that compares resistance in surrounding villages before and after nearby domains came under direct shogunal rule. The second uses a grid-year panel dataset to examine how the frequency of resistance varied with geographic access to outside authorities.

3.3.1. Tenryō Conversion DiD

I interpret conversions of domain territory to direct shogunal rule (*tenryō*) as increases in surrounding villagers' perceived exposure to shogunal authority. Conversion brought the authority closer, demonstrated the shogunate's willingness to intervene, and could make further intervention in adjacent territory easier to administer. Villagers in neighboring domains could therefore see an appeal to the shogunate as more effective.

To be clear, I predict that the tenryo conversion increased exit from neighboring domains in general, not specifically exit to the converted territory. As discussed in Section 3.2.3, direct appeal to the shogunate was often practiced in conjunction with an exit to a neighboring lord aiming for external assistance. The outcome variable here is thus the total number of exit events from neighboring domains regardless of destination, whose information is often not available in the source document. Despite this data limitation,

the result still provides evidence for the mechanism of exit through alternative authority, that greater exposure to outside authority increased exit through either or both horizontal and vertical paths, but not for exit to the converted territory itself.

Data. I construct a panel dataset of resistance events surrounding each domain capital that was active in a given year. Active domains and their capital locations for each year are identified using Kimura, Fujino, and Murakami (2015) and Kodama and Kitajima (1989). Branch domains (*shi-han*) are treated as part of their parent domains (*hon-pan*). Forty-nine conversion events between 1617 and 1807 are identified based on Kodama and Kitajima (1989). The complete list of conversions, along with their dates and causes, is provided in SI Section C.1. Each observation records the number of resistance events occurring in neighboring domains around each domain each year, together with an indicator for whether the domain was converted that year. Specifically, for each domain and year, I count the number of resistance events within a radius around the domain capital, excluding villages within the focal domain (Figure 2). I use radii of 30, 40, and 50 kilometers to demonstrate the robustness of the results to the size of the sampled area.

The resistance data are drawn from Aoki (1975), a compilation of incidents based on local histories, official records, and diaries. Aoki's chronology is widely regarded as the authoritative record of peasant resistance in Tokugawa Japan[7], serving as a foundational reference for both Japanese and international historical scholarship (Hosaka 2002, 18–19, 203–204; White 1995, 22–23). While comprehensive, the chronology cannot guarantee exhaustive coverage of exit events, particularly small-scale escapes. This paper focuses on mass departures that had implications for rulers' governance. Because such events could disrupt taxation and often necessitated official responses or negotiations, I expect them to be more thoroughly recorded in the sources Aoki consulted. However, the entries are less consistent regarding villagers' destinations or the authorities they petitioned. SI Section B.3 details all available information on destinations (Table A.7). Information about the duration of absence, payments during departure, and final settlements is also

[7]Hosaka (2002) characterizes it as the culmination of postwar Japanese research on peasant uprisings and "one of the largest contributions in Japanese historical studies after WW2."

incomplete; thus, the sources do not reliably establish typical lengths of absence or frequencies of return. Accounts of documented returns are compiled in SI Section B.2.

I construct the village sample by selecting eligible incidents and then identifying their participants. My transcription contains 3,020 event entries with 7,664 recorded village names. It is based on the peasant-uprising chronology in Aoki (1975) separately published chronologies of urban disturbances, village conflicts (*murakata sōdō*), and province-wide petitions are excluded from this sample. For each event, I record the date, participating villages, home authority, and tactic, following the classification in Section 3.2. Restricting these entries to those from 1615 onward and to those involving exit or direct confrontation leaves 1,142 episodes; non-violent petitions and other disputes that do not involve domain authority are excluded. I then separate each episode's list of participating villages into individual village-event records. Removing events without a named village or those taking place within a shogunal territory results in 940 episodes involving 2,738 distinct village names before geocoding. I locate these villages using several sources.[8] Villages whose locations remain unresolved are excluded. This process yields 910 episodes, 2,571 distinct villages, and 2,741 village-event records (Table A.5, SI Section B). I then use the coordinates and the time-varying domain capital data to identify each village's nearest outside authority at the date of the event.

Identification assumption. The DiD design requires parallel trends: in the absence of conversion, resistance in villages governed by surrounding lords should have evolved similarly near both converted and unconverted capitals. I assess the plausibility of this assumption by examining the pre-conversion trends in the event study below.

A major challenge for this comparison is the potential for the causes of conversion to also influence subsequent resistance in neighboring domains. Domain histories reveal three primary triggers for conversion: (1) succession, partition, or incapacity; (2) misconduct, misgovernment, or political sanction; and (3) administrative or strategic

[8] I use the historical village point data described by Honda, Natsume, and Nemoto (2022), together with the *Nihon Rekishi Chimei Taikei* Place-Name Dataset and the Historical Administrative Area Dataset maintained by the Center for Open Data in the Humanities. Modern settlement locations come from agricultural-settlement boundary data in the Ministry of Agriculture, Forestry and Fisheries' 2020 Census of Agriculture and Forestry database. I also use the address-search API of the Geospatial Information Authority of Japan.

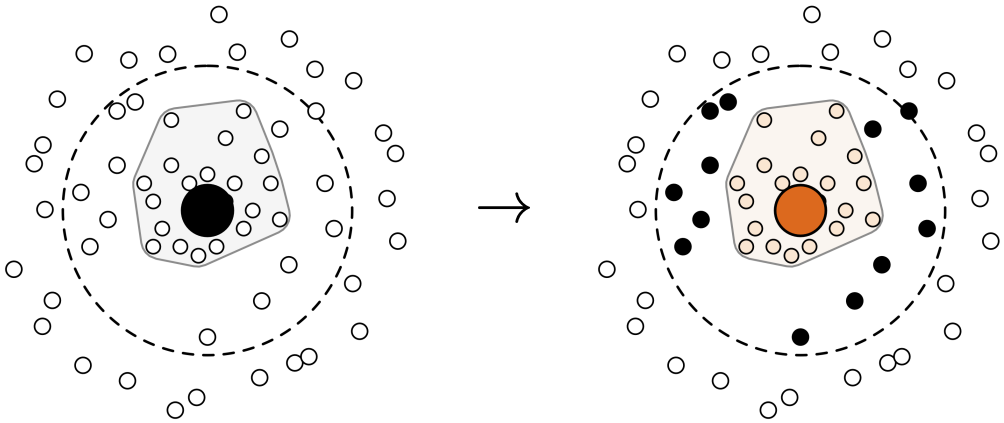


Figure 2: DiD Design: Tenryō Conversion

Note: The large black circle is the focal capital, the gray zone around it is its territory, small circles are villages, and the dashed circle is the analysis radius r. Left: the focal territory is governed by a domain lord. Right: conversion places it under direct shogunal rule (*tenryō*, orange), bringing shogunal authority closer to surrounding villages; black circles identify surrounding foreign villages inside r, while pale circles inside the gray territory are excluded from the outcomes and open circles fall outside the treated area.

reorganization (SI Section C.1). The majority, 38 of the 49 conversions (78%), arose from causes internal to the converted domain: succession failures or incapacity (21) and the lord's misconduct (17). Such internal affairs are unlikely to affect grievances among neighboring villagers. In contrast, fiscal distress or regional unrest could plausibly influence both the timing of conversion and resistance patterns in adjacent domains. In particular, if conversions were prompted by peasant revolts or fiscal distress, endogeneity would be an obvious concern, even though the source documents do not include any conversion case specifically due to such unrest. To address these issues, I test whether prior resistance in nearby villages predicts subsequent domain conversion (SI Section C.11) and, as a further robustness check, exclude conversions with documented or possible links to unrest (SI Section C.2). The results indicate that past exit and confrontation do not significantly predict conversion timing, and the estimated effect of conversion on exit remains positive and statistically significant after these exclusions.

Another challenge in identifying the mechanism is that conversion may generally cause regional unrest by altering regional administration or the information environment, which could increase nearby exit simply as a result of such general unrest. To rule out this

possibility, I also examine direct confrontation, which provides a comparison outcome for a broader rise in unrest; this helps show that the effect of conversion is specific to exit.

Model specification. For each converted domain, I include observations from five years before to five years after its first conversion, including the conversion year itself. Never-converted domains serve as controls and are included only for the same calendar years corresponding to these event windows. I estimate a linear probability model, conducting separate analyses for exit and direct confrontation:

$$\text{NonCompliance}_{j,t} = \beta_1\big(\text{Converted}_j \times \text{Post}_{j,t}\big) + \varphi_j + \omega_t + \varepsilon_{j,t}$$

Here, $\text{NonCompliance}_{j,t} \in \big\{\text{Exit}_{j,t}, \text{Confront}_{j,t}\big\}$ equals one if at least one event of the tactic occurs in villages of other domains within radius r of focal capital j in year t. Converted_j equals one for the 49 treated capitals and zero for never-converted capitals, and remains constant across years. $\text{Post}_{j,t}$ equals one in years when the converted territory is under direct shogunal rule and zero otherwise, including after a lord resumes control. Forty-eight of the 49 treated capitals were converted only once during the Tokugawa period.[9] Domain fixed effects φ_j absorb constant factors around each capital; year fixed effects ω_t absorb common shocks. I report standard errors clustered by domain and year and Conley spatial standard errors with a 70-kilometer cutoff.

Using the same sample, I estimate an event study of resistance before and after first conversion:

$$\text{NonCompliance}_{j,t} = \sum_{s \neq -1} \beta_s\big(\text{Converted}_j \times \mathbf{1}\big[t - t_j^* = s\big]\big) + \varphi_j + \omega_t + \varepsilon_{j,t}$$

The first conversion year is denoted t_j^*, with the prior year ($s = -1$) serving as the reference period. Event time measures years relative to the first conversion.

Results. Following a domain's conversion to tenryō, the probability of recording at least one exit event in nearby villages increased by approximately 4 percentage points, rising from 2.5% in comparable control cases to about 6.8% (Table 1). The estimated coefficient is 0.0427 (clustered $p = 0.011$; Conley $p = 0.003$). To the contrary, the increase in direct confrontation was far smaller (0.0126) and statistically indistinguishable from

[9] Matsumae was converted in 1807 and again in 1855; its estimation window surrounds the first conversion, and the main analysis results are robust to excluding it.

Outcome:	Any Nearby Exit			Any Nearby Confrontation		
Sample Radius:	30km	40km	50km	30km	40km	50km
Variables						
Converted × Post	0.0183	0.0292	0.0427	0.0114	0.0062	0.0126
Cluster SE	(0.0088)**	(0.0131)**	(0.0167)**	(0.0105)	(0.0102)	(0.0125)
Conley SE (70km)	(0.0090)**	(0.0120)**	(0.0144)***	(0.0118)	(0.0122)	(0.0150)
Controls						
Domain FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Fit Statistics						
Num. Obs.	31,989	31,989	31,989	31,989	31,989	31,989
Converted/Observed Domains	49/269	49/269	49/269	49/269	49/269	49/269
R2	0.034	0.046	0.057	0.070	0.088	0.114
Mean Dep. Var.	0.010	0.017	0.025	0.017	0.025	0.035

Table 1: Domain-to-Tenryo Conversion and Nearby Exit

Note: Linear probability model (LPM) estimates using an annual domain-year panel. "Nearby" refers to resistance events in villages belonging to other domains within the specified radius. Converted domains contribute observations for up to five years before and after the conversion year and are matched to never-converted controls from the same calendar years. Standard errors in parentheses are calculated as (i) two-way clustered by domain and year and (ii) Conley (70km) based on capital coordinates; significance stars correspond to each estimator's p-value. Significance codes: .***: 0.01; .**: 0.05; .*: 0.1.

zero ($p = 0.315$ and $p = 0.401$). This pattern suggests that conversion did not merely trigger a general rise in regional unrest, but affected exit specifically.

The event study shows no statistically detectable differences in pre-conversion trends (Figure 3). The pre-conversion coefficients are jointly insignificant for exit ($F = 0.95$, $p = 0.432$) and confrontation ($F = 1.16$, $p = 0.324$). Exit rises in the year after conversion ($\hat{\beta}_1 = 0.123$, $p = 0.013$); later estimates remain positive but are less precise. No individual post-conversion confrontation coefficient is significant at the 5-percent level. In the sensitivity analysis of Rambachan and Roth (2023), the confidence interval for the average post-conversion exit effect excludes zero when post-conversion deviations from parallel trends are bounded at 20 percent of the largest pre-conversion deviation, but includes zero at 25 percent (SI Section C.8).

The positive relationship between conversion and nearby exit holds up well across a range of checks and alternative samples. When I remove three conversions that coincided with direct evidence of unrest or petitioning, the estimate stays nearly the same. Including

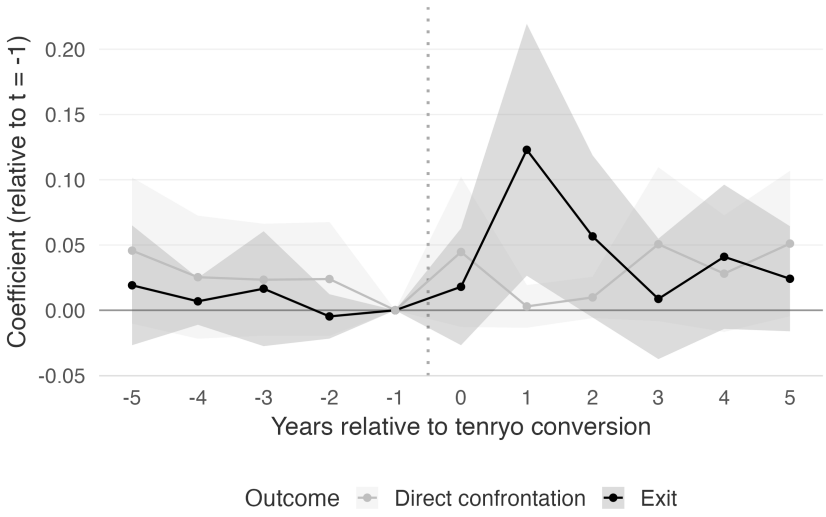


Figure 3: Tenryo-Conversion Event-Study Coefficients

Note: Yearly event-study coefficients for nearby exit (black) and direct confrontation (light gray), with $s = -1$ as the reference. Shaded bands show 95% confidence intervals; the horizontal line marks zero and the vertical dotted line marks conversion between $s = -1$ and $s = 0$. "Nearby" denotes resistance in villages of *other* domains within 50km of the focal capital. Standard errors are two-way clustered by domain and year.

two additional cases where the link to unrest is less clear gives a similar result, 0.0448 ($p = 0.0148$). The strictest test excludes four more conversions where resistance happened in the same domain or province within the previous decade, even though there's no written link between that resistance and conversion; here, the estimate is still positive at 0.0493 ($p = 0.0138$). Across all these variations, the effect on confrontation is statistically indistinguishable from zero (SI Section C.2). Looking at the timing of conversions across 29 cases and 233 domains, there's no evidence that previous nearby exit or confrontation predicted when conversion would happen ($p = 0.436$; SI Section C.11).

Other robustness checks, including the shrinked sample area (Table 1), counting only desertion events (SI Section C.4), the use of different Conley bandwidths (SI Section C.12, SI Section G.1), count outcomes (SI Section C.3), or leaving out controls that might have been exposed to the policy (SI Section C.9), also yield positive exit effects, though sometimes with wider confidence intervals. A placebo test, where conversions are reassigned at random, shows that the observed exit effect is unusually large: it lands far out in the upper tail of the placebo distribution (SI Section C.10). Finally, using the Sun–Abraham method (which is designed for staggered treatment timings) still produces

positive point estimates for both exit and confrontation, though I can't report reliable clustered standard errors because of the small number of treated groups (SI Section C.5).

3.3.2. Grid-Year Panel

Here, I exploit the variation in geographic exposure to jurisdictional overlap by developing a grid-year panel dataset of resistance events.

Distance Ratio. I measure grid-level exposure to jurisdictional overlap with a *distance ratio*:

$$\rho_{i,t}^{\text{grid}} := \frac{d_{i,t}^1}{d_{i,t}^2}$$

where $d_{i,t}^1$ and $d_{i,t}^2$ denote the straight-line distances from grid cell i to the nearest and second-nearest active capitals in year t, respectively. The ratio $\frac{d_{i,t}^1}{d_{i,t}^2}$ approaches one when either a new capital appears closer than the former second-nearest, or when the nearest capital disappears or relocates farther away. Values near zero indicate the dominance of a single capital, while values approaching one signify that two capitals are nearly equidistant.

Ideally, village-level data would provide the finest level of analysis. However, for villages that did not resist, their domain affiliation is recorded only at the end of the

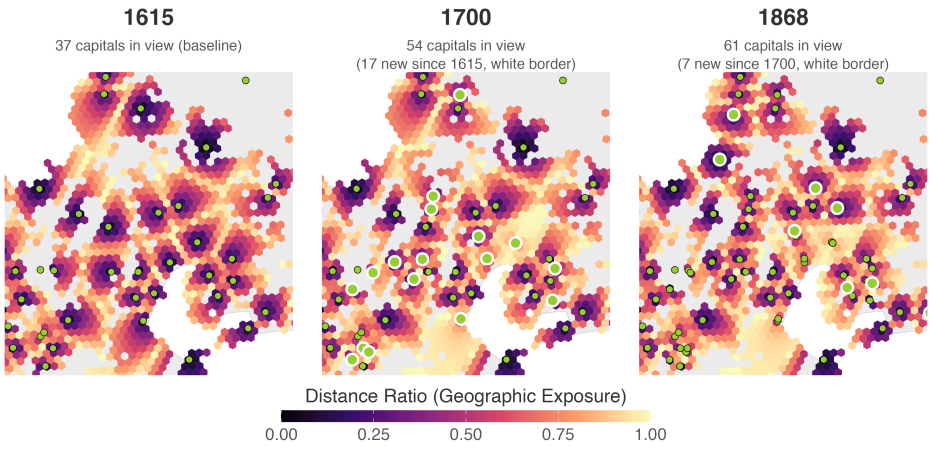


Figure 4: Time-Varying Distance Ratio across 5km Hexagonal Grid: Kinki Region

Note: Each panel maps $\rho_{i,t}^{\text{grid}}$ (nearest-capital distance / second-nearest-capital distance) in a $\approx 200\,\text{km} \times 200\,\text{km}$ window of central Japan (lat 34.4–36.2°N, lon 135.5–137.6°E). Yellowgreen circles mark active capitals; white borders mark capitals added since the previous snapshot. Yellow indicates greater geographic exposure and purple a dominant nearest capital.

Tokugawa period. To overcome this limitation, I aggregate the data at the level of fixed 5km hexagonal grid cells. Each village is assigned to a hexagon, and for each hexagon-year, I calculate the exposure ratio $\rho_{i,t}^{\text{grid}}$ using the two geographically nearest active capitals, regardless of the domain affiliations within the hexagon. Populated hexagons are identified based on the late-Tokugawa *Kyūdaka–Kyūryō Torishirabe Chō* catalog, which lists a total of 46,086 villages (Section 3.4). The resulting panel covers 2,225,802 cell-years across 8,763 hexagons, recording annual counts of exit and confrontation events for each cell. Figure 4 maps $\rho_{i,t}^{\text{grid}}$ of three different years, showing that the ratio shifts as existing domains dissolve and new ones emerge.

The distance-ratio approach has some caveats. First, because some domains held scattered, noncontiguous parcels (*tobichi*), villages within the same hexagon may not belong to either of the domains represented by the nearest two capitals. Some may be attached to a lord whose capital lies farther away, or to a third domain whose capital is not among the two closest to the centroid. The village-level analysis in the next subsection instead provides analysis with accurate domain affiliations. Second, the distance ratio does not capture every aspect of exposure to jurisdictional overlap, such as the relationship between neighboring domain lords and economic conditions of domains that could affect rulers' willingness to mediate disputes or receive migrants. While such factors are partially dealt with by the domain taxation analysis in Section 3.5, the current analysis seeks to hold those factors constant with fixed effects as explained below, isolating the variation in the geographic measure of exposure to jurisdictional overlap. Lastly, the distance ratio measure calculated with straight-line distances may not capture the actual travel costs of villagers that would vary by terrain and road conditions. SI Section D.4 is aimed to account for this issue by re-computing the ratio with terrain-weighted least-cost-path distances.

Model Specification. The estimating equation is

$$\text{NonCompliance}_{i,t} = \beta_3 \rho_{i,t}^{\text{grid}} + \lambda_i + \eta_t + \varepsilon_{i,t}$$

with hexagon and year fixed effects, where $\text{NonCompliance}_{i,t} \in \{\text{Exit}_{i,t}, \text{Confront}_{i,t}\}$ indicates whether any resistance of each type occurred in hexagon i in year t. The main estimates are linear probability models; SI Section D.2 uses the count of resistance

	Grid exposure ratio ($\rho^{\text{grid}}_{i,t}$)			Closest-capital dist. ($d^1_{i,t}$, km)		
Outcome:	(1)	(2)	(3)	(4)	(5)	(6)
	Any exit	Any confr.	Any confr. (strict)	Any exit	Any confr.	Any confr. (strict)
Coefficient	0.00035	0.00026	−0.00012	0.00000	0.00000	0.00000
Cluster SE	(0.00017)**	(0.00019)	(0.00011)	(0.00000)*	(0.00000)	(0.00000)
Conley SE (10km)	(0.00017)**	(0.00023)	(0.00012)	(0.00000)	(0.00000)	(0.00000)
Controls						
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Fit Statistics						
Num.Obs.	2,225,802	2,225,802	2,225,802	2,225,802	2,225,802	2,225,802
R^2	0.006	0.007	0.006	0.006	0.007	0.006
Mean Dep. Var.	0.00022	0.00064	0.00026	0.00022	0.00064	0.00026

Table 2: Geographic Exposure Predicts Recorded Exit

Note: LPMs on any-event indicators at the 5km hexagon-year level for all 8,763 cells. Columns (1)–(3) use the annual grid exposure ratio $\rho^{\text{grid}}_{i,t} = \frac{d^1_{i,t}}{d^2_{i,t}}$ from the two closest active capitals. Columns (4)–(6) use distance to the closest active capital $d^1_{i,t}$ (km), which measures remoteness without the second-nearest capital. "Confrontation (strict)" includes only destruction and uprising. All models include grid and year FE. SEs are (i) two-way clustered by cell and year and (ii) Conley (10km). See SI Section D.3 for ever-active cells. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

events instead. Temporary variation in $\rho^{\text{grid}}_{i,t}$ comes from capital dissolutions, transfers, and reestablishments that reorder the nearest authorities. Hexagon and year fixed effects absorb time-invariant factors and common shocks.

Results. Table 2 shows that the distance ratio is associated with exit but not confrontation, whereas the absolute distance to the closest capital predicts neither. In columns (1)–(3), a one-unit increase in the ratio is associated with a 0.035-percentage-point increase in cell-year exit probability. The estimate is significant at 5% with both cell-and-year clustering ($p = 0.048$) and Conley SE ($p = 0.037$). The smaller positive estimate for confrontation is not statistically significant. Because the baseline exit probability is only 0.022%, a one-standard-deviation increase in the ratio (SD = 0.26) corresponds to roughly 40% of that mean.

Columns (4)–(6) distinguish geographic exposure from simple remoteness by using distance to the closest active capital $d^1_{i,t}$. The exit coefficient is small and marginal with clustered standard errors ($p = 0.092$) but null with Conley standard errors ($p = 0.162$);

confrontation is also null. The contrast between the two sets of columns suggests that exit depends on the degree of jurisdictional overlap rather than its mere remoteness from any authority.

The positive association between the exposure ratio and exit is robust to a range of alternative approaches, though some specifications yield weaker statistical significance. The estimated effect remains positive when the sample is restricted to the 1,466 grid cells that ever experienced resistance (SI Section D.3), when estimated with Poisson and negative binomial fixed-effects count models (SI Section D.2), when exit is restricted to desertion (SI Section D.1), when the distance ratio is calculated using terrain-weighted least-cost-path distances rather than straight-line distances (SI Section D.4), and when the capital set is expanded to include *daikansho* (regional administrative offices of the shogunate) (SI Section D.5).

3.4. Exposure to Outside Authority and Resistance Tactics

The preceding tests ask whether exposure to outside authority increases the likelihood of exit. This section instead asks, conditional on resistance occurring, which tactic (exit or confrontation) a resisting village chose.

Data. I construct a dataset of villages recorded as participants in resistance episodes by Aoki (1975). Each data point represents a geocoded village-event, capturing the village's home authority, date, and observed tactic—either exit or direct confrontation—as assembled in Section 3.2. For each village, I identify its closest foreign domain (i.e., the domain whose capital is nearest among all capitals except the home lord's) and calculate straight-line distances from the village to both its own capital and the closest foreign capital, based on the time-varying panel of capitals from Section 3.3.1. For the main analysis, regional shogunal intendant offices (*daikansho*) are not included in the set of outside authorities. SI Section E.2.9 provides a robustness check including them when coding the closest foreign capital.

Distance Ratio. I measure each village's geographic exposure to outside authority using the distance ratio ρ_i: the distance from the village to its own lord's (home) capital (d_i^{HC}), divided by the distance to the nearest other lord's (foreign) capital (d_i^{FC}):

$$\rho_i := \frac{d_i^{\mathrm{HC}}}{d_i^{\mathrm{FC}}}$$

Unlike the grid exposure ratio $\rho_{i,t}^{\mathrm{grid}}$ in Section 3.3.2, which is calculated using the two closest active capitals, ρ_i is anchored to each village's designated home domain.

As discussed in Section 2.1 and Section 2.3, the possibility of exit through alternative authority limits extraction when villagers can reach another ruler's seat and benefit from external protection, mediation, or oversight. The ratio captures the geographic dimension of that access. A higher value of the ratio indicates greater exposure, making exit a more feasible option. Proximity to foreign capitals enhanced villagers' ability to obtain protection, mediation, or recognition: mediators or guards could more easily shelter those fleeing, villages could reach neighboring domain capitals or shogunal offices more quickly, and administrative transfer became more practical. In contrast, as villages are located farther away from the home capital, they are pushed to the administrative periphery, where the home lord's coercive reach gets weaker, making departure harder to intercept or punish. These geographic factors should encourage resistance through exit rather than direct confrontation.

A limitation of the distance ratio is that it is a composite measure that captures both proximity to an outside authority and proximity to the home capital. This makes it difficult to disentangle the jurisdictional-overlap interpretation from simpler accounts, such as decay in state capacity at the periphery. To address this problem, I supplement the analysis with regressions using the absolute distances to the home and foreign capitals separately.[10]

Model Specification. Using the repeated cross-sectional data of revolting villages, I estimate the relationship between the distance ratio and their choice of exit or confrontation. The outcome variable is a binary indicator of exit or confrontation, and the distance

[10] Other caveats and limitations discussed in Section 3.3.2 apply here as well. To account for political and economic factors that could influence jurisdictional conflict between neighboring domains, I include fixed effects as detailed below, treating the distance ratio as capturing specifically the geographic component of exposure to outside authority. To consider actual travel costs rather than straight-line distances, I also replicate the analysis using terrain-weighted least-cost-path distances.

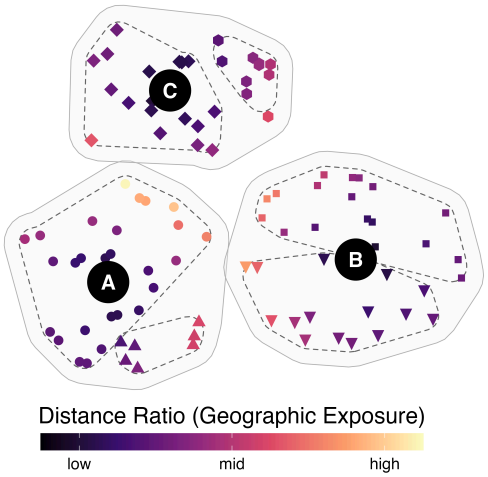


Figure 5: How Home-Neighbor Dyad Fixed Effects Isolate Within-Pair Variation

Note: A schematic, not data. Three domains (A, B, C) are drawn with their capitals as black circles and their villages as shapes shaded by the distance ratio ρ_i, the village's distance to its own lord's capital divided by its distance to the nearest other lord's capital. Darker color marks villages closer to their own capital and far from any rival, lighter color the reverse. The pair of each village's home domain and its nearest rival defines a dyad.

ratio is the main explanatory variable. To isolate the variation in the distance ratio, I employ a home-neighbor dyad fixed effects design illustrated in Figure 5.

Home-neighbor dyad fixed effects isolate within-pair variation in the distance ratio. Each dyad is an ordered pair of home and nearest-rival domains (for example, A–B and A–C in Figure 5). Dyad fixed effects absorb time-invariant home- and neighbor-domain factors, including their coercive capacity, ruler ideology, agricultural productivity, and administrative quality, as well as pair-specific factors such as cross-border ties. Within an A–B dyad, for instance, the estimate compares A villages whose nearest rival is B but whose relative position between the two capitals differs.

The main specification is:

$$\text{Exit}_i = \beta_2 \ln(\rho_i) + X_i' \boldsymbol{\gamma_2} + \underbrace{\varphi_{j,k}}_{\text{home-neighbor dyad FE}} + \underbrace{\omega_t}_{\text{decade FE}} + \varepsilon_i$$

Here, i indexes a village-event, while j and k index its home and nearest-rival domains and t its decade. Exit_i equals one for exit and zero for direct confrontation. The logged

ρ_i uses the capitals active at event i.[11] X_i contains latitude, longitude, their interaction, logged elevation and ruggedness, logged distance to the nearer of Edo or Kyoto, and logged wet-rice suitability.[12] The dyad fixed effects $\varphi_{j,k}$ absorb time-invariant features of each ordered home-rival pair, while ω_t absorbs decade-specific shocks.

Results. Table 3 shows that geographic exposure to outside authority as measured by the diatance ratio predicts exit rather than confrontation conditional on resistance. The coefficients are positive throughout and statistically significant in the preferred controlled specifications. In Model 5, with home-neighbor dyad fixed effects and full controls, a one-

Outcome:	Exit_i						
	Distance Ratio					*Abs. Distances*	
Models:	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variables							
Distance ratio ($\ln(\rho_i)$)	0.0458	0.0519	0.0201	0.0212	0.0244		
Cluster SE	(0.0154)***	(0.0141)***	(0.0071)***	(0.0072)***	(0.0068)***		
Conley SE (10km)	(0.0117)***	(0.0093)***	(0.0078)***	(0.0078)***	(0.0085)***		
$\ln(d_i^{\text{HC}})$						0.0160	0.0203
Cluster SE						(0.0104)	(0.0083)**
Conley SE (10km)						(0.0106)	(0.0113)*
$\ln(d_i^{\text{FC}})$						−0.0306	−0.0309
Cluster SE						(0.0129)**	(0.0135)**
Conley SE (10km)						(0.0138)**	(0.0148)**
Controls							
Decade FE	No	Yes	Yes	Yes	Yes	Yes	Yes
Home Domain FE	No	No	Yes	Yes	No	Yes	No
Neighbor Domain FE	No	No	Yes	Yes	No	Yes	No
Home-Nbr Dyad FE	No	No	No	No	Yes	No	Yes
Geography	No	No	No	Yes	Yes	Yes	Yes
Fit Statistics							
Num.Obs.	2,739	2,739	2,739	2,739	2,739	2,739	2,739
R2	0.028	0.225	0.621	0.626	0.678	0.627	0.678
Mean Dep. Var.	0.246	0.246	0.246	0.246	0.246	0.246	0.246

Table 3: Exposure to Alternative Authority Predicts Exit (Exit versus Confrontation)

Note: OLS estimates of each village's exit-versus-confrontation choice conditional on resisting. Models (1)–(5) use the log distance ratio; Models (6)–(7) replace it with the two absolute distances (d_i^{HC}: home capital; d_i^{FC}: nearest foreign capital). Full results in Table A.29. SEs in parentheses: (i) clustered on home, nearest-neighbor, decade; (ii) Conley (10 km) on village coordinates; stars reflect that estimator's p-value. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

[11]The logged ratio constrains the coefficients on the two logged distances to equal magnitude with opposite signs. Linear hypothesis tests find no evidence against this restriction (SI Section E.3.3).

[12]Wet rice suitability is measured as potential output under irrigation and medium input, based on the Caloric Suitability Index developed by Galor and Özak (2016).

standard-deviation increase in the log distance ratio (≈ 1.58) raises exit probability by 3.9 percentage points, or 16% of the 24.6% baseline rate.

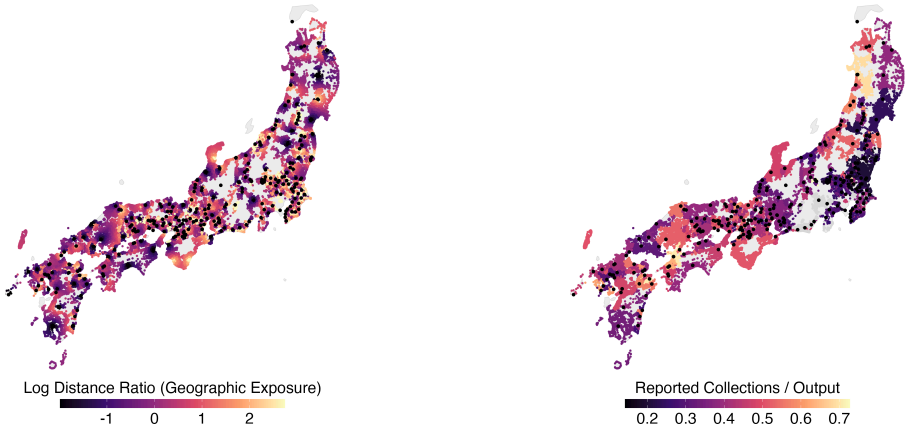
Models 6 and 7 disentangle the two distances to assess the relative importance of foreign access versus home remoteness. In Model 6 (home- and neighbor-domain fixed effects with full controls), only foreign-capital distance is statistically significant (cluster $p = 0.026$; Conley $p = 0.027$); the home-capital coefficient is positive but indistinguishable from zero (cluster $p = 0.136$). In the preferred dyad-fixed-effects specification (Model 7), the coefficient on home-capital distance is 0.0203 (cluster $p = 0.022$; Conley $p = 0.073$), and the coefficient on foreign-capital distance is -0.0309 (cluster $p = 0.031$; Conley $p = 0.037$). A one-standard-deviation decrease in $\ln\left(d_i^{\mathrm{FC}}\right)$ (approximately 0.92) increases the probability of exit by 2.8 percentage points, which is 11% of the baseline rate. Proximity to the foreign capital thus seems the active margin in Model 6 and the stronger, more robust predictor in Model 7.

To further probe the mechanism, SI Section E.3.1 replicates the analysis using the distance to the nearest domain border instead of the distance to the nearest foreign capital when computing the distance ratio. The alternative measure does not significantly predict exit choice, and controlling for this variable leaves the capital-based exposure estimate largely unchanged. This finding reinforces the rationale behind the capital-based measure: villagers secured leverage by appealing to authorities capable of providing protection, mediation, or recognition. The result suggests that the specific location of such authority mattered more than simply the ease of crossing into an adjacent domain.

The results are robust across a range of checks, including alternative control variables (SI Section E.2.1), excluding villages with repeated resistance events (SI Section E.2.2), and omitting cross-island dyads whose direct routes span sea straits (SI Section E.2.4). They are also unchanged when shogunal intendant offices (*daikansho*) are added to the set of potential authorities (SI Section E.2.9).

3.5. Exposure to Outside Authority and Taxation

To complement the behavioral evidence presented in the preceding analyses, I lastly offer suggestive evidence for fiscal traces of the exit mechanism. If the threat of mass exit disci-



(a) Village Log Distance Ratio (Geographic Exposure) (b) Reported Effective Tax Rates

Figure 6: Geographic Exposure and Tax Rates Across Domains

Note: Panel (a) plots village locations shaded by the log distance ratio $\ln(\rho_i) = \ln(d_i^{\text{HC}}/d_i^{\text{FC}})$, where HC is the home-domain capital and FC is the nearest foreign-domain capital; higher values indicate greater geographic exposure. Panel (b) shades those villages by their domain's reported collections divided by reported output, the effective tax-rate measure used in the analysis. Gray points denote villages in domains without tax data, and black dots mark domain capitals.

plines rulers, domain lords facing greater exposure to jurisdictional competition should consequently face tighter constraints on resource extraction. In Tokugawa Japan, domain lords competed intensely for scarce labor. Those plagued by labor shortages were more incentivized to accept migrants, while those with abundant labor faced weaker incentives. As a result, lords bordered by domains with greater absorptive capacity confronted more credible exit threats—and thus operated under stricter limits on extraction—than did those bordered by less absorptive neighbors.

Data. I assemble a cross-section of late-Tokugawa *daimyō* domains. The starting sample is the 275 domains for which Kodama and Kitajima (1989) compiles late-Tokugawa domain records, with capital locations from Kimura, Fujino, and Murakami (2015).[13] From the outset, I exclude non-*daimyō* territories—including shogunal *goryō* and shrine- or temple-administered *jisharyō*—because they lack comparable fiscal returns in Kodama and Kitajima (1989) and exercised little independent tax-setting authority (Murakami 1996). As tax rates are only available for the final years of the Tokugawa period, the analysis is cross-sectional. My aim here is thus to provide suggestive fiscal evidence for the exit mechanism, rather than to establish causal identification. Kodama

[13] As in the preceeding analyses, branch domains are treated as part of parent domains.

reports effective tax rates for 256 of the 275 domains; the remaining 19 lack tax rate data (Figure 6(b) shades their villages in gray). The domain-level distance ratio measure of exposure to jurisdictional overlap (discussed below) requires a list of villages under each domain's control. However, the source document for village lists omits 33 out of 275 domains. Due to the limits on tax data and the lists of villages, the effective sample of domains is reduced to 230 domains.

The outcome of interest is the effective domain tax rate, drawn from Kodama and Kitajima (1989) (pp. 425–431): *shunodaka* (reported collections in rice equivalents) divided by *uchidaka* (the domain's internal record of actual production, distinct from its public *kokudaka* assessment). The ratio captures the proportion of reported output that was collected as tax, reflecting reported collections relative to reported output rather than the ruler's intended extraction rate.

Two important features of the tax data merit attention. First, the figures were compiled by the Meiji government in 1869–1870, based on the average of each domain's production and collections over the preceding five years (1865–69). This averaging process mitigates the effects of annual harvest fluctuations, yielding a more accurate measure of underlying tax burdens. In addition, the widespread adoption of *jōmen-hō* (fixed-rate taxation) ensured the stability of tax rates throughout the final decades of the Tokugawa period (Iinuma 1974, 20–24; Smith 1958). The tax data thus can be seen as capturing each domain's persistent extraction intensity, rather than short-term fluctuations.[14][15]

Domain-Level Distance Ratio. To measure each domain's exposure to jurisdictional overlap, I calculate the average village-level distance ratio, constructed as in Section 3.4, across all villages within the domain. The complete list of 46,086 villages (as of the end of the period) and their home domains is sourced from the *Kyūdaka–Kyūryō Torishirabe*

[14] Actual tax rates varied across villages, though systematic village-level taxation records are not available. Yet the domain-level effective tax rate is informative enough here as a measure of average extraction intensity.

[15] One might wonder whether public goods provision, rather than tax reduction, served as an alternative means for rulers to address peasant grievances. If so, tax rates could be an incomplete measure of oppressiveness. This is unlikely for two reasons. First, public works projects depended on corvée labor (bueki) extracted from peasant villages, which increased rather than alleviated their burdens. Second, peasant revolt episodes recorded in Aoki (1975) indicate that peasants' grievances centered primarily on excessive taxes and unfair assessments, not on the lack of infrastructure.

Chō (Ledgers of Former Assessments and Territories, hereafter LFAT), a Meiji-era land census. If a village was under the control of multiple domains, I assign it to the domain holding the largest share of its assessed productivity (*kokudaka*, an index of aggregate agricultural output, rice-denominated but not limited to actual rice yield). Geocoding for each village is drawn from Honda, Natsume, and Nemoto (2022), which is used when computing each distance ratio. Figure 6 shows the log distance ratio for each village (panel a) side-by-side with domain-level tax rates (panel b).

One potential issue of using the capital-based distance ratio is that domain capital locations could be endogenous if rulers placed capitals or exchanged border villages for strategic reasons to deter exit. Most capitals, however, were inherited from the Sengoku (Warring-States) period (1467–1600) (Hall 1955) and were largely fixed by the 1615 one-castle-per-domain edict (*ikkoku ichijō rei*); later relocations required shogunal approval and were rare. One possibility is that each domain lord's relationship with the shogunate influenced both the likelihood of his request of such strategic territorial arrangement being approved by the shogunate and his extractive capacity. I therefore control for each daimyō's status, which indicate his relationship with the shogunate: shinpan (relatives), fudai (hereditary vassals who fought for the shogunate at Sekigahara in 1600), and tozama (outside lords who became allies only after the battle).

Second, the distance ratio might correlate with factors beyond jurisdictional overlap that also influence taxation. Given that home-capital distance is part of the measure, I address following potential confounders. (i) Domain scale: larger domains tend to be richer and may invest more in coercive capacity, and if larger armies (*hanpei*) help suppress revolt, that would push tax rates up and bias the exposure estimate conservatively. I thus control for domain population, agricultural productivity, and territory size, and in robustness checks for army share relative to population. (ii) Peripheral state capacity: distance from the capital may proxy for weaker monitoring of local tax agents and higher agency costs (Sng and Moriguchi 2014), which could lead daimyō to restrain extraction; Tokugawa institutions limited village officials' discretion, but some domains implemented country-samurai (*gōshi*) system, under which lower-ranking samurai lived in villages as local officials. I thus control for whether the system was present. (iii) Tax delivery costs:

peripheral villages might pay lower taxes if hauling rice to the capital were costly, but this concern is likely minor because collections typically went to nearby storehouses rather than to capitals[16]. (iv) Frontier development: remoter villages may be less developed, and rulers might set lower taxes in peripheral areas to attract investment. I thus control for the share of villages whose names include new rice field (*shinden*), a marker of recent agricultural development.

Pooled Benchmark. The pooled benchmark model regresses domain tax rates on the average distance ratio:

$$\tau_j = \alpha_4 + \beta_4 \ln(\rho_j) + X_j' \boldsymbol{\gamma_4} + \varepsilon_j$$

where τ_j is domain j's tax rate, ρ_j is the average distance ratio of all villages controlled by domain j, and X_j contains three groups of controls. Core controls are logged territory size, population, and wet-rice suitability,[17] the number of new-farmland villages, shogunal alliance, and the gōshi system indicator. Geographic controls are capital latitude and longitude and their interaction, logged elevation, ruggedness, and distance to the Edo–Kyoto axis, and the monthly shares of rainy and snowy days.[18] Neighbor controls are logged weighted means of adjacent domains' territory, population, and rice productivity, the weighted number of new-farmland villages, and the share of neighbors with family ties to the focal domain; weights count the focal villages for which each domain is the nearest neighbor.

Table 4 reports a negative association between the distance ratio and reported tax rates. In Models 1–4, a one-standard-deviation increase in the log distance ratio (SD = 1.34) is associated with a 1.5 to 3.2 percentage-point reduction in the tax rate. This negative association persists after including the full set of controls and holds in spatial-lag models (Anselin and Rey 2014) (Models 5–6), which account for potential spatial

[16]For instance, Fukui Prefecture (1994: 335-346) documents this practice in Fukui Domain.

[17]The wet-rice measure is an agro-climatic potential-yield surface (Galor and Özak 2016) rather than an observed annual output series; I use it only as a proxy for labor-absorption capacity.

[18]Weather variables represent proportion of rainy/snowy days per month, imputed as weighted averages using daily weather data from 40 locations, with distance to domain capital as weight. Data from Historical Weather Database by Professor Minoru Yoshimura.

correlation in tax rates among neighboring domains that might arise from inter-domain competition or collusion.

The negative association survives several robustness checks. The sign remains negative with alternative covariates, including kokudaka, samurai population share, and population relative to wet-rice suitability (SI Section F.2.1). Applying the sensitivity analysis of Diegert, Masten, and Poirier (2022) shows that, when exposure- and outcome-side confounding are bounded equally, both SD ratios must reach about 0.554 before the identified set admits a nonnegative coefficient (SI Section F.2.2). Double/debiased machine learning (Chernozhukov et al. 2018) on the same Model 4 controls with ridge learners yields a similar negative association, although more flexible learners (random forest and stacked ensembles) produce imprecise estimates near zero (SI Section F.2.3). The sign also remains negative when the shogunal intendants' offices are included when computing the distance ratio (SI Section F.2.6).

Outcome:	τ_j					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Distance ratio ($\ln(\rho_j)$)	−0.0241	−0.0203	−0.0110	−0.0112	−0.0140	−0.0094
HC SE	(0.0053)***	(0.0058)***	(0.0055)**	(0.0055)**	(0.0047)***	(0.0051)*
Conley SE (70km)	(0.0066)***	(0.0067)***	(0.0056)**	(0.0055)**	(0.0058)**	(0.0050)*
Controls						
Core	No	Yes	Yes	Yes	No	Yes
Geography	No	No	Yes	Yes	No	Yes
Neighbors	No	No	No	Yes	No	Yes
τ_j Spatial Lag	No	No	No	No	Yes	Yes
Fit Statistics						
Num.Obs.	232	230	230	230	230	230
R2	0.074	0.141	0.370	0.389	0.312[a]	0.439[a]
Mean Dep. Var.	0.392	0.394	0.394	0.394	0.394	0.394

Table 4: Pooled Benchmark: Domain-Level Geographic Exposure and Tax Rates

Note: OLS estimates (ML in Models 5–6) of reported effective tax rates (collections divided by reported output) on the log domain-average distance ratio. Full results in Table A.42. SEs in parentheses: (i) HC1 (ML SEs in 5–6); (ii) Conley (70 km) on capital coordinates, computed via two-step filtered-residual approximation (Anselin and Rey 2014, pp. 34–36); stars reflect that estimator's p-value. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1. [a] Nagelkerke pseudo-R2

Neighbor-Heterogeneity Test. To test if the negative association between the distance ratio and tax rates is sensitive to the quality of the outside option, I interact exposure with $\hat{y}_{-j}$, a discrete measure of neighboring domains' outside-option quality:

$$\tau_j = \alpha_5 + \beta_5 \ln(\rho_j) + \delta_5 \ln(\rho_j) \mathbb{1}_{\{\hat{y}_{-j} = \text{High}\}} + X_j' \boldsymbol{\gamma_5} + \varepsilon_j$$

I proxy outside-option quality with neighboring domains' demand for labor. Labor-scarce rulers attracted workers through land redistribution, tools and rations, irrigation, and other public goods (Hyugano 1957; Miyazaki 1995), with stronger incentives under greater scarcity. For instance, persistent labor shortages in the eastern part of Japan (Kantō) led domains to adopt aggressive peasant recruitment policies (*iri-byakushō seisaku*) offering tax exemptions, food and seed grants, and other settlement inducements (Gorai 1950).

The moderator operationalizes this labor-absorption capacity. $\hat{y}_{-j}$ equals "Low" when neighbors' weighted average labor-to-productivity ratio is above its median and "High" otherwise.[19] I weight each neighbor by the number of domain j villages for which its capital is nearest and divide its population by potential wet-rice yield from Galor and Özak (2016).[20] High $\hat{y}_{-j}$ thus denotes neighbors with greater capacity and incentives to absorb migrants. The full interaction specification retains neighboring rice yield and territory size among the controls in X_j.

The results show that the distance ratio is negatively associated with tax rates in the High outside-option group across all four specifications. With full controls (Model 4), the slope for the High group remains statistically significant at -0.0158 ($p = 0.023$), compared with -0.0085 ($p = 0.182$) in the Low group. The consistently more negative point estimates for the High outside-option group are consistent with the retention mechanism, although the difference in slopes between groups is not statistically precise ($p = 0.323$). The Appendix repeats the interaction analysis using neighboring domains' population density as an alternative proxy for labor-absorption capacity (SI Section F.3.3).

[19] Hainmueller, Mummolo, and Xu (2019) show that continuous multiplicative-interaction models rest on a linear-interaction assumption that frequently fails in practice, and recommend replacing the continuous moderator with a discrete one precisely for settings like this, in which the moderating relationship is inherently nonlinear.

[20] I use the potential yield measure under irrigated conditions with medium inputs.

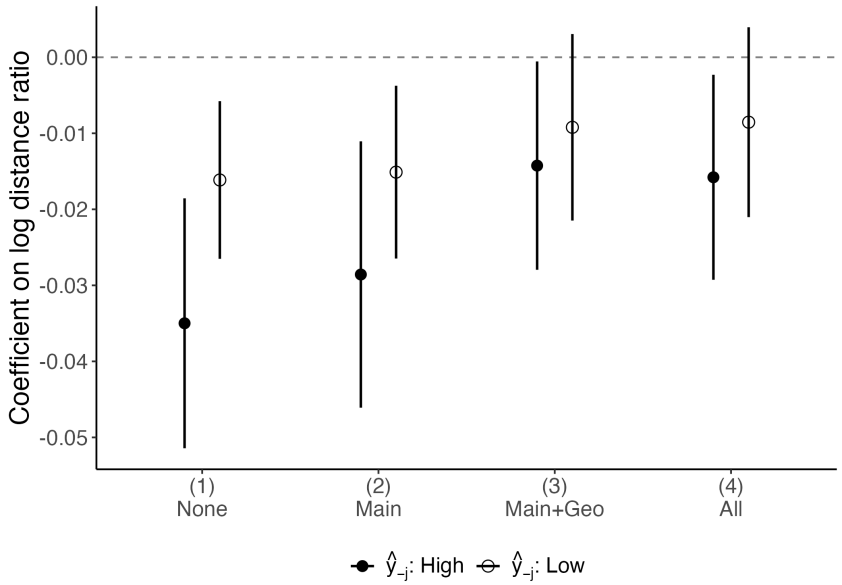


Figure 7: Tax-Exposure Slopes by Outside-Option Quality

Note: Group-specific OLS slopes on $\ln(\rho_j)$. Filled points marked High denote below-median neighbor labor-to-potential-yield ratios, hence greater capacity and incentive to absorb migrants; open points marked Low denote above-median ratios. Lines are 95% heteroskedasticity-consistent confidence intervals. Controls are cumulative: (1) none; (2) domain; (3) +geography; (4) +neighbor. Full table in Table A.43.

Additional Analyses. The null result for border distance also emerges in the domain tax analysis: proximity to the nearest border does not significantly predict tax rates, whereas exposure based on capital distance consistently remains negatively associated with them (SI Section F.3.2).[21] Combined with the village finding in Section 3.4, this pattern supports the argument that villagers required access to an administrative authority capable of offering protection and advocacy on their behalf. Incorporating the log of average home-to-capital distance—the predictor implied by state capacity decay (Sng and Moriguchi 2014)—leaves the exposure coefficient negative (SI Section F.2.4); replacing the distance ratio with remoteness likewise yields no significant negative association once geographic controls are included (SI Section F.3.1).

[21]The boundary data were constructed following the procedure described in (Honda, Natsume, and Nemoto 2022).

4. Conclusion

In this paper I develop a theory of mass exit through alternative authority for polities that combine horizontal rivalry among local rulers with vertical oversight by a common superior, and I test its behavioral implications in Tokugawa Japan. Three designs speak to the mechanism of exit through alternative authority. A difference-in-differences analysis around *tenryō* conversions finds that bringing shogunal authority closer to neighboring villages raises recorded exit among them. A grid-year panel data analysis shows the occurence of exit is associated with geographic exposure to alternative authority. Village-level analysis on the choice of resistance tactics further shows that geographic exposure to an outside authority, rather than proximity to a border, predicts the choice of exit over direct confrontation. Domain-level tax rate data adds a suggestive fiscal trace of the mechanism: domains with greater exposure to jurisdictional overlap report lower tax rates, with steeper negative associations where neighbors could plausibly absorb migrants. Together, the results provides micro-foundational evidence for a fundamental premise shared by theories of state development that access to alternative authority shapes the mode of popular resistance. To clarify, my contribution is to directly demonstrate the behavioral mechanism, and not to establish the full institutional consequences emphasized in those broader literatures.

The findings are generalizable to other contexts with a similar horizontal–vertical structure of authority, which is widespread, appearing in feudal monarchies and principalities, empires and provinces, colonial regimes of indirect rule, and federal governments alike (Gerring et al. 2011, 377–382, 405–407). Tokugawa Japan presents a strict test of the mechanism of exit through alternative authority. Villages were bound to the land, unauthorized departures met severe punishments. Unusually granular records of peasant resistance developed by valuable historical work including Aoki (1975) allows us to test the exit mechanism in this context. That we observe behavioral responses to geographic access under these restrictive conditions underscores that mass exit through alternative authority shaped outcomes even where outside options were particularly costly.

The underlying mechanism, wherein a credible outside option constrains local rulers, also speaks to other literatures. In civil wars, civilians may shift allegiance and

resources between competing governments and rebel authorities, but such movement only constrains predation when another armed actor is both able and willing to govern and protect those who relocate (Kalyvas 2006). In the context of competition for foreign investment, it is firms, not villages, that are mobile, and fiscal discipline arises to the extent that alternative jurisdictions offer credible and productive options (Acharya and Lee 2023; Kleven et al. 2020). While the Tokugawa evidence does not directly establish either relationship, it clarifies why, in both contexts, the presence of an accessible alternative authority is essential for mobile actors to anticipate protection or redress.

Data Availability Statement

Replication data and code will be deposited in the International Organization Dataverse (https://dataverse.harvard.edu/dataverse/IOJ) upon conditional acceptance. The release will include the documented intermediate geographic inputs, scripts, and codebook needed to reproduce the reported tables and figures.

Ethical Statement

This research uses publicly available historical records and archival data on Tokugawa-era Japan and does not involve human subjects. No IRB review was required.

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A. Formal Analysis

Here I present a simple framework of tax competition between two rulers governing mobile subjects. The model aims to clarify the fundamental causal relationships among exposure to outside authority, subjects' resistance decisions, and rulers' extraction, and to identify the conditions under which these relationships hold, while accounting for strategic interactions among the players in a parsimonious setting. The model thus abstracts from the full complexity of historical contexts, simplifying or omitting many relevant aspects, such as collusion between rulers, tax differentiation by individual subjects' characteristics, bargaining between refugees and rulers, intervention by higher authorities, and longer-term economic effects of migration.

A.1. Setup

Consider two neighboring domains, indexed by $j \in \{1, 2\}$. All model primitives are common knowledge. Each ruler governs a continuum of villages represented by the unit interval and chooses a domain-level tax rate $\tau_j \in [0, 1]$. The sequence of play is as follows: (i) rulers simultaneously set tax rates; (ii) villages observe both τ_j and τ_{-j} and select an action $a_{i_j} \in \{\text{comply}, \text{fight}, \text{exit}\}$.[22] For brevity, *fight* denotes direct confrontation with the ruler. Let μ_j^a represent the share of villages in domain j choosing action a. Let each village in domain j have productive yield $y_j > 0$.

A compliant village obtains $y_j(1 - \tau_j) + \varepsilon_{i_j}$, where $\varepsilon_{i_j} \sim \text{Uniform}[-\delta, \delta]$ and $\delta > 0$. A fight succeeds with probability μ_j^{fight}, yields y_j if successful and zero otherwise, and costs $k > 0$. Exit yields the neighboring domain's net-of-tax income minus village-specific exit cost: $y_{-j}(1 - \tau_{-j}) - c_{i_j}$, where $c_{i_j} \in \{c_L, c_H\}$ and $c_H > c_L > 0$. The share $\theta_j \in [0, 1]$ with cost c_L measures exposure to outside authority, reflecting geographic proximity to a rival capital; the remaining $1 - \theta_j$ face cost c_H.

Villages that exit withdraw both labor and tax contributions from their home domain, as reflected in the payoff structure. The net income of the neighboring domain represents the value of the outside option available to these villagers. To simplify, and also to model exit as bargaining tactics rather than permanent migration, I assume that rulers reach an agreement with each villager who exits, allowing them to return home after obtaining benefits equal in

[22] Ordinary petitioning is outside the scope of the game. The choice a_{i_j} can thus be considered as the village's response after unsuccessful petitioning.

value to the outside option. Accordingly, exit does not increase the tax base of the receiving domain.

Village utility is

$$u_{i_j}\left(a_{i_j};\tau_j,\tau_{-j}\right)=\begin{cases} y_j(1-\tau_j)+\varepsilon_{i_j} & \text{if } a_{i_j}=\text{comply}\\ y_j\mu_j^{\text{fight}}-k & \text{if } a_{i_j}=\text{fight}\\ y_{-j}(1-\tau_{-j})-c_{i_j} & \text{if } a_{i_j}=\text{exit}\end{cases}.$$

Ruler j collects taxes only from compliant villages and maximizes

$$u_j(\tau_j,\tau_{-j})=\left[1-\mu_j^{\text{fight}}-\mu_j^{\text{exit}}\right]y_j\tau_j=\mu_j^{\text{comply}}y_j\tau_j.$$

In the following analysis, I focus on equilibria in which low-exit-cost villages choose between compliance and exit and high-exit-cost villages choose between compliance and fighting. Assumption 1 guarantees that this continuation is an equilibrium after any pair of tax rates. It does not rule out other equilibria, such as broad mobilization in which villages with cheap exit also fight.[23] The results are comparative statics under this selection.

ASSUMPTION 1: Let $0<c_L<k<\delta$ and $0<\theta_j<1$. For each $j\in\{1,2\}$,

$$\frac{k-c_L}{\delta-c_L}>\frac{y_j(1-\theta_j)}{2\delta}\quad\text{and}\quad c_H>y_{-j}+\delta.$$

For the equilibrium comparative statics, the tax candidate derived below must satisfy $0<\tau_j^*<1$, $0<\mu_j^{\text{fight}*}<1-\theta_j$, and $0<\mu_j^{\text{exit}*}<\theta_j$ in both domains. These strict inequalities ensure interior cutoffs and persist under small parameter changes. The proof establishes global ruler best responses at such a candidate, including deviations that move a village cutoff to a boundary. Uniqueness is asserted only among interior candidates under the selected village continuation.

[23] In such a mobilization continuation, no village exits, so neither outside-option quality nor exposure to outside authority enters any chosen payoff: the fight share depends only on the home tax, and, if this continuation is selected after every tax pair, the ruler's tax does not respond to either. Also, such broad mobilization is empirically very rare: in the resistance records compiled by Aoki (1975), the median episode involves one named village, and only 11 episodes involve a quarter or more of the home domain's villages.

A.2. Equilibrium Results

PROPOSITION 1 (Exposure to Outside Authority and Exit): Under Assumption 1 and the stated interiority conditions: (a) whenever greater exposure to outside authority lowers the home ruler's equilibrium tax, the equilibrium exit share among non-compliant villages is strictly increasing in θ_j; and (b) the equilibrium exit rate is strictly increasing in θ_j regardless of the tax response.

Greater exposure to outside authority expands the low-cost exit pool and contracts the high-cost fight pool. When the ruler cuts taxes, that response offsets only part of the resulting increase in exit because further tax reductions eventually sacrifice more revenue per compliant village than they recover through retention.

PROPOSITION 2 (Outside-Option Quality and Equilibrium Taxes): Greater exposure to outside authority lowers the equilibrium tax exactly when the neighboring domain's equilibrium net income exceeds a threshold determined by home-domain primitives. Under Assumption 1 and the stated interiority conditions, let $z^*_{-j} := y_{-j}(1-\tau^*_{-j})$. Then

$$\kappa_j := c_L - k + \frac{y_j(1-\theta_j)\left[4\delta - y_j(1-\theta_j)\right]}{2\delta - y_j(1-\theta_j)^2},$$

$$\frac{\partial \tau^*_j}{\partial \theta_j} < 0 \quad \text{if and only if} \quad z^*_{-j} > \kappa_j.$$

The tax response is zero at equality and positive below the threshold. If $\kappa_j \leq 0$, it is negative throughout the interior regime.

In the model, exposure to outside authority both strengthens the exit constraint and reduces the pool of villages available for direct confrontation. When the net outside option becomes sufficiently attractive, this retention effect predominates and the ruler lowers taxes; below this threshold, the reduction in confrontation instead allows higher taxes.

A.3. Equilibrium and Proofs

Write $[x]_0^1 := \min\{1, \max\{0, x\}\}$ and $D_j := 2\delta - y_j(1-\theta_j) > 0$. The selected continuation has clipped resistance shares

$$\mu_j^{\text{fight}} = (1-\theta_j)\left[\frac{\delta - k - y_j(1-\tau_j)}{D_j}\right]_0^1,$$

$$\mu_j^{\text{exit}} = \theta_j\left[\frac{y_{-j}(1-\tau_{-j}) - c_L - y_j(1-\tau_j) + \delta}{2\delta}\right]_0^1.$$

To verify this continuation, a high-cost village's exit payoff is below $-\delta$, whereas compliance pays at least $-\delta$. Given that low-cost villages do not fight, the high-cost fight share is the fixed point of $(1-\theta_j)\left[\frac{\delta-k-y_j(1-\tau_j)+y_j\mu_j^{\text{fight}}}{2\delta}\right]_0^1$. Assumption 1 implies $y_j(1-\theta_j) < 2\delta$, so this map is a contraction. Its unique fixed point is the clipped expression above. Moreover, its maximal value, attained at $\tau_j = 1$, is bounded by $(1-\theta_j)\frac{\delta-k}{D_j}$. Consequently,

$$y_j\mu_j^{\text{fight}} - k \le \frac{y_j(1-\theta_j)(\delta-k)}{D_j} - k \leftarrow c_L.$$

The strict inequality is equivalent to the first inequality in Assumption 1. Low-cost exit pays at least $-c_L$, so low-cost villages strictly prefer exit to fight on this continuation. This verifies the selection without claiming that other village equilibria are absent.

When both cutoffs are interior, these shares become

$$\mu_j^{\text{fight}}(\tau_j) = \frac{(1-\theta_j)\left[\delta - k - y_j(1-\tau_j)\right]}{2\delta - y_j(1-\theta_j)},$$

$$\mu_j^{\text{exit}}(\tau_j, \tau_{-j}) = \frac{\theta_j}{2\delta}\left[y_{-j}(1-\tau_{-j}) - c_L - y_j(1-\tau_j) + \delta\right].$$

A.3.1. Equilibrium Tax Rates

Define

$$A_j := \frac{(1-\theta_j)y_j}{2\delta - y_j(1-\theta_j)} + \frac{\theta_j y_j}{2\delta},$$

$$B_j := \frac{(1-\theta_j)(\delta - y_j - k)}{2\delta - y_j(1-\theta_j)} + \frac{\theta_j}{2\delta}\left[y_{-j} - c_L - y_j + \delta\right],$$

$$C_j := \frac{\theta_j y_{-j}}{2\delta}.$$

Then total resistance is $B_j + A_j\tau_j - C_j\tau_{-j}$, and ruler utility is

$$u_j(\tau_j, \tau_{-j}) = y_j\tau_j\left[1 - B_j + C_j\tau_{-j} - A_j\tau_j\right].$$

On this interior segment, $\frac{\partial^2 u_j}{\partial \tau_j^2} = -2y_jA_j < 0$, and the first-order condition is

$$\tau_j = p_j + q_j\tau_{-j}, \quad p_j = \frac{1-B_j}{2A_j}, \quad q_j = \frac{C_j}{2A_j}.$$

Moreover, $A_j \geq \theta_j \frac{y_j}{2\delta}$ implies $q_j \leq \frac{y_{-j}}{2y_j}$ and hence $q_1 q_2 \leq \frac{1}{4} < 1$. The two interior first-order conditions therefore have the unique candidate

$$\tau_j^* = \frac{p_j + q_j p_{-j}}{1 - q_j q_{-j}}.$$

This interior candidate is a global best response for each ruler under the selected continuation. To see this, hold the other ruler at its candidate tax. Positive fight in the neighboring domain implies $z_{-j}^* < \delta - k$. Hence the low-cost exit cutoff is below one for every home tax in $[0,1]$. Assumption 1 also implies

$$y_j(1-\theta_j) < \frac{2\delta(k-c_L)}{\delta - c_L} < \delta + k,$$

where the last inequality follows from $(\delta - k)(\delta + c_L) > 0$. The high-cost fight cutoff is therefore below one at every home tax as well. Each resistance share is consequently a positive multiple of the maximum of zero and an affine, increasing function of τ_j. On every smooth segment, ruler revenue has second derivative $-2y_j$ times the sum of the active resistance slopes, which is nonpositive. At each cutoff's activation, its first derivative jumps downward by $y_j\tau_j$ times the newly active slope. Revenue is thus globally concave on $[0,1]$ and strictly concave around the interior candidate. The first-order condition identifies the unique global best response to the other candidate tax. Applying this argument to both rulers establishes the candidate as a Nash equilibrium under the selected continuation; it does not exclude equilibria involving other continuations or boundary cutoffs.

A.3.2. Proof of Proposition 1

PROOF: At fixed tax rates, $\frac{\partial \mu_j^{\text{exit}}}{\partial \theta_j} = \frac{\mu_j^{\text{exit}}}{\theta_j} > 0$. Writing $D_j = 2\delta - y_j(1-\theta_j) > 0$ and $N_j = \delta - k - y_j(1-\tau_j) > 0$, differentiation gives $\frac{\partial \mu_j^{\text{fight}}}{\partial \theta_j} = -2\delta \frac{N_j}{D_j^2} < 0$.

First suppose $\frac{d\tau_j^*}{d\theta_j} \geq 0$. The other ruler's interior first-order condition gives $\frac{d\tau_{-j}^*}{d\theta_j} = q_{-j}\frac{d\tau_j^*}{d\theta_j} \geq 0$. Let $W_j = y_j - y_{-j}q_{-j}$. Since $A_{-j} \geq \theta_{-j}\frac{y_{-j}}{2\delta}$, $W_j \geq \frac{y_j}{2} > 0$. For $E_j = y_{-j}(1-\tau_{-j}^*) - c_L - y_j(1-\tau_j^*) + \delta$, therefore, $\frac{dE_j}{d\theta_j} = W_j\frac{d\tau_j^*}{d\theta_j} \geq 0$, and

$$\frac{d\mu_j^{\text{exit}*}}{d\theta_j} = \frac{E_j}{2\delta} + \theta_j \frac{W_j}{2\delta} \cdot \frac{d\tau_j^*}{d\theta_j} > 0.$$

Now suppose $\frac{d\tau_j^*}{d\theta_j} < 0$. At the optimum, compliance equals $A_j\tau_j^*$. Because $A_j' < 0$,

$$\frac{d\left(\mu_j^{\text{fight}} + \mu_j^{\text{exit}}\right)}{d\theta_j} = -A_j'\tau_j^* - A_j\frac{d\tau_j^*}{d\theta_j} > 0.$$

Meanwhile,

$$\frac{d\mu_j^{\text{fight}}}{d\theta_j} = \frac{\left[-2\delta N_j + (1-\theta_j)y_j D_j \frac{d\tau_j^*}{d\theta_j}\right]}{D_j^2} < 0.$$

Thus exit increases because total resistance increases while fight decreases, proving part (b). For part (a), the quotient-rule numerator for the exit share is $\left(\frac{d\mu_j^{\text{exit}*}}{d\theta_j}\right)\mu_j^{\text{fight}*} - \mu_j^{\text{exit}*}\left(\frac{d\mu_j^{\text{fight}*}}{d\theta_j}\right) > 0$.

A.3.3. Proof of Proposition 2

PROOF: Separate the neighboring net income from the home ruler's coefficients. Define

$$\tilde{B}_j := \frac{(1-\theta_j)(\delta - y_j - k)}{2\delta - y_j(1-\theta_j)} + \frac{\theta_j}{2\delta}(\delta - c_L - y_j), \quad \tilde{p}_j := \frac{1-\tilde{B}_j}{2A_j}.$$

Then $B_j = \tilde{B}_j + \theta_j\frac{y_{-j}}{2\delta}$, so $p_j = \tilde{p}_j - q_j$. The two equilibrium first-order conditions can be written exactly as

$$\tau_j^* = \tilde{p}_j - q_j(1-\tau_{-j}^*), \quad \tau_{-j}^* = p_{-j} + q_{-j}\tau_j^*.$$

When θ_j changes, p_{-j} and q_{-j} are fixed because they depend on the other domain's exposure to outside authority and the two yields, not on θ_j. They are not fixed when y_{-j} changes.

Let a prime denote differentiation with respect to θ_j. We have

$$A_j' = \frac{y_j}{2\delta} - \frac{2\delta y_j}{D_j^2} < 0, \quad K_j := \frac{A_j - \theta_j A_j'}{4\delta A_j^2} > 0, \quad q_j' = y_{-j}K_j.$$

Differentiating both equilibrium conditions, substituting $\left(\tau_{-j}^*\right)' = q_{-j}\left(\tau_j^*\right)'$, and collecting terms gives

$$\left(1 - q_j q_{-j}\right)\left(\tau_j^*\right)' = \tilde{p}_j' - q_j'(1-\tau_{-j}^*) = \tilde{p}_j' - K_j z_{-j}^*.$$

This retains the other ruler's endogenous tax response. Since $1 - q_j q_{-j} > 0$, the sign is determined by $\frac{\tilde{p}_j'}{K_j} - z_{-j}^*$.

To calculate the threshold, write $a_j := 1 - \theta_j$ and $H_j := 2\delta - y_j \theta_j a_j > 0$. Combining terms in A_j gives $A_j = y_j \frac{H_j}{2\delta D_j}$, and direct differentiation yields

$$K_j = \frac{\delta\big(2\delta - y_j a_j^2\big)}{y_j H_j^2},$$

$$\tilde{p}_j' = \frac{\delta\big[(c_L - k)\big(2\delta - y_j a_j^2\big) + y_j a_j\big(4\delta - y_j a_j\big)\big]}{y_j H_j^2}.$$

Thus $\frac{\tilde{p}_j'}{K_j} = \kappa_j$, with κ_j as stated in the proposition, and

$$\frac{\partial \tau_j^*}{\partial \theta_j} = \frac{K_j}{1 - q_j q_{-j}}\big(\kappa_j - z_{-j}^*\big).$$

The multiplier is strictly positive, proving all three sign cases. If $\kappa_j \leq 0$, then $z_{-j}^* > 0 \geq \kappa_j$, so the derivative is strictly negative. The comparison is local in exposure to outside authority along the interior equilibrium: no claim of existence or uniqueness of a cutoff in gross neighboring productivity is required.

A.4. Non-Vacuousness

Both signs of the tax response occur on non-empty open sets. For tax restraint, take $\delta = 10$, $y_1 = y_2 = 18$, $k = \frac{7}{2}$, $c_L = \frac{1}{2}$, $c_H = 29$, and $\theta_1 = \theta_2 = \frac{5}{6}$. The interior coefficients are $A_j = \frac{63}{68}$, $B_j = \frac{77}{272}$, $C_j = \frac{3}{4}$, $p_j = \frac{65}{168}$, and $q_j = \frac{17}{42}$. The equilibrium has $\tau_j^* = \frac{13}{20}$, $\mu_j^{\text{fight}*} = \frac{1}{510}$, and $\mu_j^{\text{exit}*} = \frac{19}{48}$. Neighboring net income is $z_{-j}^* = \frac{63}{10}$, exceeding $\kappa_j = \frac{35}{13}$, and $\frac{\partial \tau_j^*}{\partial \theta_j} = -\frac{5628}{36875} < 0$.

For a tax increase, take $\delta = 10$, $y_1 = y_2 = 12$, $k = 7$, $c_L = \frac{1}{2}$, $c_H = 23$, and $\theta_1 = \theta_2 = \frac{1}{3}$. Then $\tau_j^* = \frac{7}{8}$, $\mu_j^{\text{fight}*} = \frac{1}{12}$, and $\mu_j^{\text{exit}*} = \frac{19}{120}$. Here $z_{-j}^* = \frac{3}{2} < \kappa_j = \frac{241}{22}$, so greater exposure to outside authority raises the tax. In both examples, all primitive and interiority inequalities are strict, and the global-best-response argument above applies. Continuity therefore preserves the respective signs and equilibrium conditions on open neighborhoods of these parameter vectors.

B. Descriptive Statistics

B.1. Village Resistance Data

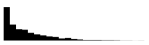
	Unique	Missing %	Mean	SD	Min	Median	Max	Histogram	Source
Basic									
Home Ruler	168	0	–	–	–	–	–	–	Aoki (1975)
Nbr Ruler	224	0	–	–	–	–	–	–	Aoki (1975)
Year	218	0	1,777.3	59.0	1,615.0	1,782.0	1,867.0		Aoki (1975)
Decade	26	0	1,772.7	59.2	1,610.0	1,780.0	1,860.0		Aoki (1975)
Outcome									
Exit	2	0	0.2	0.4	0.0	0.0	1.0		Aoki (1975)
Direct Confrontation	2	0	0.8	0.4	0.0	1.0	1.0		Aoki (1975)
Predictor									
ρ_i	2,529	0	7.1	61.3	0.0	1.0	2,792.0		–
$\ln(\rho_i)$	2,529	0	0.2	1.5	−5.6	0.0	7.9		–
Dist. to Home Cap	2,517	0	45.8	85.5	0.0	21.2	1,096.0		–
Dist. to Nbr Cap	2,508	0	24.0	29.7	0.0	16.4	494.5		–
ρ_i^{LCP}	168	0	7.2	28.1	0.2	1.7	601.7		–
Controls									
Latitude	2,458	0	35.7	2.0	27.7	35.4	42.0		Honda, Natsume, and Nemoto (2022) and other sources
Longitude	2,445	0	136.5	3.3	128.9	136.6	142.0		Honda, Natsume, and Nemoto (2022) and other sources
Dist. to Edo	2,484	0	413.6	235.6	29.4	413.2	1,356.8		–
Dist. to Edo or Kyoto	2,484	0	234.0	152.4	0.0	191.1	1,041.7		–
Elevation	103	0	19.9	20.9	0.0	12.0	108.0		Geospatial Information Authority of Japan
Ruggedness	146	0	3.7	3.8	0.0	2.5	22.6		–
Wet-Rice Suitability	716	0	18,549.9	3,417.3	0.0	19,003.2	22,570.0		Galor and Özak (2016)
Avg. Caloric Suitability	1,133	0	2,221.3	242.7	941.1	2,297.5	2,531.2		Galor and Özak (2016)

Table A.5: Summary Statistics of the Village Resistance Data ($N = 2,741$)

Note: The distance ratios and capital distances summarize the measures recorded for each village. The preferred regressions instead identify the nearest active foreign capital at each event date from the dated tenure of each domain, so their exposure measure and estimation sample differ from these descriptive statistics.

B.2. Recorded Return after Exit

Year	Home	Village	Recorded end	Aoki p.
1759	松山	久妙寺	Returned after reconciliation	123
1788	宇和島	惣川村西	Required to return; later punishment	166
1800	松山	波万	Persuaded and returned that day	181
1813	宇和島	岩木	Domain persuasion; return and imprisonment	196
1818	吉田	俵津蒲	Returned on the twenty-fifth	204
1820	吉田	奥野川	Stopped or persuaded; return and repression	207
1827	宇和島	富野川	Persuaded to return; later exile or relocation	218

Table A.6: Exit Events Explicitly Recording Return

Note: These seven entries in the analysis sample explicitly record return after flight or attempted flight. Source: (Aoki 1975).

B.3. Audit of Destination Codes

Tactic	Foreign	Edo shogunate	Kyoto	Other	Uncoded	Total
Panel A: Unique events						
Desertion	55	1	0	0	126	182
End-run appeal	7	112	1	0	26	146
Total	62	113	1	0	152	328
Panel B: Village records						
Desertion	86	1	0	0	222	309
End-run appeal	18	365	1	0	72	456
Total	104	366	1	0	294	765

Table A.7: Destination Codes Are More Complete for End-Run Appeals Than for Desertions

Note: Counts cover exit records dated 1615–1868. Panel A counts each event once; Panel B counts every participating village. Column headings report the recorded destinations; Edo denotes the Edo shogunate. Other includes recorded destinations outside Foreign, Edo, and Kyoto; Uncoded denotes no recorded destination. Each tactic includes attempted actions, consistent with the pooled exit outcome in the main analyses. Every category, including zero counts, is displayed. Source: (Aoki 1975).

B.4. Domain Data

	Unique	NA %	Mean	SD	Min	Median	Max	Hist.	Source
Outcome									
Exits	23	0	2.2	5.5	0.0	0.0	37.0		Aoki (1975)
Direct Confrontations	39	0	7.0	14.4	0.0	1.0	164.0		Aoki (1975)
Resistance Events	45	0	9.2	17.7	0.0	2.0	201.0		Aoki (1975)
Exit Rate	56	40	0.3	0.4	0.0	0.1	1.0		Aoki (1975)
Tax Rate (τ_j)	256	7	0.4	0.1	0.1	0.4	0.7		Kodama and Kitajima (1989)
Predictor									
ρ_j	241	12	7.3	15.3	0.1	2.2	138.5		–
$\ln(\rho_j)$	241	12	1.0	1.3	−2.4	0.8	4.9		–
Dist. to Home Cap	241	12	49.0	66.7	2.5	22.4	442.5		–
Dist. to Nbr Cap	242	12	16.1	10.8	3.9	12.5	80.2		–
ρ_j^{LCP}	242	12	239.0	1,988.8	0.1	2.7	24,956.3		–
Controls: Main									
Wet-Rice Suitability	242	12	3,011,700.3	4,937,787.1	48,040.8	1,232,259.1	41,226,133.2		Galor and Özak (2016)
Territory Size	258	6	811.0	1,785.9	9.5	257.1	16,262.8		Honda, Natsume, and Nemoto (2022) and other sources
Population	255	7	87,781.0	159,087.3	2,466.0	30,833.0	1,065,910.0		LFAT
Pop. Density	249	9	166.5	125.1	11.9	150.2	1,285.0		–
# New Land	32	12	7.8	27.9	0.0	1.0	232.0		–
Shogunate Kin (0/1)	2	0	0.6	0.5	0.0	1.0	1.0		Kimura, Fujino, and Murakami (2015)
Goushi System (0/1)	2	0	0.1	0.3	0.0	0.0	1.0		Kimura, Fujino, and Murakami (2015)

Table A.8: Summary Statistics of the Domain Data ($N = 275$)

	Unique	Missing %	Mean	SD	Min	Median	Max	Histogram	Source
Controls: Geography									
Latitude	265	1	35.5	1.7	31.6	35.3	41.4		Kimura, Fujino, and Murakami (2015)
Longitude	265	1	136.5	3.3	128.8	136.6	141.5		Kimura, Fujino, and Murakami (2015)
Mean Elevation	242	12	16.1	13.6	2.6	12.0	88.4		–
Mean TRI	240	12	3.2	1.8	0.2	2.9	8.9		–
Dist. to Edo or Kyoto	242	12	188.5	148.9	25.2	135.0	671.3		–
% Rainy Days	251	3	0.1	0.0	0.1	0.1	0.2		Yoshimura HWDB[24]
% Snowy Days	251	3	0.0	0.0	0.0	0.0	0.1		Yoshimura HWDB
Controls: Neighbor									
Prop. Same Fam.	65	12	0.1	0.2	0.0	0.0	1.0		Kimura, Fujino, and Murakami (2015)
Avg. Count of New-Farmland Villages	78	12	32.9	69.2	0.0	7.5	421.0		–
Avg. Wet-Rice Suitability	242	12	2,135,287.7	3,139,273.3	288,682.9	1,350,670.1	41,057,071.8		Galor and Özak (2016)
Avg. Territory Size	242	12	433.8	602.5	12.4	278.0	6,926.3		Honda, Natsume, and Nemoto (2022) and other sources
Avg. Population	239	13	56,178.5	84,147.0	6,264.6	31,723.2	1,061,486.4		LFAT
Avg. Pop. Density	239	13	175.5	107.1	39.9	161.0	1,028.7		–
Avg. Pop. Density (Prod.)	239	13	0.0	0.0	0.0	0.0	0.2		–
Controls: Alternative									
Samurai Pop.	230	16	70,945.3	134,463.8	614.0	21,519.5	896,808.0		Kimura, Fujino, and Murakami (2015)
# Villages	170	6	196.5	304.5	1.0	82.5	2,243.0		LFAT
Dist. to Edo	242	12	370.8	244.7	42.2	309.4	1,039.5		–

Table A.9: Summary Statistics of the Domain Data (Cont.)

[24] Historical Weather Database made available online by Minoru Yoshimura: http://tk2-202-10627.vs.sakura.ne.jp/

C. Tenryō Conversion Difference-in-Differences

C.1. Conversion Events

The domain conversion to the shogunate's territory typically followed succession issues and misconduct (Kimura, Fujino, and Murakami 2015; Kodama and Kitajima 1989; Ravina 1999, 17–21). The analysis uses 49 events from domain histories (Table A.10).

Year	Domain	Province	Formal reason	Local source
1617	Itabashi 板橋藩	下野	Administrative transfer	HD v.2 p.194; HS p.102
1618	Nagamine 長峰藩	越後	Administrative relocation	HD v.3 p.105
1619	Musashi Komuro 武蔵小室藩	武蔵	No heir	HD v.2 p.632
1632	Omori 大森藩	近江	Succession restriction	HD v.5 p.30; HS pp.258-259
1639	Fukozu 深溝藩	三河	Family partition and relocation	HD v.4 p.314
1642	Nariwa 成羽藩	備中	Administrative transfer	HD v.6 p.255
1645	Fuchu 府中藩	常陸	No heir	HD v.2 p.39
1650	Kaibara 柏原藩	丹波	No heir	HD v.5 p.429; HS p.275
1653	Toyooka 豊岡藩	但馬	No heir	HD v.5 p.463
1656	Funai 府内藩	豊後	No heir	HD v.7 p.402
1657	Marugame 丸亀藩	讃岐	No heir	HD v.6 p.447; chronology p.451
1658	Kakegawa 掛川藩	遠江	No heir	HD v.4 p.177
1658	Yashima 矢島藩	出羽	Testamentary family partition	HD v.1 p.389
1660	Sakura 佐倉藩	下総	Official misconduct	HD v.2 p.453; Aoki p.33
1665	Fukumoto 福本藩	播磨	Family partition	HD v.5 p.530; HS p.295
1665	Saijo 西条藩	伊予	Official dereliction	HD v.6 p.469; Aoki p.38
1666	Miyazu 宮津藩	丹後	Household conflict and misgovernment	HD v.5 p.443; Aoki p.39

Table A.10: 49 Tenryō Conversions Used in Table 1 (Part 1 of 3)

Note: The table lists the first qualifying conversion of each treated domain, with corrected dates and source-stated reasons. HD denotes *Hanshi Daijiten* (Kimura, Fujino, and Murakami 2015); HS denotes *Hanshi Sōran* (Kodama and Kitajima 1989). Locators give volume and printed page.

Year	Domain	Province	Formal reason	Local source
1668	Shimabara 島原藩	肥前	Misgovernment	HD v.7 p.212; Aoki p.40
1670	Harima Shingu 播磨新宮藩	播磨	Succession restriction	HD v.5 p.583
1676	Aso 麻生藩	常陸	No heir	HD v.2 p.110
1678	Yamazaki 山崎藩	播磨	No heir	HD v.5 p.562; narrative p.557
1679	Kururi 久留里藩	上総	Mental incapacity	HD v.2 p.530; chronology p.533
1679	Niwase 庭瀬藩	備中	No heir	HD v.6 p.199
1681	Numata 沼田藩	上野	Misgovernment and official failure	HD v.2 p.257; Aoki pp.47,49
1681	Takada 高田藩	越後	Household conflict	HD v.3 p.113
1683	Tatebayashi 館林藩	上野	Succession termination	HD v.2 p.280; chronology p.288; Aoki p.45
1684	Sanuki 佐貫藩	上総	Official misconduct	HD v.2 p.546
1687	Karasuyama 烏山藩	下野	Succession-rule violation	HD v.2 p.165
1687	Soumi 沢海藩	越後	Misconduct	HD v.3 p.54; HS p.154
1689	Fukami 深見藩	相模	Official dereliction	HD v.2 p.681
1689	Kitami 喜多見藩	武蔵	Collective punishment after violence	HD v.2 p.675
1692	Hachiman 八幡藩	美濃	No heir	HD v.4 p.33
1692	Takayama 高山藩	飛騨	Administrative and political transfer	HD v.4 p.3
1693	Takahashi 備中松山藩	備中	No heir	HD v.6 p.243

Table A.11: 49 Tenryō Conversions Used in Table 1 (Part 2 of 3)

Note: Continuation of Table A.10. Source abbreviations and page conventions are unchanged.

Year	Domain	Province	Formal reason	Local source
1698	Fukuyama 福山藩	備後	No heir	HD v.6 p.267; Aoki p.57
1698	Sano 佐野藩	下野	Administrative relocation	HD v.2 p.236
1698	Yoshii 吉井藩	上野	Administrative relocation	HD v.2 p.336
1701	Ako 赤穂藩	播磨	Violence	HD v.5 p.608
1702	Nagashima 長島藩	伊勢	Violence and mental derangement	HD v.4 p.385
1703	Tamanawa 玉縄藩	相模	Administrative relocation	HD v.2 p.683; HS p.146
1704	Yamura 谷村藩	甲斐	Administrative relocation	HD v.3 p.322; HS p.181
1712	Hojo 北条藩	安房	Misgovernment and extraction	HD v.2 p.569; chronology p.572; Aoki p.66
1724	Kofu 甲府藩	甲斐	Administrative transfer	HD v.3 pp.331–332
1725	Matsumoto 松本藩	信濃	Violence and mental derangement	HD v.3 p.451; Aoki p.70
1739	Takayanagi 高柳藩	越後	Appointment-related land exchange	HD v.3 p.122; HS p.156
1767	Obata 小幡藩	上野	Political punishment	HD v.2 p.361; chronology p.367
1786	Sagara 相良藩	遠江	Political fall	HD v.4 p.161; chronology p.165
1788	Omi Komuro 近江小室藩	近江	Official misconduct and corruption	HD v.5 p.5; HS pp.259-260; Aoki p.146
1807	Matsumae 松前藩	蝦夷	Strategic and military transfer	HD v.1 p.5; chronology p.16

Table A.12: 49 Tenryō Conversions Used in Table 1 (Part 3 of 3)

Note: Continuation of Table A.10. Source abbreviations and page conventions are unchanged.

C.2. Potentially Unrest-Related Conversions

Conversion endogeneity motivates a separate exclusion exercise. A review of local histories and Aoki's chronology identifies three cases with documented unrest or petitioning near conversion —Saijō (1665), Shimabara (1668), and Hōjō (1712)—and two further cases, Numata (1681) and Ōmi Komuro (1788). Removing the first three increases the exit estimate from 0.0427 to 0.0472 (two-way cluster $p = 0.0076$); removing all five yields 0.0448 ($p = 0.0148$). A maximal screen also removes four conversions preceded within ten years by a revolt event recorded in Aoki (1975) in the same named domain or province, even though the histories do not identify those events as causes; the exit estimate is 0.0493 ($p = 0.0138$). Confrontation remains statistically indistinguishable from zero in all three samples (Table A.13). These screens show that the exit result does not depend on cases with documented or potential links to unrest, although domain histories cannot establish the absence of unrecorded unrest.

Specification	Outcome	Estimate	Cluster SE	Cluster p	Conley p	Treated
Baseline	Exit	0.0427	0.0167	0.0113	0.0031	49
Baseline	Confrontation	0.0126	0.0125	0.3147	0.4013	49
Exclude 3 documented cases	Exit	0.0472	0.0174	0.0076	0.0011	46
Exclude 3 documented cases	Confrontation	0.0129	0.0133	0.3338	0.4339	46
Exclude 5 documented/mixed cases	Exit	0.0448	0.0182	0.0148	0.0013	44
Exclude 5 documented/mixed cases	Confrontation	0.0112	0.0142	0.4292	0.4995	44
Maximal 10-year history screen	Exit	0.0493	0.0198	0.0138	0.0012	40
Maximal 10-year history screen	Confrontation	0.0140	0.0153	0.3632	0.4333	40

Table A.13: Sensitivity to Excluding Potentially Unrest-Related Conversions

Note: The narrow set excludes Saijō (1665), Shimabara (1668), and Hōjō (1712), with documented unrest or petitioning near conversion. This timing does not establish that unrest caused each conversion. The broader set also excludes Numata (1681) and Ōmi Komuro (1788) as mixed cases. The maximal screen additionally removes Sakura, Miyazu, Tatebayashi, and Matsumoto because Aoki records an event in the same named domain or province during the preceding ten years, even though the local histories do not identify those events as causes of conversion. Cluster standard errors are two-way by domain and year; Conley p-values use a 70 km cutoff. Silence in a short domain history is not treated as proof that no unrest occurred. Sources: (Aoki 1975; Kimura, Fujino, and Murakami 2015; Kodama and Kitajima 1989).

C.3. Count Models

Outcome:	Nearby Exit Count			Nearby Confrontation Count		
Radius:	30km	40km	50km	30km	40km	50km
Converted × Post	0.0387	0.0635	0.0835	0.0179	0.0275	0.0346
Cluster SE	(0.0264)	(0.0359)*	(0.0440)*	(0.0155)	(0.0207)	(0.0263)
Conley SE (70km)	(0.0292)	(0.0395)	(0.0460)*	(0.0122)	(0.0193)	(0.0292)
Num. Obs.	31,989	31,989	31,989	31,989	31,989	31,989
Domain FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Converted/Observed Domains	49/269	49/269	49/269	49/269	49/269	49/269

Table A.14: Count Robustness for Domain-to-Tenryo Conversions

Note: OLS fixed-effects estimates with event counts (rather than binary indicators) as outcomes, using the same annual domain-year panels as Table 1. Tenryo-converted domains contribute available observations within five years before and after conversion and are matched to never-converted controls observed in the same calendar years represented by the conversion window. Two SE rows: (i) two-way clustered on domain and year; (ii) Conley spatial standard errors with a 70 km cutoff on domain capital coordinates. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

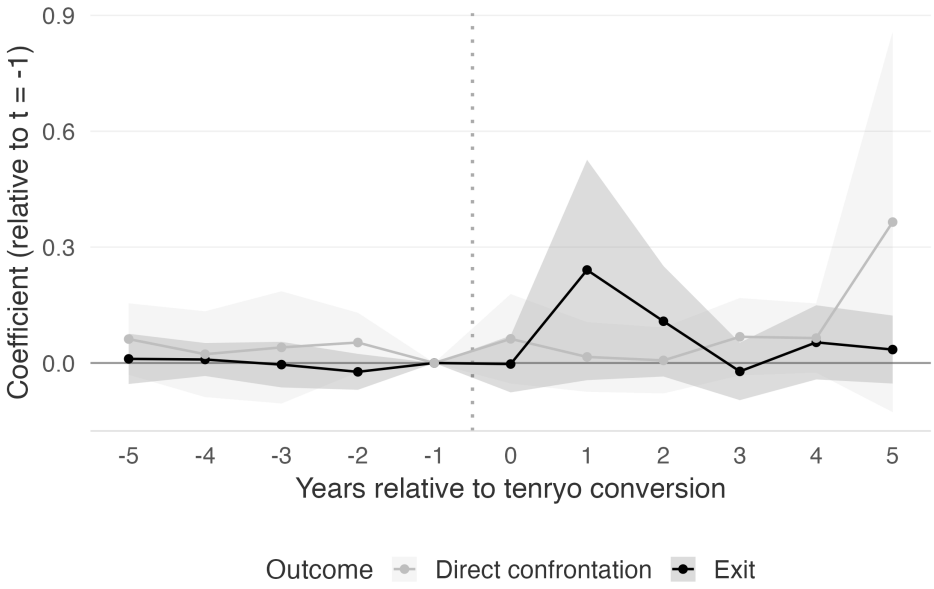


Figure A.8: OLS Count Event-Study Coefficients (50km)

Note: Year-by-year OLS coefficients from the specification $\text{count}_{j,t} = \sum_{s \neq -1} \beta_s \big(\mathbf{1}\big[t - t_j^* = s\big] \big) + \varphi_j + \omega_t + \varepsilon_{j,t}$, where the outcome is the count of nearby exit (black) or direct-confrontation (light gray) events within 50km of domain j's capital and $s = -1$ is the reference. Shaded bands show 95% confidence intervals; the horizontal line marks zero and the vertical dotted line marks conversion between $s = -1$ and $s = 0$. Standard errors are two-way clustered by domain and year.

C.4. Disaggregating Exit into Desertion and End-Run Appeal

Table A.15 replicates Table 1 with the exit outcome restricted to desertion only (*chōsan*), excluding end-run appeals.

Outcome:	Any Nearby Desertion		
Sample Radius:	30km	40km	50km
Variables			
Converted × Post	0.0127	0.0196	0.0295
Cluster SE	(0.0072)*	(0.0114)*	(0.0138)**
Controls			
Domain & Year FE	Yes	Yes	Yes
Fit Statistics			
Num. Obs.	31,989	31,989	31,989
R2	0.031	0.043	0.054
Mean Dep. Var.	0.004	0.007	0.010

Table A.15: Desertion-Only DiD

Note: Replicates Table 1 with the exit outcome restricted to desertion (*chōsan*), excluding end-run appeals. Specification matches Table 1. Two-way clustered (domain & year) standard errors. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

C.5. Alternative First-Conversion Estimates

Table A.16 compares the pooled estimate with the interaction-weighted estimator of Sun and Abraham (2021). The latter follows first-conversion timing, while the main exposure indicator can switch off after re-enfeoffment. Its exit point estimate is 0.0454, compared with the main 0.0427, and its confrontation point estimate is 0.0314. The unadjusted two-way clustered covariance is not positive semidefinite; the aggregate confrontation variance is negative. I therefore report these point estimates without cluster standard errors or p-values. With 49 treated domains distributed over 40 conversion cohorts, this sparse specification does not provide an independent inferential check on the main contrast between tactics.

Estimator	Nearby Exit			Nearby Confrontation		
	ATT	SE	p	ATT	SE	p
TWFE	0.0427	(0.0167)**	0.0113	0.0126	(0.0125)	0.3147
Sun & Abraham (2021)	0.0454	—	—	0.0314	—	—

Table A.16: Pooled Exposure and First-Conversion Point Estimates

Note: The TWFE row reproduces the main 50km specification from Table 1. The Sun and Abraham row uses first-conversion cohorts and never-converted controls; exposure timing therefore differs when domains were re-enfeoffed. Two-way clustered (domain & year) inference is reported for TWFE. Dashes mark unavailable Sun–Abraham inference because the unadjusted clustered covariance is invalid; no ad hoc covariance correction is substituted. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

C.6. Goodman-Bacon Decomposition of the TWFE Estimator

Table A.17 reports the Goodman-Bacon (2021) decomposition for a balanced, full-history panel with 30 converted and 157 never-converted domains (50,303 observations). This diagnostic codes treatment as beginning at first conversion and remaining on thereafter. It therefore examines a different time window and exposure definition from the main specification, which permits re-enfeoffment. Treated-versus-never-treated comparisons receive 94.9% of the weight; already-treated and not-yet-treated comparisons receive 3.5% and 1.6%, respectively. These weights describe this balanced diagnostic and do not decompose the main windowed coefficient or establish homogeneous effects.

Comparison type	Weight	Avg. 2×2 estimate
Treated vs. never-treated	94.9%	0.0102
Treated vs. already-treated	3.5%	0.0013
Treated vs. yet-to-be-treated	1.6%	0.0182

Table A.17: Goodman-Bacon Decomposition of the 50 km Tenryo DiD

Note: The decomposition uses a balanced 50 km domain-year panel. Weights describe the contribution of each type of two-by-two comparison to the two-way fixed-effects estimate. A large weight on comparisons with never-treated domains limits the role of comparisons between differently timed treated cohorts; it does not establish homogeneous effects across cohorts.

C.7. Cohort-Split Sun and Abraham ATT

To inspect heterogeneity, Table A.18 estimates the Sun and Abraham (2021) first-conversion specification separately for pre-1750, 1750–1799, and post-1800 conversions. The groups contain 45, 3, and 1 treated domains, respectively. Their exit point estimates are 0.0482, −0.0064, and 0.0708. Most identifying cases belong to the first group; the small later groups, especially the single post-1800 domain, do not support a general comparison across historical periods. The covariance and timing qualifications in SI Section C.5 also apply here.

Cohort bin	Cohorts	Treated units	ATT	SE	p
All cohorts	40	49	0.0454	—	—
pre-1750	36	45	0.0482	—	—
1750–1799	3	3	−0.0064	—	—
post-1800	1	1	0.0708	—	—

Table A.18: Cohort-Split Sun and Abraham (2021) ATT on Nearby Exit (50 km)

Note: Each row re-runs the interaction-weighted estimator on the sub-sample of treated domains whose tenryo-conversion year falls in the indicated bin (plus all never-treated controls). Dashes mark inference not reported because the covariance matrix clustered by domain and year is invalid for that sample; point estimates are shown.

C.8. Rambachan-Roth Relative-Magnitudes Breakdown

Table A.19 reports the relative-magnitudes sensitivity curve of Rambachan and Roth (2023) for the average post-conversion exit effect at 50 km. $\overline{M}$ bounds the largest post-period deviation from parallel trends relative to the largest pre-period deviation. The 95% robust interval excludes zero at tested values through $\overline{M} = 0.20$, where it is $[0.0047, 0.1040]$, and includes zero at 0.25. Thus $[0.20, 0.25]$ brackets the breakdown on the tested grid; 0.20 is not an exact threshold. Under the smoothness restriction Δ^{SD}, the interval excludes zero up to $M = 0.0018$; at the empirical dispersion of the pre-period coefficients, $\sigma_{\text{pre}} = 0.0108$, it is $[-0.1453, 0.1849]$. The pre-period joint test gives $F = 0.95$, $p = 0.432$. Nonrejection of that test does not establish parallel trends, which is why the sensitivity bounds are reported separately.

Restriction	Bound	95% robust CI	Zero in CI?
Δ^{RM}	0.00	$[0.0133, 0.0773]$	No
Δ^{RM}	0.20	$[0.0047, 0.1040]$	No
Δ^{RM}	0.50	$[-0.0407, 0.1567]$	Yes
Δ^{RM}	1.00	$[-0.1313, 0.2493]$	Yes
Δ^{RM}	2.00	$[-0.3153, 0.4333]$	Yes
Δ^{SD}	0.0000	$[0.0242, 0.0793]$	No
Δ^{SD}	0.0010	$[0.0129, 0.0882]$	No
Δ^{SD}	0.0018	$[0.0004, 0.0988]$	No
Δ^{SD}	0.0020	$[-0.0027, 0.1014]$	Yes
Δ^{SD}	0.0030	$[-0.0182, 0.1143]$	Yes
Δ^{SD}	0.0050	$[-0.0488, 0.1391]$	Yes
Δ^{SD}	0.0100	$[-0.1318, 0.1822]$	Yes

Table A.19: Sensitivity of the Average Post-Conversion Exit Effect

Note: Rambachan–Roth (2023) intervals for the average post-conversion exit effect at 50 km. Under Δ^{RM}, $\overline{M}$ bounds post-period deviations relative to the largest pre-period deviation. Under Δ^{SD}, M bounds the second difference of the trend. The largest tested bounds retaining a positive lower confidence limit are $\overline{M} = 0.20$ and $M = 0.0018$. These are grid-based thresholds. The exit pre-trend joint Wald test gives $F = 0.95$ and $p = 0.432$.

C.9. Spatial Interference

Because conversions cluster in time and space, I examine possible interference by removing converted domains whose capitals lie within 100 km and five years of another conversion, removing never-converted controls within 50 km of a converted capital, and imposing both restrictions. Table A.20 reports the results. Exit coefficients remain positive, ranging from 0.0285 to 0.0427. Excluding potentially exposed controls alone leaves a statistically significant estimate; removing overlapping treated domains produces wider intervals that include zero. Confrontation estimates remain statistically indistinguishable from zero in all four specifications.

	Nearby Exit		Nearby Confrontation	
Sample	$\hat{\beta}$	SE	$\hat{\beta}$	SE
Full sample ($N = 31{,}989$)	0.0427	(0.0167)**	0.0126	(0.0125)
	(0.0144)***			(0.0150)
Drop overlap ($N = 31{,}673$)	0.0337	(0.0256)	0.0333	(0.0215)
	(0.0260)			(0.0230)
Clean controls ($N = 13{,}110$)	0.0421	(0.0178)**	0.0085	(0.0123)
	(0.0148)***			(0.0156)
Strictest ($N = 12{,}794$)	0.0285	(0.0264)	0.0266	(0.0207)
	(0.0269)			(0.0250)
Domain FE: Yes Year FE: Yes				

Table A.20: SUTVA / Spatial Interference Robustness

Note: Each row re-estimates the main 50 km DiD specification on progressively restricted samples. "Drop overlap" removes converted domains within 100 km and 5 years of another conversion. "Clean controls" removes never-converted domains within 50 km of any converted capital. "Strictest" applies both exclusions. Two SE rows are reported in parentheses: (i) two-way clustered on domain and year and (ii) Conley spatial standard errors with a 70 km cutoff. Significance stars on each SE row reflect that estimator's p-value. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

C.10. Permutation Test

I randomly reassign the 49 conversions across the 288 eligible domains, drawing years from the observed conversion dates, and re-estimate the main exit specification 500 times. No placebo estimate is as large in absolute value as the observed estimate of 0.0427 ($p \approx 0.002$). The placebo estimates center on zero (mean -0.0004; 95% range $[-0.0302, 0.0296]$). Because actual conversions were not random, this test shows that the observed effect is unusual.

Full-pool permutation test (50 km, nearby exit)	Result
Actual coefficient (Converted × Post)	0.0427
Actual clustered p-value	0.0113
Valid / requested iterations	500 / 500
Eligible domains	288
Placebo events per iteration	49
Permutation mean	-0.0004
Permutation SD	0.0150
Permutation 2.5% quantile	−0.0302
Permutation 97.5% quantile	0.0296
Two-sided extreme draws / valid draws	0 / 500
Two-sided Monte Carlo p-value	0.0020
Right-tail Monte Carlo p-value	0.0020

Table A.21: The Observed Exit Estimate Lies Beyond the Permutation Distribution

Note: Each iteration draws 49 placebo conversion events without replacement from the 288 eligible domains, including the actually treated domains. Dates are drawn from the empirical conversion-year distribution. The same five-year event window and matched calendar years used in Table 1 are applied before estimating the 50 km two-way fixed-effects exit coefficient. The two-sided tail counts draws whose absolute estimate is at least as large as the actual absolute estimate. Monte Carlo p-values use $\frac{k+1}{B+1}$, where k is the number of extreme draws and B is the number of valid draws.

C.11. Hazard Model for Conversion Timing

A remaining concern is that the shogunate converted domains in response to recent nearby unrest. I test this with a hazard model of conversion timing, covering 29 conversions across 233 domains (51,313 domain-years). The predictors are five-year lagged averages of nearby exit and confrontation within 50 km, with territory size, distance to the Edo–Kyoto axis, and alliance status as controls. Neither lagged exit ($p = 0.2066$ Cox, $p = 0.4912$ logit) nor confrontation ($p = 0.8093$ Cox, $p = 0.3432$ logit) predicts conversion (joint $p = 0.4355$). Only alliance status predicts conversion, consistent with accounts of attainder politics (Ravina 1999).

Model	Variable	$\hat{\beta}$	SE	p	Ratio
Cox	Nearby exit MA(5)	-4.4807	3.5476	0.2066	0.0113
Cox	Nearby confrontation MA(5)	0.6075	2.5178	0.8093	1.8359
Cox	Log area	-0.2176	0.1728	0.2078	0.8044
Cox	Log distance to Edo–Kyoto axis	0.2430	0.3158	0.4416	1.2750
Cox	Fudai/kin (vs. tozama)	1.2120	0.5163	0.0189	3.3602
Logit	Nearby exit MA(5)	-2.4002	3.4863	0.4912	0.0907
Logit	Nearby confrontation MA(5)	-2.2383	2.3614	0.3432	0.1066
Logit	Log area	-0.1556	0.1685	0.3558	0.8559
Logit	Log distance to Edo–Kyoto axis	0.2100	0.3110	0.4997	1.2336
Logit	Fudai/kin (vs. tozama)	1.1734	0.5198	0.0240	3.2331
Cox	Joint Wald: exit and confrontation	$\chi^2 = 1.663$		0.4355	—

Table A.22: Lagged Nearby Resistance Does Not Predict Conversion Timing

Note: Cox proportional-hazards and discrete-time logit estimates. The complete-case timing sample contains 51313 domain-years across 233 domains, including 29 observed conversions from the 49 treated domains in the primary analysis. Resistance variables are lagged five-year moving averages within 50 km of the focal capital. SEs are from the fitted Cox and logit models; ratios are hazard ratios for Cox and odds ratios for logit. The Cox joint Wald test has two degrees of freedom.

C.12. Conley Standard Error Bandwidth Sensitivity

The 70 km Conley cutoff used for the main DiD standard errors is a researcher choice. Table A.23 reports cutoffs of 30, 50, 70, 100, and 150 km. The point estimate remains 0.0427; Conley standard errors range from 0.0117 to 0.0144, with all p-values below 0.005.

Inference	Cutoff	$\hat{\beta}$	SE	p	N
Cluster (domain+year)	–	0.0427	0.0167	0.0113	31989
Conley	30 km	0.0427	0.0138	0.0020	31989
Conley	50 km	0.0427	0.0140	0.0023	31989
Conley (paper)	70 km	0.0427	0.0144	0.0031	31989
Conley	100 km	0.0427	0.0121	0.0004	31989
Conley	150 km	0.0427	0.0117	0.0003	31989

Table A.23: Conley Bandwidth Sweep for the Tenryo DiD (50 km)

Note: Tenryo DiD exit coefficient with standard errors computed under the cluster estimator (the main-text default) and Conley spatial HAC standard errors at five cutoffs. The point estimate does not depend on the inference method; only the SE and p-value vary. The hand-picked 70 km cutoff used in Table 1 sits near the middle of the sweep.

D. Grid-Year Panel Analysis

D.1. Desertion-Only Outcome

Table A.24 replicates Table 2 with exit restricted to desertion (*chōsan*) only. The desertion-specific coefficients track the pooled exit estimates closely in both sign and significance, across the full panel and the ever-active subsample, confirming that the within-cell exit response is driven by the prototypical desertion cases rather than by end-run appeals.

	Full sample		Ever-active cells	
Outcome:	(1)	(2)	(3)	(4)
	Desertion	Pooled exit	Desertion	Pooled exit
Grid exposure ratio ($\rho^{\text{grid}}_{i,t}$)	0.00024	0.00035	0.00152	0.00198
Cluster SE	(0.00014)*	(0.00017)**	(0.00087)*	(0.00109)*
Conley SE (10km)	(0.00012)**	(0.00017)**	(0.00075)**	(0.00103)*
Controls				
Grid & Year FE	Yes	Yes	Yes	Yes
Fit Statistics				
Num.Obs.	2,225,802	2,225,802	372,364	372,364
R^2	0.005	0.006	0.006	0.008
Mean Dep. Var.	0.00009	0.00022	0.00054	0.00131

Table A.24: Desertion-Only Grid-Year Panel

Note: LPM on binary any-event-of-type indicators. Replicates Table 2 with exit restricted to desertion (*chōsan*), excluding end-run appeals. Pooled exit columns reproduce the main specification for comparison. Columns (1)–(2) use the full panel of 8,763 cells; columns (3)–(4) restrict to the 1,466 ever-active hexagons. Grid and year FE. SEs: (i) two-way clustered (cell, year); (ii) Conley (10 km). Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

D.2. Count Models: Poisson and Negative Binomial

Table A.25 replicates the main-text Table 2 using two count-data estimators with hexagon and year fixed effects: Poisson (columns 1–3) and the negative binomial (columns 4–6). Because both estimators use a log link ($\log \mathbb{E}[y_{i,t} \mid \mathrm{X}_{i,t}] = \beta \mathrm{X}_{i,t} + \alpha_i + \tau_t$), the raw coefficient $\hat{\beta}$ sits on the log-count scale. The *incidence rate ratio*, $\mathrm{IRR} \equiv \exp(\hat{\beta})$, back-transforms this to the count scale: a one-unit increase in the regressor multiplies the expected count by IRR. Both estimators give positive exit coefficients, although the Poisson estimate is imprecise (clustered $p = 0.1472$; Conley $p = 0.1239$). The exit coefficient is positive and corresponds to a multiplicative association of roughly 4–5× per unit distance ratio (Poisson IRR = 4.020; NB IRR = 5.357). The confrontation coefficient is small and statistically indistinguishable from zero under both estimators (Poisson $p = 0.5676$; NB $p = 0.4891$), and the strict-confrontation coefficient turns negative (IRRs below one). Both estimators drop cells with no events of the relevant outcome. Their retained samples therefore vary by outcome and contain fewer cells than the full 8,763-cell panel.

	Poisson			Neg. Binom.		
Outcome:	(1)	(2)	(3)	(4)	(5)	(6)
	Exit count	Confrontation count	Confrontation count (strict)	Exit count	Confrontation count	Confrontation count (strict)
Grid exposure ratio ($\rho^{\text{grid}}_{i,t}$)	1.3912	0.3240	−0.4231	1.6784	0.4453	−0.5370
IRR	4.020	1.383	0.655	5.357	1.561	0.584
Cluster SE	(0.9599)	(0.5668)	(0.5299)	(0.8860)*	(0.6437)	(0.7985)
Conley SE (10km)	(0.9043)	(0.5257)	(0.6052)	(0.8601)*	(0.6620)	(0.7938)
Controls						
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Fit Statistics						
Num.Obs.	62,900	202,130	40,810	62,900	202,130	40,810
Pseudo R^2	0.135	0.130	0.127	0.093	0.092	0.087
Mean Dep. Var.	0.0003	0.0009	0.0004	0.0003	0.0009	0.0004

Table A.25: Grid-Panel Count Models: Poisson and Negative Binomial with FE

Note: Columns (1)–(3) estimate a Poisson model and columns (4)–(6) a negative binomial, both with hexagon and year fixed effects on the 5 km grid-year panel (1615–1868). Coefficients are on the log-count scale; IRR = $\exp(\hat{\beta})$ gives the multiplicative effect on the expected count. Both estimators drop all-zero outcome groups, so Num. Obs. reflects the retained sample for each outcome rather than the full 8,763-cell panel. SEs: (i) two-way clustered on cell and year; (ii) Conley (10 km). Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

D.3. Ever-Active Cells Subsample

Table A.26 replicates Table 2 on the 1,466 "ever-active" hexagons — cells with at least one exit or confrontation event between 1615 and 1868. This restriction filters out the 83% of cells that recorded neither form of resistance and sharpens within-cell variation among active locales while using the same binary-outcome LPM and fixed-effects structure. Columns (1)–(3) use the grid exposure ratio $\rho^{\text{grid}}_{i,t}$; columns (4)–(6) use the closest-capital distance $d^1_{i,t}$ (km).

The distance-ratio coefficient on exit rises to 0.00198 ($p = 0.054$ Conley, $p = 0.070$ clustered), while the confrontation and strict-confrontation coefficients remain small and statistically indistinguishable from zero. The closest-capital regressor remains a null predictor of exit on this subsample ($p = 0.179$ clustered, $p = 0.259$ Conley). It carries a marginal negative signal for strict confrontation (Conley $p = 0.028$), plausibly because uprisings and castle attacks require a nearby state target. Remoteness therefore contains some information about where direct confrontation is mechanically feasible but not where exit is chosen; relative geographic exposure is associated with exit.

	Grid exposure ratio ($\rho^{\text{grid}}_{i,t}$)			Closest-capital dist. ($d^1_{i,t}$, km)		
Outcome:	(1)	(2)	(3)	(4)	(5)	(6)
	Any exit	Any confrontation	Any confrontation (strict)	Any exit	Any confrontation	Any confrontation (strict)
Coefficient	0.00198	0.00096	−0.00097	0.00004	−0.00004	−0.00004
Cluster SE	(0.00109)*	(0.00105)	(0.00067)	(0.00003)	(0.00003)	(0.00002)*
Conley SE (10km)	(0.00103)*	(0.00105)	(0.00062)	(0.00004)	(0.00002)*	(0.00002)**
Controls						
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Fit Statistics						
Num.Obs.	372,364	372,364	372,364	372,364	372,364	372,364
R^2	0.008	0.011	0.011	0.008	0.011	0.011
Mean Dep. Var.	0.00131	0.00384	0.00155	0.00131	0.00384	0.00155

Table A.26: Grid-Year Panel on Ever-Active Cells: Distance Ratio vs. Closest-Capital Distance

Note: Replicates Table 2 on the 1,466 "ever-active" hexagons — cells with at least one exit or confrontation event between 1615 and 1868 — which sharpens within-variation among active locales while filtering out cells with no recorded resistance. Columns (1)–(3) use the time-varying grid exposure ratio $\rho^{\text{grid}}_{i,t} = \frac{d^1_{i,t}}{d^2_{i,t}}$. Columns (4)–(6) replace it with the raw closest-capital distance $d^1_{i,t}$ (km). Grid and year FE. SEs: (i) two-way clustered (cell, year); (ii) Conley (10 km). Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

D.4. Least-Cost-Path Distance Ratio

Table A.27 presents a version of Table 2 in which the straight-line distance ratio is replaced by a least-cost-path (LCP) ratio accounting for terrain slope. The two measures are moderately correlated ($r = 0.44$). The estimated coefficient for LCP measure retains its sign and a comparable standardized magnitude: a one-standard-deviation increase in the LCP ratio is associated with approximately 0.011 percentage-point increase in exit probability, compared to 0.009 in Table 2 column (1). Confrontation coefficients remain near zero. The exit coefficient is less precisely estimated ($p = 0.278$ clustered; $p = 0.203$ Conley), likely because incorporating terrain increases the similarity of paths to each capital and thus reduces variation in the regressor.

	Full sample			*Ever-active cells*		
Outcome:	(1)	(2)	(3)	(4)	(5)	(6)
	Any exit	Any confrontation	Any confrontation (strict)	Any exit	Any confrontation	Any confrontation (strict)
Grid exposure ratio (LCP, $\rho_{i,t}^{\text{grid,LCP}}$)	0.00045	0.00053	−0.00008	0.00173	0.00177	−0.00051
Cluster SE	(0.00041)	(0.00060)	(0.00031)	(0.00169)	(0.00221)	(0.00122)
Conley SE (10km)	(0.00035)	(0.00068)	(0.00026)	(0.00141)	(0.00192)	(0.00074)
Controls						
Grid FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Fit Statistics						
Num.Obs.	2,225,802	2,225,802	2,225,802	372,364	372,364	372,364
R^2	0.006	0.007	0.006	0.008	0.011	0.011
Mean Dep. Var.	0.00022	0.00064	0.00026	0.00131	0.00384	0.00155

Table A.27: Grid-Year Panel with Least-Cost-Path Distance Ratio

Note: LPM on binary any-event-of-type indicators. Replicates Table 2 with the haversine grid exposure ratio replaced by a terrain-weighted least-cost-path (LCP) variant: $\rho_{i,t}^{\text{grid,LCP}}$ equals the ratio of nearest-to-second-nearest LCP cost from each hexagon centroid to any active domain capital in year t, where cost is cumulative across pixels of a $(1 + \text{slope})$ cost surface derived from the SRTM elevation raster. Columns (1)–(3) use the full panel; (4)–(6) restrict to ever-active hexagons. Grid and year FE. SEs: (i) two-way clustered (cell, year); (ii) Conley (10 km). Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

D.5. Daikansho-Inclusive Distance Ratio

Table A.28 replicates Table 2 with the daikansho-inclusive distance ratio. Adding daikansho to the capital set compresses the ratio toward unity and reduces within-cell variation; all grid-level coefficients attenuate and lose statistical significance. The village-level and tax results are unaffected by this inclusion (SI Section E.2.9; SI Section F.2.6).

	Full sample			Ever-active cells		
Outcome:	(1)	(2)	(3)	(4)	(5)	(6)
	Any exit	Any confront.	Any confront. (strict)	Any exit	Any confront.	Any confront. (strict)
Grid exposure ratio ($\rho^{\text{grid}}_{i,t}$)	0.00019	0.00023	−0.00006	0.00095	−0.00012	−0.00096
Cluster SE	(0.00019)	(0.00021)	(0.00012)	(0.00111)	(0.00111)	(0.00068)
Conley SE (10km)	(0.00018)	(0.00021)	(0.00012)	(0.00107)	(0.00097)	(0.00060)
Controls						
Grid & Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Fit Statistics						
Num.Obs.	2,225,802	2,225,802	2,225,802	372,364	372,364	372,364
R^2	0.006	0.007	0.006	0.008	0.011	0.011
Mean Dep. Var.	0.00022	0.00064	0.00026	0.00131	0.00384	0.00155

Table A.28: Grid-Year Panel with Daikansho-Inclusive Distance Ratio

Note: LPM on binary any-event-of-type indicators. Replicates Table 2 with the daikansho-inclusive distance ratio. Columns (1)–(3) use the full panel of 8,763 cells; columns (4)–(6) restrict to the 1,466 ever-active hexagons. See Table 2 notes for details. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E. Village Exit Choice Analysis

E.1. Full Table

Outcome:	exit_i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
$\ln(\rho_i)$	0.0458	0.0235	0.0143	0.0201	0.0212	0.0244
	(0.0154)***	(0.0117)*	(0.0089)	(0.0071)***	(0.0072)***	(0.0068)***
lat					−1.4681	−2.0087
					(0.6101)**	(1.4745)
lon					−0.2692	−0.4046
					(0.1515)*	(0.3957)
ln(Dist. Edo/Kyoto)					0.2237	0.0784
					(0.0419)***	(0.0793)
ln(Rice)					−0.0055	−0.0018
					(0.0045)	(0.0046)
ln(Elev.)					−0.0228	−0.0335
					(0.0198)	(0.0193)*
ln(Rugg.)					0.0289	0.0330
					(0.0152)*	(0.0146)**
Lat × Lon					0.0100	0.0141
					(0.0047)**	(0.0114)
Controls						
Home (j) Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor (k) Domain FE	No	No	Yes	Yes	Yes	No
Decade (t) FE	No	No	No	Yes	Yes	Yes
Home-Neighbor Dyad (j, k) FE	No	No	No	No	No	Yes
Fit Statistics						
Num.Obs.	2,739	2,739	2,739	2,739	2,739	2,739
R2	0.028	0.390	0.543	0.621	0.626	0.678

Table A.29: Villages Exposed to Alternative Authority Opt for Exit Over Confrontation (Full Table)

Note: The coefficients represent OLS estimates of the relationship between each village's geographic exposure and its choice of exit conditional on recorded resistance. Clustered (home ruler, nearest neighboring ruler, and decade) standard errors are in parentheses. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E.2. Robustness Checks

E.2.1. Alternative Covariates

Outcome:	exit_i							
	Cal. Suit.				Dist. to Edo			
Models:	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Variables								
$\ln(\rho_i)$	**0.0434**	**0.0162**	**0.0209**	**0.0234**	**0.0439**	**0.0135**	**0.0199**	**0.0240**
	(0.0150)***	**(0.0090)***	**(0.0072)*****	**(0.0069)*****	**(0.0149)*****	**(0.0092)**	**(0.0075)****	**(0.0068)*****
lat	−1.3123	−0.1980	−1.6336	−2.0788	−1.7772	−2.7623	−2.9088	−2.9748
	(0.3162)***	(0.7180)	(0.5745)***	(1.3142)	(0.6161)***	(1.0481)**	(0.7516)***	(1.3385)**
lon	−0.2880	0.1762	−0.2967	−0.4006	−0.4525	−0.6267	−0.7051	−0.6908
	(0.0781)***	(0.2098)	(0.1291)**	(0.3399)	(0.1775)**	(0.3681)	(0.2625)**	(0.3908)*
ln(Dist. Edo/Kyoto)	−0.0187	0.4531	0.2295	0.1037				
	(0.0451)	(0.1668)**	(0.0576)***	(0.0744)				
ln(Cal.)	−0.0614	0.0953	0.1490	0.2386				
	(0.1949)	(0.1834)	(0.1591)	(0.1564)				
ln(Elev.)	0.0121	−0.0146	−0.0261	−0.0410	−0.0005	−0.0198	−0.0257	−0.0350
	(0.0260)	(0.0193)	(0.0184)	(0.0175)**	(0.0203)	(0.0200)	(0.0194)	(0.0191)*
ln(Rugg.)	0.0328	0.0222	0.0288	0.0324	0.0441	0.0302	0.0329	0.0349
	(0.0241)	(0.0161)	(0.0155)*	(0.0146)**	(0.0209)**	(0.0153)*	(0.0149)**	(0.0141)**
Lat × Lon	0.0091	0.0001	0.0112	0.0146	0.0127	0.0197	0.0210	0.0215
	(0.0022)***	(0.0056)	(0.0044)**	(0.0101)	(0.0045)***	(0.0084)**	(0.0061)***	(0.0104)**
ln(Dist. Edo)					−0.1374	−0.3334	−0.1806	−0.2381
					(0.0755)*	(0.3733)	(0.2859)	(0.4100)
ln(Rice)					−0.0085	−0.0077	−0.0050	−0.0018
					(0.0067)	(0.0062)	(0.0044)	(0.0047)
Const.	42.3502				64.0829			
	(11.2821)***				(24.2964)**			
Controls								
Home (j) Domain FE	No	Yes	Yes	No	No	Yes	Yes	No
Neighbor (k) Domain FE	No	Yes	Yes	No	No	Yes	Yes	No
Decade (t) FE	No	No	Yes	Yes	No	No	Yes	Yes
Home-Neighbor Dyad (j,k) FE	No	No	No	Yes	No	No	No	Yes
Fit Statistics								
Num.Obs.	2,739	2,739	2,739	2,739	2,739	2,739	2,739	2,739
R2	0.061	0.551	0.627	0.679	0.069	0.549	0.626	0.678

Table A.30: Villages Exposed to Alternative Authority

Opt for Exit Over Direct Confrontation (Alternative Covariates)

Note: Clustered (Home & Neighbor Domains & Decade) standard errors in parentheses.
Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E.2.2. Excluding Villages with More Than One Recorded Resistance Episode

Outcome:	exit_i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
$\ln(\rho_i)$	**0.0488**	**0.0270**	**0.0228**	**0.0229**	**0.0234**	**0.0231**
	(0.0153)***	**(0.0119)****	**(0.0104)****	**(0.0094)****	**(0.0094)****	**(0.0080)*****
lat					−0.8936	−0.4750
					(0.7595)	(1.5455)
lon					−0.0921	0.0445
					(0.2281)	(0.4405)
ln(Dist. Edo/Kyoto)					0.1885	0.1062
					(0.0942)*	(0.1155)
ln(Rice)					−0.0031	−0.0003
					(0.0039)	(0.0047)
ln(Elev.)					−0.0077	−0.0227
					(0.0221)	(0.0203)
ln(Rugg.)					0.0211	0.0309
					(0.0161)	(0.0159)*
Lat × Lon					0.0061	0.0029
					(0.0057)	(0.0118)
Const.	0.2255					
	(0.0373)***					
Controls						
Home (j) Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor (k) Domain FE	No	No	Yes	Yes	Yes	No
Home-Neighbor Dyad (j, k) FE	No	No	No	No	No	Yes
Decade (t) FE	No	No	No	Yes	Yes	Yes
Geography	No	No	No	No	Yes	Yes
Fit Statistics						
Num.Obs.	2,295	2,295	2,295	2,295	2,295	2,295
R2	0.032	0.431	0.590	0.672	0.675	0.732

Table A.31: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation (Excluding Repeat Villages)

Note: Clustered (Home & Neighbor Domains & Decade) standard errors in parentheses. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E.2.3. Absolute Distances

Outcome:	exit_i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
ln(Dist. to Home Capital$_i$)	**0.0399**	**0.0248**	**0.0127**	**0.0231**	**0.0160**	**0.0203**
	(0.0223)*	**(0.0161)**	**(0.0109)**	**(0.0096)****	**(0.0104)**	**(0.0083)****
ln(Dist. to Foreign Capital$_i$)	**−0.0596**	**−0.0203**	**−0.0172**	**−0.0147**	**−0.0306**	**−0.0309**
	(0.0140)***	**(0.0218)**	**(0.0199)**	**(0.0147)**	**(0.0129)****	**(0.0135)****
lat					−1.5650	−2.0841
					(0.6458)**	(1.5054)
lon					−0.2878	−0.4180
					(0.1658)*	(0.4041)
ln(Dist. Edo/Kyoto)					0.2343	0.0869
					(0.0430)***	(0.0839)
ln(Rice)					−0.0055	−0.0017
					(0.0046)	(0.0047)
ln(Elev.)					−0.0211	−0.0319
					(0.0206)	(0.0199)
ln(Rugg.)					0.0305	0.0340
					(0.0163)*	(0.0153)**
Lat × Lon					0.0107	0.0147
					(0.0050)**	(0.0116)
Const.	0.2984					
	(0.0788)***					
Controls						
Home (j) Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor (k) Domain FE	No	No	Yes	Yes	Yes	No
Home-Neighbor Dyad FE	No	No	No	No	No	Yes
Decade (t) FE	No	No	No	Yes	Yes	Yes
Geography	No	No	No	No	Yes	Yes
Fit Statistics						
Num.Obs.	2,739	2,739	2,739	2,739	2,739	2,739
R2	0.030	0.390	0.543	0.621	0.627	0.678

Table A.32: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation (Abs. Dist. Measures)

Note: Clustered (Home & Neighbor Domains & Decade) standard errors in parentheses. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E.2.4. Water-Crossing Subsample

The main village distance-ratio regression uses straight-line great-circle distances from each village to its home-domain capital and nearest foreign-domain capital, and these lines can traverse sea straits such as Kanmon, Bungo, Naruto, or Sado. To check if these cases drive the results, I drop every resistance-event observation whose village, home capital, or nearest foreign capital lies on a different main island (Honshu, Kyushu, Shikoku, Sado, or Awaji), classified from latitude and longitude. This removes 150 of 2,739 observations (5.5%). On this water-safe subsample the coefficient on $\log(\rho_i)$ remains positive across all six specifications, including the dyadic home-neighbor fixed-effects column that mirrors Table 3 Model 5 ($\hat{\beta} = 0.0253$, clustered $p < 0.001$; baseline 0.0244, $p = 0.001$).

Outcome:	exit_i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
Distance ratio ($\ln(\rho_i)$)	0.0460	0.0232	0.0160	0.0209	0.0207	0.0253
Cluster SE	(0.0156)***	(0.0127)*	(0.0105)	(0.0082)**	(0.0080)**	(0.0063)***
Conley SE (10km)	(0.0121)***	(0.0113)**	(0.0094)*	(0.0079)***	(0.0079)***	(0.0086)***
Controls						
Home Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor Domain FE	No	No	Yes	Yes	Yes	No
Home-Nbr Dyad FE	No	No	No	No	No	Yes
Decade FE	No	No	No	Yes	Yes	Yes
Geography	No	No	No	No	Yes	Yes
Fit Statistics						
Num. Obs.	2,589	2,589	2,589	2,589	2,589	2,589
R^2	0.029	0.391	0.552	0.635	0.639	0.683
Mean Dep. Var.	0.251	0.251	0.251	0.251	0.251	0.251

Table A.33: Village Distance-Ratio Regression: Water-Crossing Subsample

Note: Re-runs the six village distance-ratio specifications on the subsample that drops resistance-event observations whose straight-line village-to-home-capital or village-to-foreign-capital line crosses a sea strait (i.e., the village, the home capital, and the nearest foreign capital are not all on the same main island). Island membership (Honshu, Kyushu, Shikoku, Sado, Awaji) is assigned from latitude and longitude. 150 (5.5%) of 2,739 observations are classified as sea-crossing and dropped. The coefficient on $\log(\rho_i)$ remains positive across all six specifications and statistical significance is preserved in Model 6 (dyadic home-neighbor FE plus geography controls), which corresponds to Table 3 Model 5. SE rows: (i) three-way clustered on (home, neighbor, decade); (ii) Conley spatial SE with a 10 km cutoff on village coordinates. Stars reflect the corresponding estimator's p-value. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E.2.5. Conley Standard Error Bandwidth Sensitivity

Table A.34 reports the bandwidth sensitivity of Conley spatial standard errors for the main village distance-ratio regression (Table 3 Model 5, dyadic home-neighbor FE). The coefficient on $\log(\rho_i)$ is stable across all bandwidths from 25 to 150 km and remains significant at the 5% level in every specification.

Standard error	Estimate	SE	p
Cluster (home + nbr + decade, paper)	0.0244	(0.0068)	0.001
Conley 25 km	0.0244	(0.0089)	0.006
Conley 50 km	0.0244	(0.0097)	0.012
Conley 75 km	0.0244	(0.0102)	0.017
Conley 100 km	0.0244	(0.0099)	0.014
Conley 150 km	0.0244	(0.0078)	0.002

Table A.34: Conley SE Bandwidth Sensitivity: Village Distance-Ratio Regression

Note: The coefficient on $\log(\rho_i)$ from Table 3 Model 5 (dyadic home-neighbor FE) is stable across all bandwidths from 25 to 150 km and remains significant at the 5% level in every specification. The first row reports the three-way cluster standard error used in the main table.

E.2.6. Small-G Cluster Sensitivity of the Decade Dimension

Table A.35 reports a small-G cluster-sensitivity check for the village distance-ratio regression (Table 3 Model 5, dyadic home-neighbor FE). The decade dimension has $G = 26$ clusters, below the conventional safe threshold for multi-way cluster-robust inference, motivating a check of whether this dimension is load-bearing for the reported significance. Dropping the decade dimension entirely and clustering only on the two higher-G dimensions (home ruler and neighbor ruler) yields $p = 0.0024$, close to the three-way number ($p = 0.0013$); clustering on decade alone yields $p = 0.0014$. The main-text significance is therefore preserved under every cluster specification considered.

Cluster specification	Estimate	SE	p
Three-way (home + nbr + decade, paper)	0.0244	(0.0068)	0.001
Two-way (home + nbr, dropping decade)	0.0244	(0.0079)	0.002
One-way on decade only ($G = 26$)	0.0244	(0.0068)	0.001

Table A.35: Small-G Cluster Sensitivity of the Decade Dimension

Note: Re-runs Table 3 Model 5 (dyadic home-neighbor FE) under three cluster specifications. The decade dimension has $G = 26$ clusters, below the conventional safe threshold for multi-way clustering. The key finding is that the paper's three-way cluster SE is robust to dropping the decade dimension entirely — two-way clustering on home and neighbor gives nearly identical inference — indicating the decade cluster is not load-bearing for the reported p -value.

E.2.7. Excluding Tenure-Related Narratives

Aoki's entries name participating villages but rarely identify the tenure status of the households involved. Removing every event whose demand narrative mentions tenants, landlords, or *komae* smallholders leaves the preferred village-choice estimate positive and statistically significant (Table A.36). The estimates therefore do not depend on events with explicit tenancy content.

Sample	Estimate	Cluster p	Conley p	Events	Villages
Baseline	0.0244	0.0013	0.0041	909	2739
Exclude tenure-keyword events	0.0218	0.0008	0.0113	890	2672

Table A.36: Village-Choice Sensitivity to Tenure-Related Narratives

Note: The restricted sample removes entire resistance events whose demand narrative contains *kosaku* (tenant), *jinushi* (landlord), or *komae* (smallholder). This excludes nineteen events and sixty-seven village records, only one of which is coded as exit. Both specifications use the preferred home-neighbor dyad and decade fixed effects with the main geographic controls. Cluster inference is three-way by home ruler, neighboring ruler, and decade; Conley inference uses a 10 km cutoff. The screen is a conservative content proxy, not a household-level tenure classification.

E.2.8. Desertion-Only Village Cross-Section

Table A.37 re-estimates the preferred specifications with the exit outcome restricted to desertion only (*chōsan*), excluding end-run appeals. The preferred coefficient is positive, marginal under three-way clustering ($p = 0.051$), and significant with Conley inference ($p = 0.041$), so the narrower definition preserves the direction and most of the pooled magnitude with some inference sensitivity.

Outcome:	Desertion_i	
Models:	(4)	(5)
Variables		
Distance ratio ($\ln(\rho_i)$)	0.0127	0.0141
Cluster SE	(0.0060)**	(0.0068)*
Conley SE (10km)	(0.0062)**	(0.0069)**
Controls		
Home + Neighbor + Decade FE + Geography	Yes	No
Dyadic + Decade FE + Geography	No	Yes
Fit Statistics		
Num.Obs.	2,727	2,727
R2	0.598	0.674

Table A.37: Desertion-Only Village Cross-Section

Note: Replicates Table 3 Models 4–5 with exit restricted to desertion (*chōsan*), excluding end-run appeals. Two SE rows: (i) three-way clustered; (ii) Conley spatial SE (10 km). Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E.2.9. Daikansho-Inclusive Distance Ratio

Table A.38 replicates Table 3 with d_i^{FC} redefined as the distance to the nearest non-home administrative center of any type, including shogunal intendant offices (*daikansho*). The preferred dyadic-FE specification (Model 5) is virtually unchanged ($\hat{\beta} = 0.0268$ vs. 0.0244 in Table 3).

Outcome:	Exit_i				
Models:	(1)	(2)	(3)	(4)	(5)
Variables					
Distance ratio ($\ln(\rho_i)$)	0.0422	0.0496	0.0201	0.0202	0.0268
Cluster SE	(0.0148)***	(0.0137)***	(0.0074)**	(0.0075)**	(0.0067)***
Conley SE (10km)	(0.0117)***	(0.0093)***	(0.0077)***	(0.0079)**	(0.0085)***
Controls					
Decade FE	No	Yes	Yes	Yes	Yes
Home Domain FE	No	No	Yes	Yes	No
Neighbor Domain FE	No	No	Yes	Yes	No
Dyadic FE	No	No	No	No	Yes
Geography	No	No	No	Yes	Yes
Fit Statistics					
Num.Obs.	2,739	2,739	2,739	2,739	2,739
R2	0.024	0.222	0.633	0.638	0.686
Mean Dep. Var.	0.246	0.246	0.246	0.246	0.246

Table A.38: Village Exit Choice with Daikansho-Inclusive Distance Ratio

Note: Replicates Table 3 with d_i^{FC} redefined to include shogunal intendant offices (*daikansho*). Two SE rows: (i) clustered on home, nearest neighboring domain, and decade; (ii) Conley spatial standard errors (10 km). Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E.3. Additional Analyses

E.3.1. Controlling for Distance to the Border

Outcome:	exit_i					
Models:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
$\ln(\rho_i)$	**0.0487**	**0.0225**	**0.0152**	**0.0206**	**0.0206**	**0.0264**
	(0.0161)***	**(0.0124)***	**(0.0098)**	**(0.0068)*****	**(0.0065)*****	**(0.0075)*****
$\ln(\text{Shortest Distance to Border}_i)$	**0.0160**	**0.0073**	**0.0010**	**0.0057**	**0.0058**	**0.0037**
	(0.0103)	**(0.0085)**	**(0.0106)**	**(0.0089)**	**(0.0086)**	**(0.0086)**
Controls						
Home (j) Domain FE	No	Yes	Yes	Yes	Yes	No
Neighbor (k) Domain FE	No	No	Yes	Yes	Yes	No
Decade (t) FE	No	No	No	Yes	Yes	Yes
Home-Neighbor Dyad (j, k) FE	No	No	No	No	No	Yes
Fit Statistics						
Num.Obs.	2,391	2,391	2,391	2,391	2,391	2,391
R2	0.037	0.377	0.535	0.632	0.638	0.676

Table A.39: Villages Exposed to Alternative Authority Opt for Exit Over Direct Confrontation

(Shortest Distance to Border)

Note: The coefficients relate the distance ratio (home-capital distance divided by distance to the nearest active non-home capital in the baseline daimyō/former-domain-capital geography) to exit choice conditional on recorded resistance, controlling for shortest distance to another domain's border. Border data match 2,391 of the 2,739 observations in the preferred panel. On this covered sample, the preferred dyadic estimate is 0.0265 without and 0.0264 with the border control; the border coefficient is 0.0037 ($p = 0.668$). Clustered (home ruler, nearest neighboring ruler, and decade) standard errors are in parentheses. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E.3.2. Sequential Decomposition of the Village Choice

The main analysis conditions on recorded resistance, so it cannot show whether exposure also predicts whether a village resists at all. I therefore add compliant villages from the end-period village ledger, dropping ledger entries that match resisting villages by name or, when names do not resolve, by nearest location. The combined data contain 46,387 observations.

I split each village's choice among compliance, exit, and direct confrontation into two binary steps:

$$\text{Step 1:} \quad \mathbb{P}(R_i = 1 \mid X_i) = \Lambda(X_{i'}\beta_1),$$
$$\text{Step 2:} \quad \mathbb{P}(E_i = 1 \mid R_i = 1, Z_i) = \Lambda(Z_{i'}\beta_2),$$

where R_i indicates resistance, E_i indicates exit, and $\Lambda(\cdot)$ is the logistic CDF. This is a nested logit with exit and confrontation in one nest. Its likelihood factors into the two steps, so they can be estimated separately and, unlike a flat multinomial logit, without the IIA restriction. The sign of β_2 is directly interpretable; its scale depends on the nest's dissimilarity parameter.

Table A.40 reports a flat multinomial logit (Columns 1–2) and the two steps estimated by logit (Columns 3–4) and LPM (Columns 5–6). Step 1 uses one observation per village and no fixed effects, because compliant villages have no neighboring ruler or event date. Step 2 matches Table 3 Model 5.

Exposure does not predict resistance: the Step 1 estimates are small and insignificant (logit $\beta = 0.028$, $p = 0.58$; LPM $\beta = -0.00078$, $p = 0.12$). It does predict the tactic: the Step 2 estimates are positive (logit $\beta = 0.276$, $p = 0.009$; LPM $\beta = 0.0244$, $p = 0.001$). Exposure is thus associated with how villages resist, without a detectable association with whether they resist. One caveat: compliant villages receive the end-period (1868) distance ratio, whereas resisting villages receive the ratio at the event date. Using each village's average ratio over 1615–1868 for both groups (with earlier neighbor assignments) yields a small negative Step 1 estimate (logit β = -0.0303, $p = 0.0402$) and a positive Step 2 estimate (LPM $\beta = 0.01436$, $p = 0.0089$).

	Multinomial logit		*Seq. binary logit*		*Seq. LPM*	
Outcome:	Exit	Direct confrontation	Non-compliance	Exit | non-compliance	Non-compliance	Exit | non-compliance
Models:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Distance ratio ($\ln(\rho_i)$)	0.2823	−0.0055	0.0281	0.2762	−0.00078	0.0244
Primary SE (see note)	(0.0149)***	(0.0167)	(0.0514)	(0.1054)***	(0.00050)	(0.0068)***
Conley SE (see note)	—	—	(0.0771)	(0.1159)**	(0.00096)	(0.0085)***
Controls						
Geo controls	Yes	Yes	Yes	Yes	Yes	Yes
Home × Neighbor FE	No	No	No	Yes	No	Yes
Decade FE	No	No	No	Yes	No	Yes
Fit Statistics						
Num. Obs.	46,365	46,365	46,274	1,310	46,274	2,739
Mean Dep. Var.	0.014	0.042	0.054	0.246	0.054	0.246

Table A.40: Sequential Decomposition of the Village Choice: Logit and LPM

Note: Columns (1)–(2) report one multinomial logit with compliance as the baseline. Columns (3)–(6) report the two steps by logit and LPM: Step 1 (Columns 3 and 5) compares resistance with compliance, one observation per village; Step 2 (Columns 4 and 6) compares exit with direct confrontation among resisting villages, using the specification of Table 3 Model 5. Primary SEs are model-based in Columns (1)–(3), heteroskedasticity-robust in Column (5), and clustered by home ruler, neighbor, and decade in Columns (4) and (6). Conley SEs use a 10 km cutoff and are not available for the multinomial logit; for Column (3), they come from a logit that omits the latitude and longitude main effects (coefficient 0.0261, $p = 0.734$). Means of the dependent variable are computed before estimation exclusions. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

E.3.3. Linear Hypothesis Testing

In the village-level analysis for validating the village mobility measure, I use the log-ratio of home-capital distance to nearby-capital distance:

$$\ln\left(\frac{\text{Dist. to Home Cap.}}{\text{Dist. to Foreign Cap.}}\right) = \ln(\text{Dist. to Home Cap.}) - \ln(\text{Dist. to Foreign Cap.})$$

Implicit in this choice is the restriction: $\beta_{\ln(\text{Dist. to Home Cap.})} + \beta_{\ln(\text{Dist. to Foreign Cap.})} = 0$

To check whether the data are consistent with this restriction, I estimate the unrestricted model including both ln(Dist. to Home Cap.) and ln(Dist. to Foreign Cap.) separately, then conduct a Wald test of the linear hypothesis $\beta_{\ln(\text{Dist. to Home Cap.})} + \beta_{\ln(\text{Dist. to Foreign Cap.})} = 0$.

Corresponding Model	Outcome	Restriction	Df	F	p-value
Table 3 (7)	exit_i	$\beta_{\ln(\text{DHC})} + \beta_{\ln(\text{DFC})} = 0$	(1, 25)	0.4035	0.5311

Table A.41: Wald Test of the Log-Ratio Restriction

The Wald test uses the preferred dyadic model and standard errors clustered by home domain, neighboring domain, and decade. It does not reject the equal-and-opposite restriction ($p = 0.531$); nonrejection does not establish that the restriction holds.

F. Domain Taxation Analysis

F.1. Full Tables

F.1.1. Main Model

Outcome:	τ_j											
Models:	(1)		(2)		(3)		(4)		(5: SPL)		(6: SPL)	
Variables												
ln(ρ_j)	**−0.0241**	**(0.0053)*** **	**−0.0203**	**(0.0058)*** **	**−0.0110**	**(0.0055)** **	**−0.0112**	**(0.0055)** **	**−0.0140**	**(0.0047)*** **	**−0.0094**	**(0.0051)***
ln(Rice)			−0.0010	(0.0158)	0.0076	(0.0157)	0.0116	(0.0163)			0.0004	(0.0135)
ln(Area)			−0.0260	(0.0112)**	−0.0318	(0.0144)**	−0.0340	(0.0151)**			−0.0227	(0.0130)*
ln(Pop.)			0.0392	(0.0167)**	0.0263	(0.0144)*	0.0232	(0.0150)			0.0222	(0.0147)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)			0.0004	(0.0002)
Shog. Ally			−0.0316	(0.0177)*	0.0059	(0.0161)	0.0011	(0.0164)			0.0025	(0.0143)
Goshi			0.0217	(0.0256)	0.0196	(0.0261)	0.0274	(0.0271)			0.0242	(0.0215)
lat					−0.5279	(0.2934)*	−0.4747	(0.3077)			−0.3535	(0.2855)
lon					−0.1758	(0.0754)**	−0.1639	(0.0796)**			−0.1252	(0.0743)*
ln(Elev.)					−0.0118	(0.0160)	−0.0137	(0.0165)			−0.0100	(0.0134)
ln(Rugg.)					0.0398	(0.0160)**	0.0373	(0.0158)**			0.0238	(0.0152)
ln(Dist. Edo/Kyoto)					−0.0370	(0.0158)**	−0.0441	(0.0171)**			−0.0335	(0.0167)**
Rainy					−1.6629	(0.8969)*	−1.5887	(0.9078)*			−1.0143	(0.7369)
Snowy					0.7322	(1.8580)	0.6920	(1.8597)			−0.2419	(1.4193)
Lat × Lon					0.0042	(0.0021)*	0.0038	(0.0023)*			0.0029	(0.0021)
Kin Nbs.							0.0396	(0.0296)			0.0188	(0.0257)
Nb. FL							0.0002	(0.0001)			0.0001	(0.0001)
ln(Nb. Rice)							−0.0244	(0.0189)			−0.0221	(0.0172)
ln(Nb. Area)							0.0180	(0.0176)			0.0116	(0.0159)
ln(Nb. Pop.)							−0.0047	(0.0207)			−0.0015	(0.0196)
Const.	0.4152	(0.0091)***	0.1760	(0.1247)	23.1744	(10.2659)**	21.8425	(10.7990)**	0.1761	(0.0274)***	16.4558	(10.0938)
Fit Statistics												
Num.Obs.	232		230		230		230		230		230	
R2	0.074		0.141		0.370		0.389		0.312[a]		0.439[a]	

Table A.42: Geographic Exposure and Reported Domain Tax Rates

Note: OLS estimates (maximum likelihood in Models 5–6) of reported effective tax rates on the log domain-average distance ratio. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Models 5–6. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1. [a] Nagelkerke pseudo-R2

F.1.2. Interaction Model

Outcome:	τ_j			
Models:	(1)	(2)	(3)	(4)
Variables				
Distance ratio $(\ln(\rho_j) \times \hat{y}_{-j})$: **High**	−0.0350	−0.0286	−0.0143	−0.0158
	(0.0084)***	(0.0089)***	(0.0070)**	(0.0069)**
Distance ratio $(\ln(\rho_j) \times \hat{y}_{-j})$: **Low**	−0.0161	−0.0151	−0.0092	−0.0085
	(0.0053)***	(0.0058)***	(0.0063)	(0.0064)
Controls				
Main	No	Yes	Yes	Yes
Geography	No	No	Yes	Yes
Neighbors	No	No	No	Yes
Fit Statistics				
Num.Obs.	232	230	230	230
R2	0.091	0.150	0.371	0.391
Mean Dep. Var.	0.392	0.394	0.394	0.394

Table A.43: Geographic Exposure and Tax Rates, Interacted with Outside-Option Quality

Note: The table reports group-specific OLS slopes for the log distance ratio. High denotes a below-median weighted labor-to-potential-yield ratio; Low denotes an above-median ratio. HC1 standard errors in parentheses.

F.1.3. Interaction Model (Full Table)

Outcome:	τ_j							
Models:	(1)		(2)		(3)		(4)	
Variables								
$\ln(\rho_j)$ × High	**−0.0350**	**(0.0084)*****	**−0.0286**	**(0.0089)*****	**−0.0143**	**(0.0070)****	**−0.0158**	**(0.0069)****
$\ln(\rho_j)$ × Low	**−0.0161**	**(0.0053)*****	**−0.0151**	**(0.0058)*****	**−0.0092**	**(0.0063)**	**−0.0085**	**(0.0064)**
ln(Rice)			0.0014	(0.0156)	0.0083	(0.0153)	0.0122	(0.0159)
ln(Area)			−0.0256	(0.0111)**	−0.0314	(0.0144)**	−0.0336	(0.0151)**
ln(Pop.)			0.0360	(0.0171)**	0.0253	(0.0145)*	0.0221	(0.0151)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)
Shog. Ally			−0.0300	(0.0180)*	0.0064	(0.0163)	0.0016	(0.0166)
Goshi			0.0221	(0.0257)	0.0197	(0.0262)	0.0278	(0.0271)
lat					−0.5362	(0.2946)*	−0.4835	(0.3095)
lon					−0.1785	(0.0758)**	−0.1668	(0.0802)**
ln(Elev.)					−0.0111	(0.0160)	−0.0125	(0.0165)
ln(Rugg.)					0.0381	(0.0163)**	0.0348	(0.0161)**
ln(Dist. Edo/Kyoto)					−0.0373	(0.0159)**	−0.0445	(0.0172)**
Rainy					−1.6210	(0.9021)*	−1.5333	(0.9156)*
Snowy					0.6224	(1.8655)	0.5548	(1.8808)
Lat × Lon					0.0042	(0.0022)*	0.0038	(0.0023)*
Kin Nbs.							0.0404	(0.0297)
Nb. FL							0.0002	(0.0001)
ln(Nb. Rice)							−0.0224	(0.0190)
ln(Nb. Area)							0.0191	(0.0174)
ln(Nb. Pop.)							−0.0081	(0.0206)
Const.	0.4181	(0.0092)***	0.1754	(0.1211)	23.4988	(10.3154)**	22.2086	(10.8672)**
Fit Statistics								
Num.Obs.	232		230		230		230	
R2	0.091		0.150		0.371		0.391	

Table A.44: Geographic Exposure and Reported Tax Rates by Outside-Option Quality

(Full Table)

Note: The coefficients represent OLS estimates of the relationship between the average distance ratio of each domain interacted with $h@(y)_{\{-j\}}$ (indicator of whether the weighted average labor-to-potential-yield ratio of neighboring domains is above its median) and the domain tax rate. Heteroskedasticity-robust standard errors in parentheses. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

F.2. Robustness Checks

F.2.1. Alternative Covariates

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5)	
Variables										
$\ln(\rho_j)$	**−0.0099**	**(0.0049)****	**−0.0106**	**(0.0057)***	**−0.0112**	**(0.0058)***	**−0.0108**	**(0.0055)****	**−0.0131**	**(0.0063)****
ln(Kokudaka)	−0.0788	(0.0160)***								
ln(Pop.)	0.0820	(0.0168)***							0.0282	(0.0156)*
ln(Area)	−0.0050	(0.0127)	−0.0328	(0.0128)**	−0.0251	(0.0142)*	−0.0052	(0.0064)	−0.0350	(0.0166)**
New FL	0.0003	(0.0004)	0.0003	(0.0004)	0.0003	(0.0004)	0.0004	(0.0004)	0.0003	(0.0003)
Shog. Ally	0.0068	(0.0154)	−0.0057	(0.0165)	0.0009	(0.0171)	0.0032	(0.0167)	0.0069	(0.0160)
Goshi	0.0245	(0.0260)	0.0261	(0.0272)	0.0309	(0.0269)	0.0267	(0.0273)	0.0321	(0.0248)
lat	−0.3791	(0.2791)	−0.4525	(0.3160)	−0.5680	(0.3505)	−0.3697	(0.3181)	0.1399	(0.2658)
lon	−0.1412	(0.0729)*	−0.1581	(0.0828)*	−0.1879	(0.0909)**	−0.1392	(0.0834)*	0.0166	(0.0673)
ln(Elev.)	−0.0064	(0.0157)	−0.0144	(0.0161)	−0.0155	(0.0165)	−0.0203	(0.0163)	−0.0081	(0.0167)
ln(Rugg.)	0.0143	(0.0156)	0.0352	(0.0159)**	0.0338	(0.0164)**	0.0296	(0.0157)*	0.0250	(0.0152)
ln(Dist. Edo/Kyoto)	−0.0418	(0.0165)**	−0.0423	(0.0173)**	−0.0435	(0.0172)**	−0.0481	(0.0168)***		
Snowy	1.8194	(1.7508)	0.4931	(1.8551)	0.9999	(1.9673)	1.0110	(1.9217)	−0.8507	(2.0177)
Rainy	−2.0637	(0.8820)**	−1.4692	(0.8923)	−1.7103	(0.9270)*	−1.6807	(0.9375)*	−1.0287	(0.9469)
Kin Nbs.	0.0266	(0.0301)	0.0500	(0.0283)*	0.0418	(0.0292)	0.0468	(0.0290)	0.0177	(0.0304)
Nb. FL	0.0001	(0.0001)	0.0002	(0.0001)	0.0002	(0.0001)	0.0002	(0.0001)	0.0002	(0.0001)*
ln(Nb. Rice)	−0.0050	(0.0174)	−0.0296	(0.0189)	−0.0291	(0.0199)	−0.0222	(0.0186)	−0.0190	(0.0187)
ln(Nb. Area)	−0.0019	(0.0170)	0.0176	(0.0167)	0.0177	(0.0176)	0.0100	(0.0163)	0.0073	(0.0182)
ln(Nb. Pop.)	−0.0073	(0.0186)	0.0018	(0.0203)	0.0005	(0.0215)	0.0011	(0.0202)	−0.0039	(0.0206)
Lat × Lon	0.0031	(0.0021)	0.0036	(0.0023)	0.0044	(0.0026)*	0.0030	(0.0023)	−0.0008	(0.0020)
ln(Rice)			−0.0017	(0.0250)	0.0239	(0.0151)			0.0047	(0.0181)
ln(Num. Vils.)			0.0373	(0.0250)						
% Samurai Pop					−0.1259	(0.1834)				
Population/wet-rice suitability							−0.0040	(0.2763)		
ln(Dist. Edo)									0.0677	(0.0261)**
Const.	18.8458	(9.8989)*	21.3229	(11.1659)*	25.2642	(12.3202)**	18.7213	(11.2921)*	−3.1359	(8.9956)
Fit Statistics										
Num.Obs.	230		231		226		230		230	
R2	0.446		0.397		0.383		0.374		0.399	

Table A.45: Geographic Exposure and Reported Domain Tax Rates

(Alternative Covariates)

Note: OLS estimates relating each domain's average distance ratio to its reported tax rate. The distance ratio measures geographic exposure, not observed competition or bargaining between rulers. Heteroskedasticity-robust standard errors are in parentheses. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

F.2.2. Sensitivity Analysis

I assess sensitivity to omitted variables with the method of Diegert, Masten, and Poirier (2022). Unlike Oster (2019), it allows the included controls to be correlated with the omitted confounders.

The sensitivity parameter $\bar{r}_X$ measures how strongly unobservables predict geographic exposure relative to observables. The exposure coefficient remains negative as long as $\bar{r}_X < 0.213$ (Figure A.9).

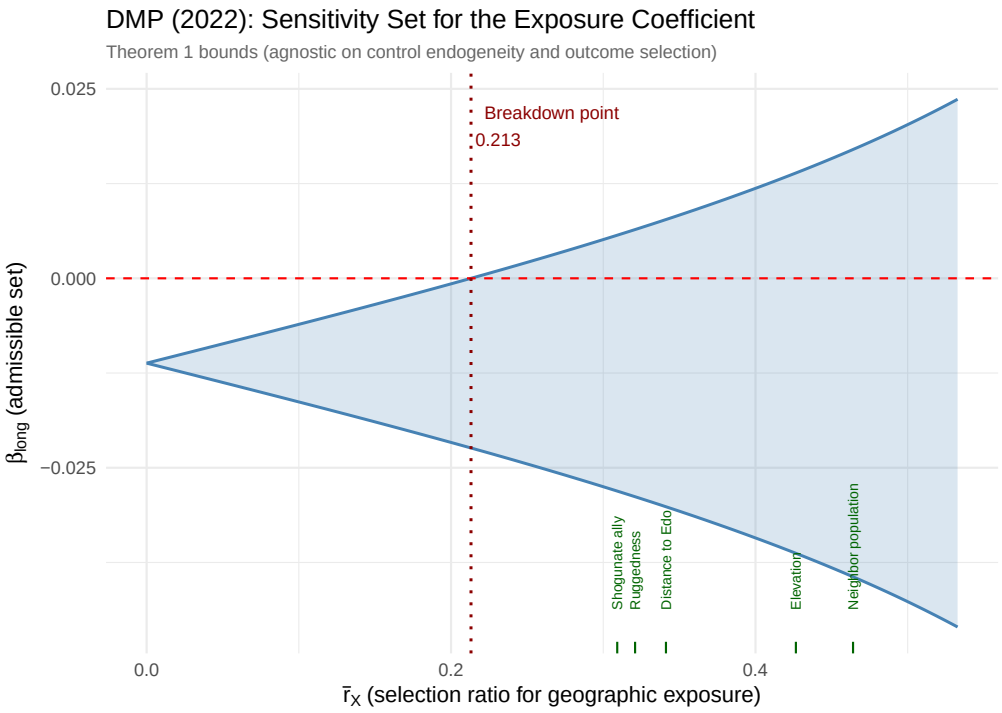


Figure A.9: DMP Sensitivity Set for the Exposure Coefficient

Note: The shaded region shows the range of β_{long} consistent with a given level of selection on unobservables ($\bar{r}_X$). The dashed red line marks $\beta = 0$; the dotted vertical line marks the breakdown point ($\bar{r}_X = 0.213$). Green ticks calibrate $\bar{r}_X$ to each labeled covariate's importance for predicting geographic exposure.

If unobservables are also assumed to matter no more for the tax rate than for exposure ($\bar{r}_Y \leq \bar{r}_X$), the coefficient remains negative up to $\bar{r} = 0.554$ (Figure A.10).

These thresholds concern the sign of the coefficient, not its statistical significance. Holding the standard error fixed, the estimate loses significance at the 5% level at $\bar{r}_X = 0.007$, or at $\bar{r}_X = \bar{r}_Y = 0.096$ under equal importance (Table A.46). The association thus loses precision much sooner than it loses its sign.

Sensitivity criterion	$\bar{r}$	Interpretation
Sign identification: $\bar{r}_X$ only	0.213	Upper endpoint of the DMP identified set first includes zero
Sign identification: $\bar{r}_X = \bar{r}_Y$	0.554	Same criterion under the equal-importance restriction
Fixed-SE significance: $\bar{r}_X$ only	0.007	Upper endpoint plus 1.96 baseline HC1 SEs first includes zero
Fixed-SE significance: $\bar{r}_X = \bar{r}_Y$	0.096	Same auxiliary criterion under equal importance

Table A.46: Diegert-Masten Sensitivity Thresholds for the Tax Regression

Note: Thresholds for Table 4 Model 4 (fully controlled). $\bar{r}_X$ and $\bar{r}_Y$ are ratios of standard deviations, not shares of explained variance. The first two rows are DMP population sign-identification breakdowns: the observed estimate remains fixed while the identified set widens to include zero. The last two rows answer the separate significance question by adding and subtracting 1.96 times the baseline HC1 SE from the DMP set while holding that SE fixed. They are transparent auxiliary thresholds, not fully bias-adjusted DMP confidence sets.

Excluding Chōshū, Satsuma, Mito, Tosa, and Aizu, whose late-period tax rates may reflect bakumatsu fiscal distress, changes the Model 4 estimate from −0.0112 to −0.0106 (p from 0.045 to 0.055; Table A.47).

Sample	Num. Obs.	Estimate	SE	p
Full sample (paper)	230	−0.0112	(0.0055)	0.045
Excluding Chōshū, Satsuma, Mito, Tosa, Aizu	225	−0.0106	(0.0055)	0.055

Table A.47: Domain Tax Regression: Excluding Bakumatsu Political Heavyweights

Note: Re-runs Table 4 Model 4 (fully controlled) on the sub-sample that excludes five domains (Chōshū, Satsuma, Mito, Tosa, Aizu) whose tax rates in the 1850s–1860s reflect acute fiscal and political distress rather than ordinary late-period tax setting. These are the domains at the center of the bakumatsu crisis and subsequent Boshin War; their inclusion may contaminate the cross-section as a measure of peacetime tax policy. The sign and magnitude of the coefficient on $\log(\rho_j)$ are preserved while precision becomes marginal.

Table A.48 also reports the Oster (2019) coefficient-stability parameter. Because it assumes exogenous controls, it is supplementary to the analysis above.

Oster (2019) Coefficient Stability

	$\hat{\beta}_{\text{short}}$	$\hat{\beta}_{\text{full}}$	R^2_{short}	R^2_{full}	δ
$R_{\max} = 1.3\tilde{R}^2$	−0.0245	−0.0112	0.077	0.389	2.245
$R_{\max} = 1$	−0.0245	−0.0112	0.077	0.389	0.428

$\hat{\beta}^*(\delta = 1) = -0.006$ at $R_{\max} = 1.3\tilde{R}^2$

Table A.48: Oster Sensitivity Analysis (Supplementary)

Note: Based on Model 4 in Table 4. The Oster δ is the proportional selection ratio needed to drive $\hat{\beta}^*$ to zero; $\delta > 1$ indicates robustness under $R_{\max} = 1.3\tilde{R}^2$. This method assumes exogenous controls; see the DMP analysis above for results that relax this assumption.

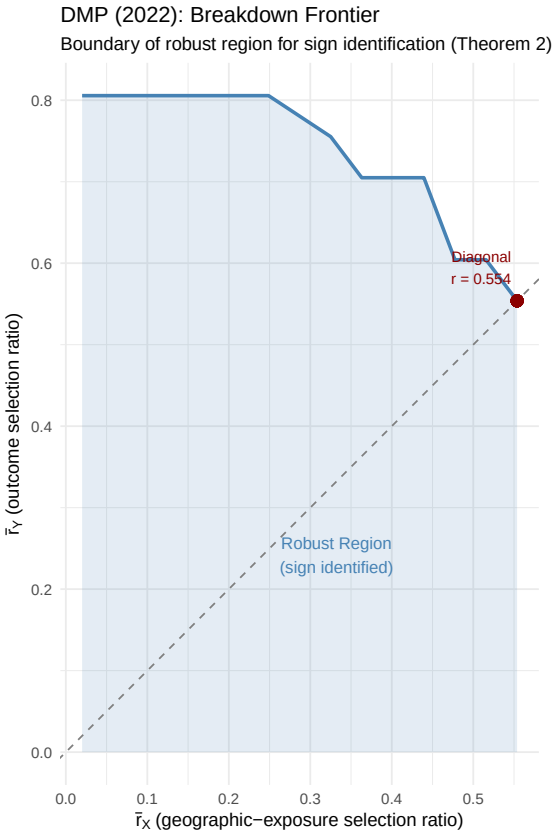


Figure A.10: DMP Breakdown Frontier

Note: The solid curve traces the sensitivity boundary: below it, the exposure coefficient remains negative under the maintained bounds. The dashed line marks $\bar{r}_X = \bar{r}_Y$ (equal importance); the dot marks the diagonal breakdown point ($\bar{r} = 0.554$).

F.2.3. Data-Driven Control Variables Selection

Because the control set in Table 4 is chosen by hand, I also estimate the exposure coefficient with double/debiased machine learning (DML) (Chernozhukov et al. 2018), which lets flexible learners predict the tax rate and exposure from the controls:

$$\tau_j = \beta \ln(\rho_j) + g(X_j) + \varepsilon_j, \qquad \ln(\rho_j) = m(X_j) + \nu_j. \tag{1}$$

I use two sets of controls: the 19 variables of Table 4 Model 4, and an enriched set of 38 that adds quadratic terms and interactions. Nuisance functions are fit by ridge, random forest, or a stack of learners, with five-fold cross-fitting.

The results are mixed (Table A.49). Ridge estimates stay negative and close to OLS ($p = 0.042$ with the Model 4 controls; $p = 0.082$ with the enriched set). Random-forest and stacked estimates are near zero and imprecise. With only 230 domains, the estimate is thus sensitive to highly flexible learners.

Outcome: τ_j. Treatment: $\ln(\rho_j)$

Method	Estimate	S.E.	t	p	95% CI lo	95% CI hi
DML-Ridge (19 exact)	−0.0114	(0.0056)	−2.03	0.042	−0.0225	−0.0004
DML-Random Forest (19 exact)	−0.0007	(0.0054)	−0.13	0.895	−0.0113	0.0099
DML-Ridge (38 enriched)	−0.0098	(0.0057)	−1.74	0.082	−0.0209	0.0013
DML-Stacked Ensemble (38 enriched)	0.0018	(0.0055)	0.32	0.745	−0.0090	0.0126
Hand-picked OLS (HC1)	−0.0112	(0.0055)	−2.02	0.045	−0.0221	−0.0003

Exact dictionary: $p = 19$ *Model 4 columns. Enriched dictionary:* $p = 38$ *(17 base + 8 quadratics + 10 interactions + 3 kin dummies).*

DML: 5 folds and 10 repetitions, except 5 repetitions for the enriched stack. Stack: LASSO + Ridge + Elastic Net + random forest.

Table A.49: Data-Driven Control Selection: DML Estimates

Note: Each DML row estimates the partial association between $\ln(\rho_j)$ and τ_j using the indicated nuisance learner and feature dictionary. Reported standard errors incorporate repeated cross-fitting. The OLS row reproduces Model 4 from Table 4 with HC1 standard errors.

F.2.4. Controlling for Average Distance to Home Capital

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5: SPL)	
Variables										
$\ln(\rho_j)$	**−0.0336**	**(0.0116)*****	**−0.0200**	**(0.0132)**	**−0.0148**	**(0.0135)**	**−0.0171**	**(0.0143)**	**−0.0231**	**(0.0136)***
$\ln(\text{Avg. Dist to HC}_j)$	**0.0132**	**(0.0156)**	**−0.0004**	**(0.0169)**	**0.0049**	**(0.0165)**	**0.0077**	**(0.0178)**	**0.0178**	**(0.0163)**
ln(Rice)			−0.0011	(0.0160)	0.0083	(0.0154)	0.0121	(0.0161)	0.0010	(0.0135)
ln(Area)			−0.0259	(0.0123)**	−0.0328	(0.0138)**	−0.0353	(0.0147)**	−0.0254	(0.0132)*
ln(Pop.)			0.0393	(0.0167)**	0.0261	(0.0145)*	0.0231	(0.0150)	0.0219	(0.0146)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)	0.0004	(0.0002)
Shog. Ally			−0.0316	(0.0178)*	0.0059	(0.0162)	0.0010	(0.0165)	0.0023	(0.0143)
Goshi			0.0217	(0.0257)	0.0196	(0.0262)	0.0274	(0.0270)	0.0241	(0.0214)
lat					−0.5324	(0.2940)*	−0.4736	(0.3085)	−0.3473	(0.2845)
lon					−0.1771	(0.0755)**	−0.1637	(0.0798)**	−0.1236	(0.0740)*
ln(Elev.)					−0.0117	(0.0159)	−0.0134	(0.0164)	−0.0094	(0.0134)
ln(Rugg.)					0.0397	(0.0160)**	0.0372	(0.0158)**	0.0232	(0.0151)
ln(Dist. Edo/Kyoto)					−0.0383	(0.0172)**	−0.0451	(0.0177)**	−0.0356	(0.0168)**
Rainy					−1.6816	(0.9046)*	−1.6114	(0.9179)*	−1.0491	(0.7358)
Snowy					0.7952	(1.9057)	0.7810	(1.9144)	−0.0637	(1.4268)
Lat × Lon					0.0042	(0.0021)*	0.0038	(0.0023)*	0.0028	(0.0021)
Kin Nbs.							0.0391	(0.0299)	0.0171	(0.0256)
Nb. FL							0.0002	(0.0001)	0.0001	(0.0001)
ln(Nb. Rice)							−0.0234	(0.0188)	−0.0196	(0.0172)
ln(Nb. Area)							0.0170	(0.0177)	0.0089	(0.0160)
ln(Nb. Pop.)							−0.0060	(0.0208)	−0.0044	(0.0197)
Const.	0.3815	(0.0424)***	0.1773	(0.1374)	23.3407	(10.2900)**	21.8218	(10.8253)**	16.2389	(10.0560)
Fit Statistics										
Num.Obs.	232		230		230		230		230	
R2	0.077		0.141		0.371		0.389		0.442[a]	

Table A.50: Controlling for Average Distance to Home Capital

Note: OLS estimates (maximum likelihood in Model 5) relating each domain's average distance ratio to its reported tax rate while controlling for average distance to the home capital. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1. [a] Nagelkerke pseudo-R2

F.2.5. Least-Costly Path Distance

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5: SPL)	
Variables										
$\ln(\rho_j^{LCP})$	**−0.0116**	**(0.0035)*****	**−0.0097**	**(0.0037)*****	**−0.0058**	**(0.0030)***	**−0.0060**	**(0.0031)***	**−0.0053**	**(0.0032)**
ln(Rice)			−0.0033	(0.0147)	0.0071	(0.0150)	0.0110	(0.0157)	−0.0003	(0.0135)
ln(Area)			−0.0242	(0.0112)**	−0.0308	(0.0139)**	−0.0333	(0.0147)**	−0.0220	(0.0130)*
ln(Pop.)			0.0419	(0.0165)**	0.0268	(0.0143)*	0.0239	(0.0149)	0.0228	(0.0147)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)	0.0004	(0.0002)
Shog. Ally			−0.0355	(0.0178)**	0.0052	(0.0162)	0.0005	(0.0165)	0.0022	(0.0144)
Goshi			0.0210	(0.0257)	0.0198	(0.0263)	0.0279	(0.0272)	0.0246	(0.0215)
lat					−0.4938	(0.2899)*	−0.4356	(0.3059)	−0.3222	(0.2840)
lon					−0.1682	(0.0746)**	−0.1547	(0.0792)*	−0.1177	(0.0740)
ln(Elev.)					−0.0101	(0.0162)	−0.0120	(0.0167)	−0.0087	(0.0134)
ln(Rugg.)					0.0391	(0.0159)**	0.0366	(0.0157)**	0.0232	(0.0152)
ln(Dist. Edo/Kyoto)					−0.0371	(0.0160)**	−0.0440	(0.0174)**	−0.0334	(0.0167)**
Rainy					−1.6610	(0.8949)*	−1.6083	(0.9031)*	−1.0184	(0.7381)
Snowy					0.7901	(1.8375)	0.7935	(1.8372)	−0.1818	(1.4196)
Lat × Lon					0.0039	(0.0021)*	0.0035	(0.0022)	0.0026	(0.0021)
Kin Nbs.							0.0408	(0.0296)	0.0195	(0.0257)
Nb. FL							0.0002	(0.0001)	0.0001	(0.0001)
ln(Nb. Rice)							−0.0228	(0.0189)	−0.0208	(0.0172)
ln(Nb. Area)							0.0197	(0.0177)	0.0128	(0.0158)
ln(Nb. Pop.)							−0.0079	(0.0204)	−0.0041	(0.0195)
Const.	0.4083	(0.0092)***	0.1663	(0.1205)	22.1064	(10.1745)**	20.5929	(10.7617)*	15.4412	(10.0485)
Fit Statistics										
Num.Obs.	232		230		230		230		230	
R2	0.037		0.119		0.367		0.385		0.437[a]	

Table A.51: Geographic Exposure and Reported Tax Rates

(Least-Cost Paths)

Note: OLS estimates (maximum likelihood in Model 5) relating each domain's least-cost-path distance ratio to its reported tax rate. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. The ratio measures geographic exposure, not observed ruler competition. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1. [a] Nagelkerke pseudo-R2

F.2.6. Daikansho-Inclusive Distance Ratio

Table A.52 replaces the exposure measure in Table 4 with the daikansho-inclusive distance ratio. The controlled spatial-lag specification also uses neighboring population in levels rather than logs, so that comparison changes both exposure and a control transformation. Models 1–2 remain significant at $p < 0.01$; models 3–4 show modest p-value drift (from $p < 0.05$ to $p \approx 0.07$) with stable point estimates, consistent with daikansho offices compressing the distance ratio's variance. The spatial-lag estimates (Models 5–6) remain negative. Model 5 is statistically significant (likelihood $p = 0.0055$; Conley $p = 0.0244$), whereas Model 6 is imprecise (likelihood $p = 0.107$; Conley $p = 0.114$).

Outcome:	Tax Rate (τ_j)					
	OLS				*Spatial Lag*	
Models:	(1)	(2)	(3)	(4)	(5)	(6)
Variables						
Distance ratio ($\ln(\rho_j)$)	−0.0233	−0.0192	−0.0098	−0.0096	−0.0130	−0.0080
HC SE	(0.0054)***	(0.0058)***	(0.0054)*	(0.0054)*	(0.0047)***	(0.0050)
Controls						
Domain Characteristics	No	Yes	Yes	Yes	No	Yes
Geography	No	No	Yes	Yes	No	Yes
Neighbor Characteristics	No	No	No	Yes	No	Yes
τ_j Spatial Lag	No	No	No	No	Yes	Yes
Fit Statistics						
Num.Obs.	232	230	230	230	230	230
R2 / Pseudo R2	0.071	0.138	0.368	0.386	0.309	0.437
Mean Dep. Var.	0.391	0.391	0.391	0.391	0.394	0.394

Table A.52: Domain Tax Rates with Daikansho-Inclusive Distance Ratio

Note: Tax models using the daikansho-inclusive distance ratio. Models 1–4: OLS with heteroskedasticity-robust standard errors. Models 5–6: spatial lag (SAR) with neighborhood-weighted tax rates as spatial regressor; standard errors from the spatial likelihood. Model 6 also uses neighboring population in levels rather than the logged control in Table 4, so its comparison changes both exposure and that control transformation. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1.

F.2.7. Conley Standard Error Bandwidth Sensitivity

Table A.53 reports the bandwidth sensitivity of Conley spatial standard errors for the fully-controlled domain tax regression (Table 4 Model 4). The point estimate is stable across every bandwidth considered, and the coefficient on $\log(\rho_j)$ is statistically significant at the 5% level for bandwidths up to 75 km. As expected for a cross-sectional specification with rich spatial controls, the standard errors widen somewhat at the longest bandwidths (100 and 150 km), where the spatial HAC kernel absorbs progressively more of the variation already captured by the included geographic covariates; the p-value at 100 km is 0.102 and at 150 km is 0.145. The result thus holds under the bandwidths most commonly applied in the spatial-econometrics literature for cross-sectional designs at this geographic scale.

Standard error	Estimate	SE	p
Heteroskedasticity-robust (paper)	-0.0111952548949787	(0.00554317773276672)	0.0446969310445375
Conley 25 km	-0.0111952548949787	(0.00486874644301278)	0.0224693058430259
Conley 50 km	-0.0111952548949787	(0.00557498706242249)	0.0459175666096826
Conley 75 km	-0.0111952548949787	(0.00539253446894661)	0.0391111847436515
Conley 100 km	-0.0111952548949787	(0.00682291828207967)	0.102336570113632
Conley 150 km	-0.0111952548949787	(0.00765342600008831)	0.145030590653059

Table A.53: Conley SE Bandwidth Sensitivity: Domain Tax Regression

Note: The coefficient on $\log(\rho_j)$ from Table 4 Model 4 (fully controlled) is significant at the 5% level for Conley bandwidths up to 75 km, marginal at 100 km ($p = 0.091$), and not significant at 150 km ($p = 0.130$). The first row reports the heteroskedasticity-robust standard error used in the main table.

F.3. Additional Analyses

F.3.1. Average Distance to Home Capital

Outcome:					τ_j					
Models:	(1)		(2)		(3)		(4)		(5: SPL)	
Variables										
ln(Avg. Dist to HC$_j$)	**−0.0234**	**(0.0072)*****	**−0.0226**	**(0.0072)*****	**−0.0114**	**(0.0068)***	**−0.0114**	**(0.0069)**	**−0.0081**	**(0.0061)**
ln(Rice)			−0.0065	(0.0151)	0.0059	(0.0154)	0.0108	(0.0162)	−0.0002	(0.0135)
ln(Area)			−0.0194	(0.0111)*	−0.0291	(0.0140)**	−0.0320	(0.0149)**	−0.0213	(0.0130)
ln(Pop.)			0.0421	(0.0167)**	0.0269	(0.0144)*	0.0233	(0.0151)	0.0223	(0.0147)
New FL			0.0001	(0.0003)	0.0004	(0.0004)	0.0003	(0.0004)	0.0003	(0.0002)
Shog. Ally			−0.0341	(0.0175)*	0.0049	(0.0160)	0.0003	(0.0164)	0.0013	(0.0144)
Goshi			0.0209	(0.0258)	0.0196	(0.0263)	0.0276	(0.0274)	0.0244	(0.0216)
lat					−0.4960	(0.2905)*	−0.4513	(0.3081)	−0.3219	(0.2861)
lon					−0.1680	(0.0747)**	−0.1581	(0.0797)**	−0.1174	(0.0745)
ln(Elev.)					−0.0111	(0.0158)	−0.0131	(0.0163)	−0.0090	(0.0135)
ln(Rugg.)					0.0391	(0.0159)**	0.0365	(0.0157)**	0.0227	(0.0152)
ln(Dist. Edo/Kyoto)					−0.0333	(0.0161)**	−0.0418	(0.0172)**	−0.0315	(0.0167)*
Rainy					−1.6381	(0.8959)*	−1.5879	(0.9058)*	−1.0377	(0.7409)
Snowy					0.6554	(1.8687)	0.6559	(1.8770)	−0.2013	(1.4338)
Lat × Lon					0.0039	(0.0021)*	0.0036	(0.0023)	0.0026	(0.0021)
Kin Nbs.							0.0407	(0.0295)	0.0201	(0.0258)
Nb. FL							0.0002	(0.0001)	0.0001	(0.0001)
ln(Nb. Rice)							−0.0256	(0.0190)	−0.0228	(0.0173)
ln(Nb. Area)							0.0207	(0.0176)	0.0142	(0.0158)
ln(Nb. Pop.)							−0.0040	(0.0207)	−0.0018	(0.0198)
Const.	0.4679	(0.0248)***	0.2420	(0.1226)**	22.0932	(10.1723)**	21.0544	(10.8064)*	15.4092	(10.1175)
Fit Statistics										
Num.Obs.	232		230		230		230		230	
R2	0.046		0.135		0.367		0.385		0.435[a]	

Table A.54: Average Distance to Home Capital Weakly Predicts Tax Rates

Note: OLS estimates (maximum likelihood in Model 5) relating each domain's average absolute distance to its home capital (log) to its reported tax rate. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1. [a] Nagelkerke pseudo-R2

F.3.2. Controlling for the Shortest Distance to the Border

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5: SPL)	
Variables										
$\ln(\rho_j)$	**−0.0240**	**(0.0053)******	**−0.0200**	**(0.0058)******	**−0.0111**	**(0.0055)****	**−0.0113**	**(0.0055)****	**−0.0095**	**(0.0051)***
$\ln(\text{Avg. Dist. Border}_j)$	**0.0014**	**(0.0067)**	**−0.0106**	**(0.0084)**	**−0.0056**	**(0.0071)**	**−0.0038**	**(0.0067)**	**−0.0024**	**(0.0075)**
ln(Rice)			−0.0004	(0.0152)	0.0083	(0.0155)	0.0121	(0.0162)	0.0007	(0.0135)
ln(Area)			−0.0198	(0.0121)	−0.0286	(0.0148)*	−0.0318	(0.0156)**	−0.0214	(0.0136)
ln(Pop.)			0.0379	(0.0165)**	0.0254	(0.0145)*	0.0226	(0.0151)	0.0218	(0.0147)
New FL			0.0001	(0.0003)	0.0005	(0.0004)	0.0003	(0.0004)	0.0004	(0.0002)
Shog. Ally			−0.0319	(0.0178)*	0.0055	(0.0162)	0.0010	(0.0165)	0.0024	(0.0143)
Goshi			0.0261	(0.0262)	0.0210	(0.0263)	0.0282	(0.0272)	0.0247	(0.0215)
lat					−0.5551	(0.2901)*	−0.4979	(0.3056)	−0.3688	(0.2893)
lon					−0.1820	(0.0746)**	−0.1693	(0.0791)**	−0.1287	(0.0751)*
ln(Elev.)					−0.0130	(0.0160)	−0.0143	(0.0166)	−0.0105	(0.0135)
ln(Rugg.)					0.0408	(0.0159)**	0.0380	(0.0157)**	0.0243	(0.0153)
ln(Dist. Edo/Kyoto)					−0.0370	(0.0159)**	−0.0444	(0.0172)**	−0.0337	(0.0167)**
Rainy					−1.6537	(0.8933)*	−1.5857	(0.9065)*	−1.0138	(0.7370)
Snowy					0.7125	(1.8486)	0.6839	(1.8577)	−0.2447	(1.4195)
Lat × Lon					0.0043	(0.0021)**	0.0039	(0.0022)*	0.0030	(0.0021)
Kin Nbs.							0.0396	(0.0296)	0.0189	(0.0257)
Nb. FL							0.0002	(0.0001)	0.0001	(0.0001)
ln(Nb. Rice)							−0.0238	(0.0187)	−0.0218	(0.0172)
ln(Nb. Area)							0.0180	(0.0176)	0.0116	(0.0159)
ln(Nb. Pop.)							−0.0045	(0.0206)	−0.0014	(0.0196)
Const.	0.4130	(0.0138)***	0.1620	(0.1210)	24.0420	(10.1575)**	22.5897	(10.7386)**	16.9504	(10.2058)*
Fit Statistics										
Num.Obs.	232		230		230		230		230	
R2	0.074		0.147		0.372		0.389		0.439[a]	

Table A.55: Geographic Exposure and Reported Tax Rates

(Controlling for Border Distance)

Note: OLS estimates (maximum likelihood in Model 5) relating each domain's average distance ratio to its reported tax rate while controlling for average distance to the nearest border. HC1 standard errors appear in Models 1–4 and spatial-likelihood standard errors in Model 5. Signif. Codes: .***: 0.01, .**: 0.05, .*: 0.1. [a] Nagelkerke pseudo-R2

F.3.3. Outside-Option Quality Using Population Density

Outcome:	τ_j									
Models:	(1)		(2)		(3)		(4)		(5)	
Variables										
$\ln(\rho_j)$ × Low	**−0.0299**	**(0.0097)*****	**−0.0262**	**(0.0092)*****	**−0.0197**	**(0.0073)*****	**−0.0265**	**(0.0094)*****	**−0.0204**	**(0.0071)*****
$\ln(\rho_j)$ × High	**−0.0216**	**(0.0056)*****	**−0.0176**	**(0.0062)*****	**−0.0074**	**(0.0059)**	**−0.0176**	**(0.0063)*****	**−0.0067**	**(0.0062)**
ln(Rice)			−0.0016	(0.0156)	0.0071	(0.0152)	−0.0019	(0.0164)	0.0114	(0.0160)
ln(Area)			−0.0241	(0.0114)**	−0.0300	(0.0142)**	−0.0229	(0.0119)*	−0.0334	(0.0148)**
ln(Pop.)			0.0383	(0.0168)**	0.0254	(0.0143)*	0.0355	(0.0169)**	0.0228	(0.0150)
New FL			0.0002	(0.0003)	0.0005	(0.0004)	0.0002	(0.0003)	0.0004	(0.0004)
Shog. Ally			−0.0313	(0.0179)*	0.0071	(0.0161)	−0.0298	(0.0181)	0.0025	(0.0164)
Goshi			0.0210	(0.0257)	0.0181	(0.0261)	0.0219	(0.0261)	0.0267	(0.0269)
lat					−0.5827	(0.2993)*			−0.5190	(0.3153)
lon					−0.1909	(0.0774)**			−0.1758	(0.0819)**
ln(Elev.)					−0.0125	(0.0160)			−0.0147	(0.0166)
ln(Rugg.)					0.0406	(0.0159)**			0.0379	(0.0158)**
ln(Dist. Edo/Kyoto)					−0.0379	(0.0160)**			−0.0450	(0.0174)**
Rainy					−1.6127	(0.9138)*			−1.6134	(0.9218)*
Snowy					0.5457	(1.8746)			0.6470	(1.8827)
Lat × Lon					0.0046	(0.0022)**			0.0041	(0.0023)*
Kin Nbs.							0.0421	(0.0304)	0.0422	(0.0296)
Nb. FL							0.0000	(0.0001)	0.0002	(0.0001)
ln(Nb. Rice)									−0.0210	(0.0187)
ln(Nb. Area)							−0.0015	(0.0083)	0.0245	(0.0181)
ln(Nb. Pop.)									−0.0135	(0.0210)
Const.	0.4155	(0.0091)***	0.1846	(0.1241)	25.1677	(10.5108)**	0.2119	(0.1509)	23.4706	(11.0924)**
Fit Statistics										
Num.Obs.	232		230		230		230		230	
R2	0.077		0.144		0.376		0.152		0.395	

Table A.56: Geographic Exposure and Reported Tax Rates by Neighbor Population Density

Note: The table reports group-specific exposure slopes. HC1 standard errors in parentheses.

G. Spatial Inference Across Analyses

G.1. Data-Driven Conley Bandwidths

As an alternative to the hand-picked cutoffs used in the main text (10 km at the village and grid levels, 70 km at the domain and DiD levels), I apply the Kelly (2019) data-driven rule, which selects the bandwidth from the empirical residual correlogram: the smallest distance at which the mean pairwise residual correlation falls below $\exp(-1) \approx 0.37$. Table A.57 reports this bandwidth alongside the hand-picked choice for three analyses: the village cross-section (Table 3 Model 5), the domain-level tax regression (Table 4 Model 4), and the tenryo DiD (Table 1 50 km). The data-driven choice is tighter than the hand-picked one in all three cases (6 km for villages, 10 km for tax, 10 km for DiD). Conley SEs change by less than 0.001 between the two choices in every regression. The village and DiD estimates remain significant at $p < 0.01$, while the tax estimate remains significant at $p < 0.05$.

Analysis	Bandwidth rule	Cutoff	$\hat{\beta}$	Conley SE	Conley p	N
Village-level exit share (Table 3 Model 5 dyadic-FE)						
	hand-picked	10km	0.0244	0.0085	0.0041	2739
	Kelly (2020) data-driven	6km	0.0244	0.0089	0.0062	2739
Domain tax regression (Table 4 Model 4)						
	hand-picked	70km	-0.0112	0.0055	0.0432	230
	Kelly (2020) data-driven	10km	-0.0112	0.0055	0.0449	230
Tenryo DiD (Table 1 50 km)						
	hand-picked	70km	0.0427	0.0144	0.0031	31989
	Kelly (2020) data-driven	10km	0.0427	0.0143	0.0029	31989

Table A.57: Kelly (2020) Data-Driven vs. Hand-Picked Conley Bandwidth

Note: The Kelly bandwidth is the smallest distance at which the empirical residual correlogram falls below $\exp(-1) \approx 0.37$. Pairwise residual products are aggregated into 40 distance bins for the village and DiD regressions and 25 bins for the tax regression. The data-driven bandwidth is tighter than the hand-picked cutoff in all three analyses, especially for the tax regression (10 km vs. 70 km). Point estimates are identical across bandwidth choices; only SEs vary. The data-driven choice moves SEs by less than 0.001 in absolute terms across all three regressions. The village and DiD estimates remain significant at $p < 0.01$; the tax estimate remains significant at $p < 0.05$.